

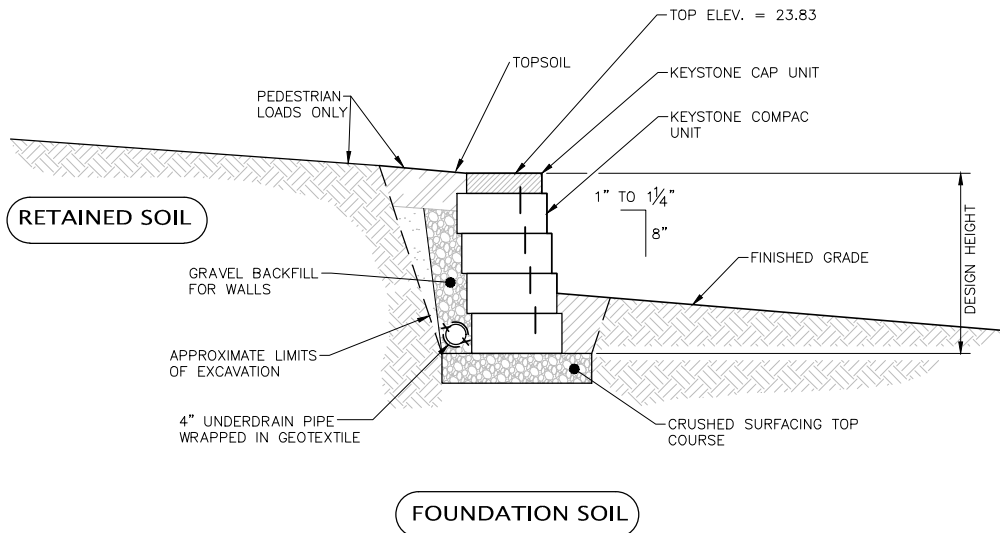


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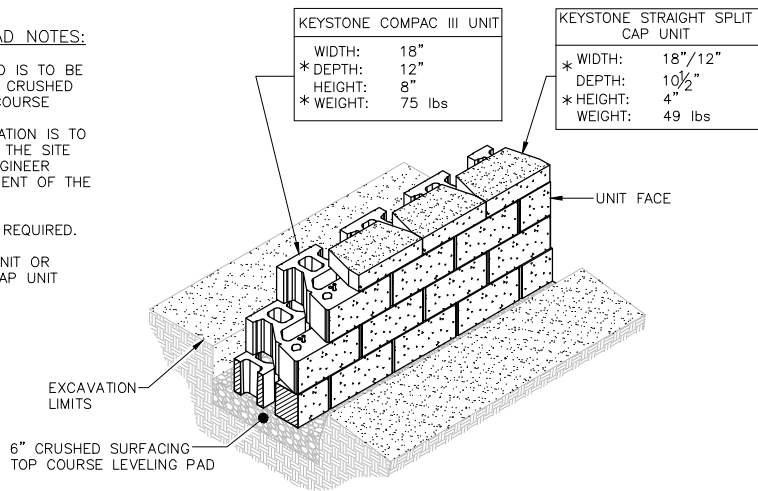
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W:\2020\2020-070 EASTSOUND WWTP DESIGN\DWG\2020-070 C7.1-C7.8 CIVIL DETAILS.DWG - 11/2/2022 3:26 PM - Jeff Smith



TYPICAL GRAVITY WALL SECTION
STANDARD UNIT - 1" SETBACK

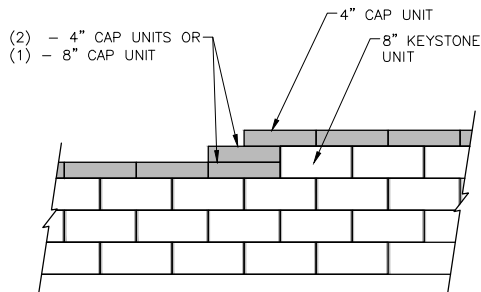
BASE LEVELING PAD NOTES:

1. THE LEVELING PAD IS TO BE CONSTRUCTED OF CRUSHED SURFACING TOP COURSE
2. THE BASE FOUNDATION IS TO BE APPROVED BY THE SITE GEOTECHNICAL ENGINEER PRIOR TO PLACEMENT OF THE LEVELING PAD.
3. FIBERGLASS PINS REQUIRED.
4. UNIVERSAL CAP UNIT OR STRAIGHT SPLIT CAP UNIT OPTION ALLOWED.



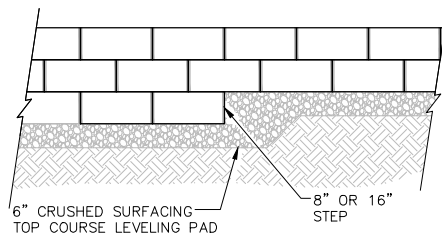
COMPAC III UNIT/BASE PAD ISOMETRIC SECTION VIEW

* DIMENSIONS & WEIGHT MAY VARY BY REGION



NOTE:
1. SECURE ALL CAP UNITS WITH KEYSTONE KAPSEAL OR EQUAL.

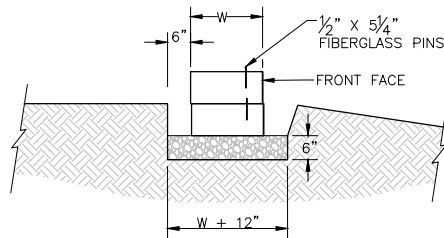
TOP OF WALL STEPS
(IF NEEDED)



ELEVATION

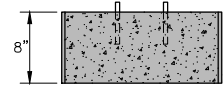
NOTE:

1. THE LEVELING PAD IS TO BE CONSTRUCTED OF CRUSHED CRUSHED SURFACING TOP COURSE.

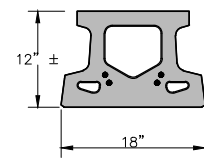


SECTION

LEVELING PAD DETAIL



COMPAC III ELEVATION

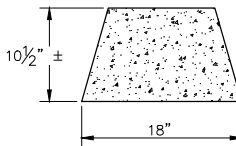


COMPAC III PLAN

COMPAC III UNIT
*Dimensions May Vary by Region

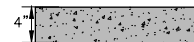


CAP UNIT ELEVATION

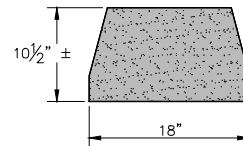


CAP UNIT PLAN

UNIVERSAL
CAP UNIT OPTION



CAP UNIT ELEVATION



CAP UNIT PLAN

STRAIGHT SPLIT
CAP UNIT OPTION

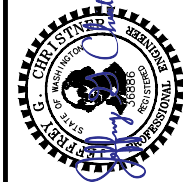


BID SET

11-02-2022



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DESIGNED BY	JCC
DRAWN BY	JCS
CHECKED BY	AWL

EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

SAN JUAN COUNTY

CIVIL DETAILS

SHEET	DATE	SCALE	JOB NUMBER
C7.8	NOV 2, 2022	AS SHOWN	2020-070
PAGE			
50			96

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W:\2020\2020-070 EASTSOUND WWP DESIGN\DWG\2020-070 C7.9-C7.10 FLOW METER & VALVE VAULT DETAILS.DWG - 11/2/2022 3:27 PM - Jeff Smith

NO.	REVISIONS	BY	DATE
6	ECOLOGY REVIEW COMMENTS	JGC	8/9/22



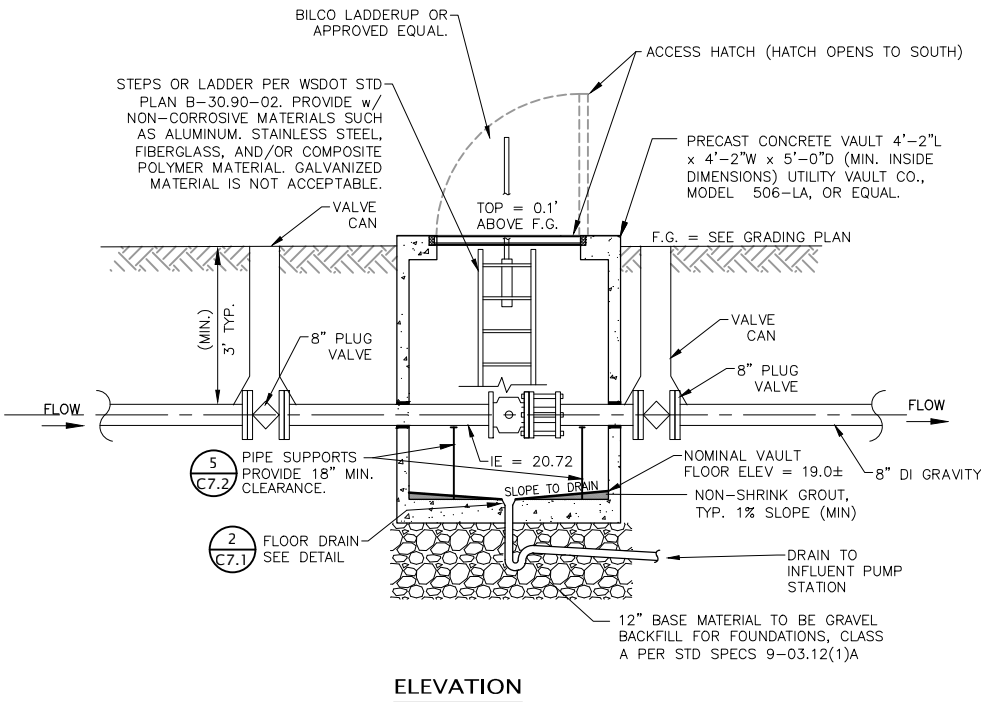
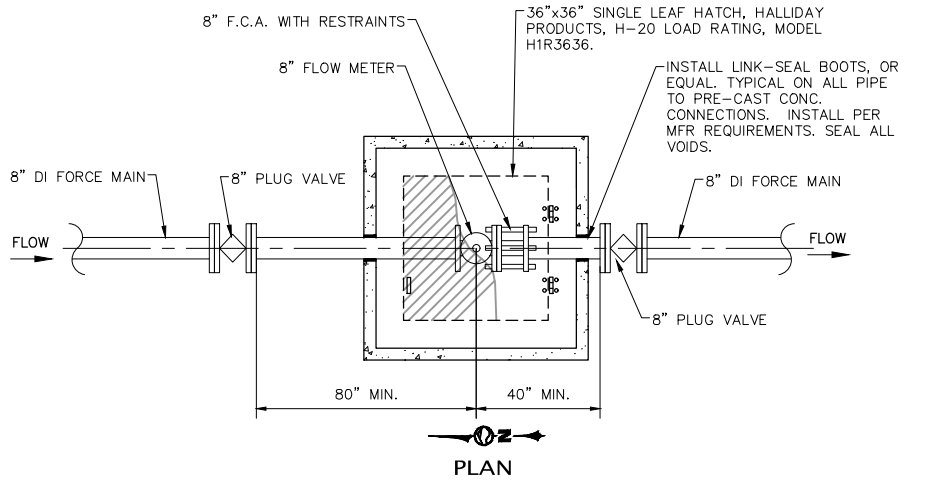
DESIGNED BY	JGC
DRAWN BY	JGS
CHECKED BY	AWL

EASTSOUND SEWER AND WATER DISTRICT
WASHINGTON
SAN JUAN COUNTY
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I
FLOW METER & VALVE VAULT DETAILS

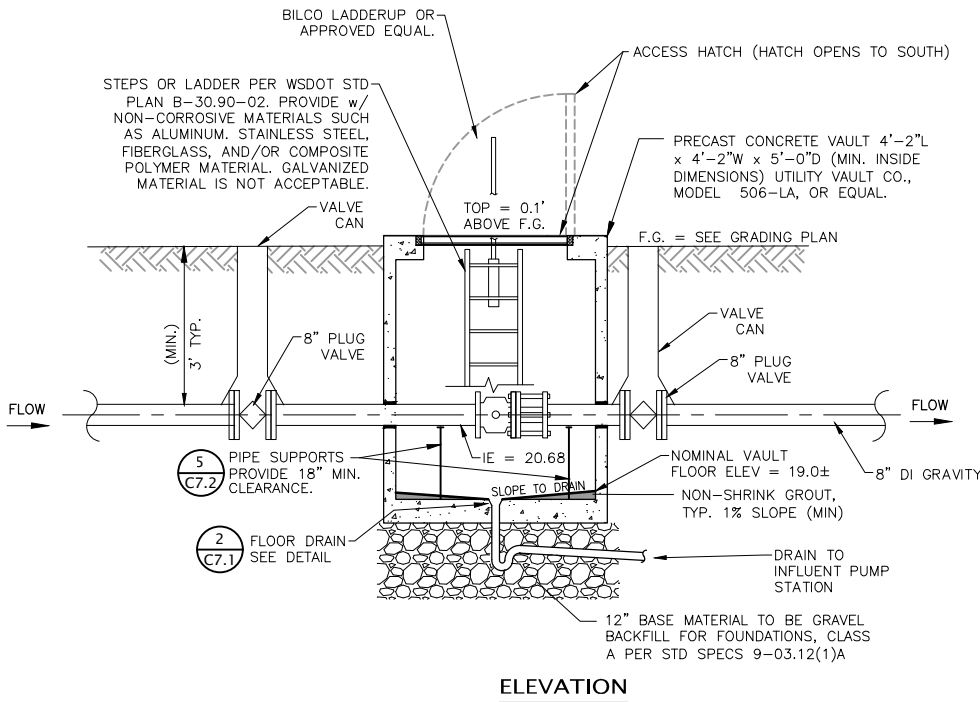
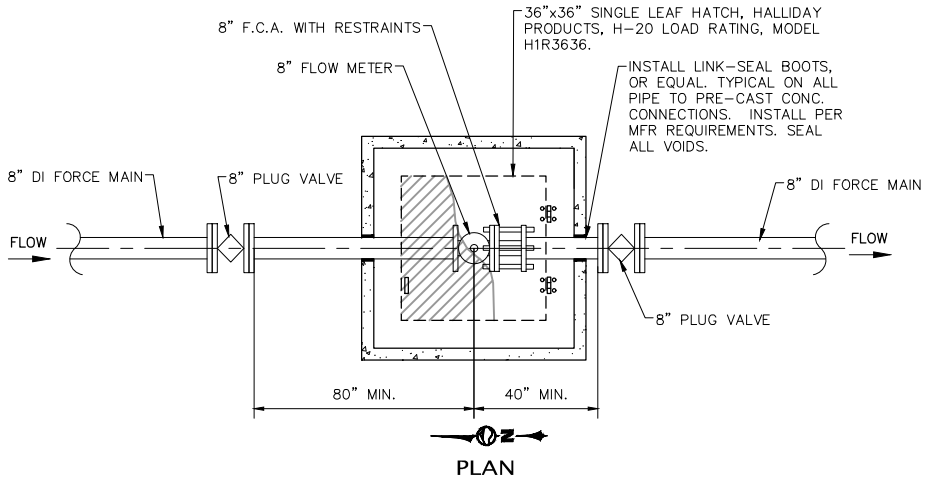
SHEET	C7.10
DATE	NOV 2, 2022
SCALE	AS SHOWN
JOB NUMBER	2020-070
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BID SET



1 TRAIN 2 FLOW METER VAULT, PLAN
NOT TO SCALE



2 TRAIN 3 FLOW METER VAULT, PLAN
NOT TO SCALE

PLOT SETTINGS: Adobe PDF1.pc3, ANSI full bleed B (17.00 x 11.00 inches), Portrait, 1:2, WE APWA_UNSCREENED.ctb
W:\2020\2020-070 EASTSOUND WWTP DESIGN\DWG\2020-070 A1.1-A1.3 BLOWER BUILDING.DWG - 11/2/2022 3:27 PM - Jeff Smith

NO.	REVISIONS	BY	DATE
6	ECOLOGY REVIEW COMMENTS	JGC	8/9/22

GENERAL NOTES:
1. PROCESS AIR PIPING AND 2W PIPING NOT SHOWN FOR CLARITY, SEE CIVIL SHEETS C1.1, C6.4, AND C6.5 FOR PIPING INFORMATION.

FINISH SCHEDULE										
ABBREVIATIONS		ROOM NO.	FLOOR		BASE	WALL		CEILING		
R.B. = RUBBER BASE			MAT'L	FINISH	MAT'L	MAT'L	FINISH	MAT'L	FINISH	HEIGHT
EXP STR = EXPOSED STRUCTURE										
GWB = GYPSUM WALL BOARD										
AC = ACOUSTICAL WALL BOARD										
FCS = FIBER CEMENT SIDING										
TILE = SUSPENDED CEILING TILE										
PLY = 1/2" PLYWOOD										
ROOM NAME										
BLOWER ROOM										
101	CONCRETE	SEALER	6" R.B.	GWB		PAINT	EXP STR	PAINT	VARIES	

DOOR SCHEDULE

DOOR NO.	DOOR SIZE WxHxT	TYPE	MATERIAL	FINISH	HARDWARE GROUP #
101A	10'-0"x8'-0"x1 3/4"	B	INS HM	PT	-
101B	3'-0"x6'-8"x1 3/4"	A	INS HM	PT	-
101C	5'-0"x8'-0"x1 3/4"	A	INS HM	PT	-

FLUSH PANEL

TYPE A

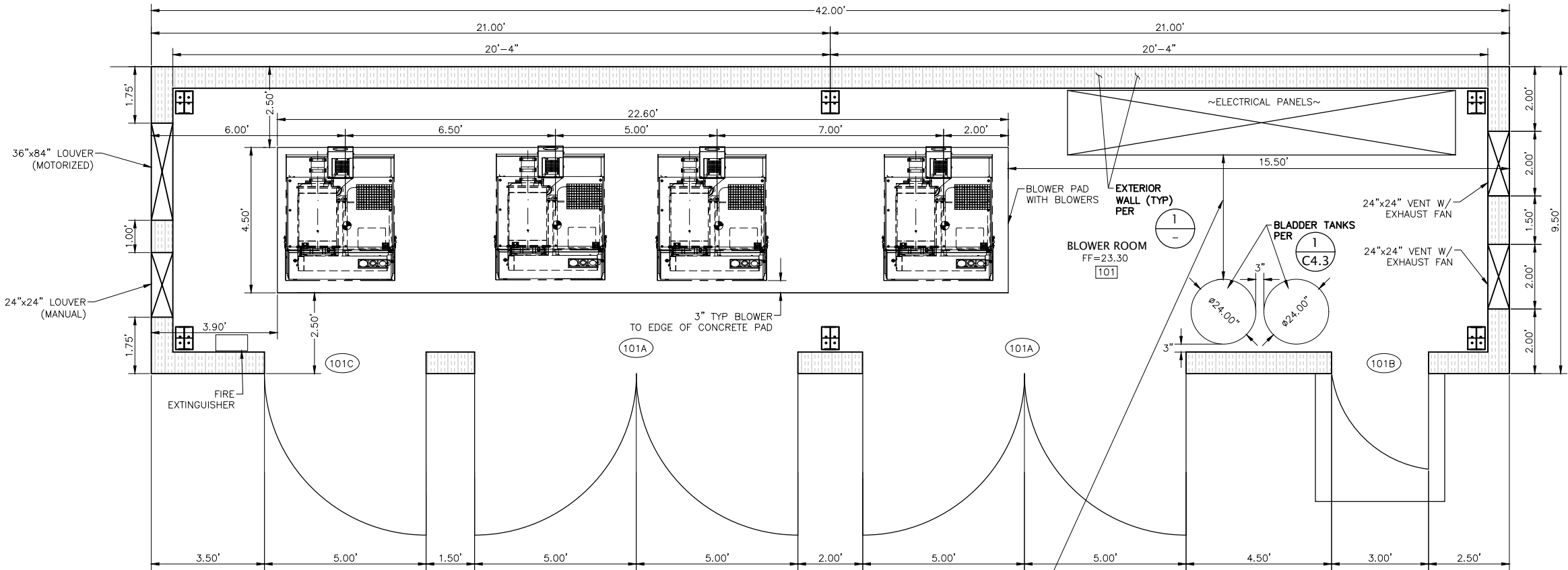
FLUSH PANEL

TYPE B

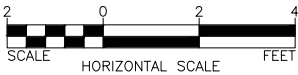
ABBREVIATIONS

INS HM = INSULATED HOLLOW METAL

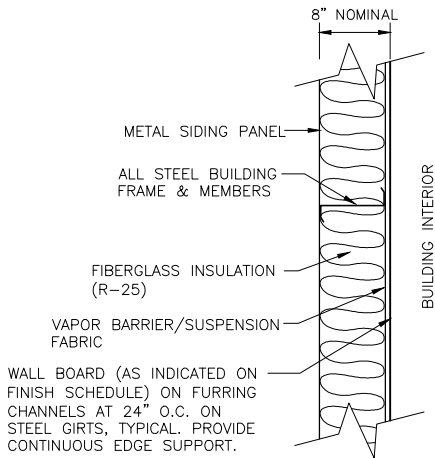
PT = PAINT



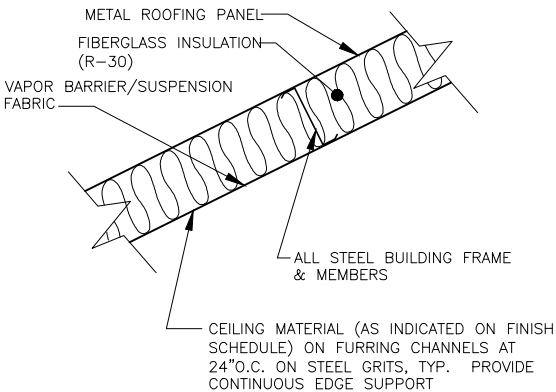
BLOWER BUILDING - PLAN



CONTRACTOR TO COORDINATE AND ENSURE ALL ELECTRICAL PANELS ARE CONFIGURED WITH 3.5 FOOT CLEARANCE, AS NEEDED, PER ELECTRICAL REQUIREMENTS.



1 TYPICAL EXTERIOR WALL
NOT TO SCALE



2 TYPICAL ROOF SECTION
NOT TO SCALE



BID SET

11-02-2022



DESIGNED BY	JGC
DRAWN BY	SNR
CHECKED BY	AWL

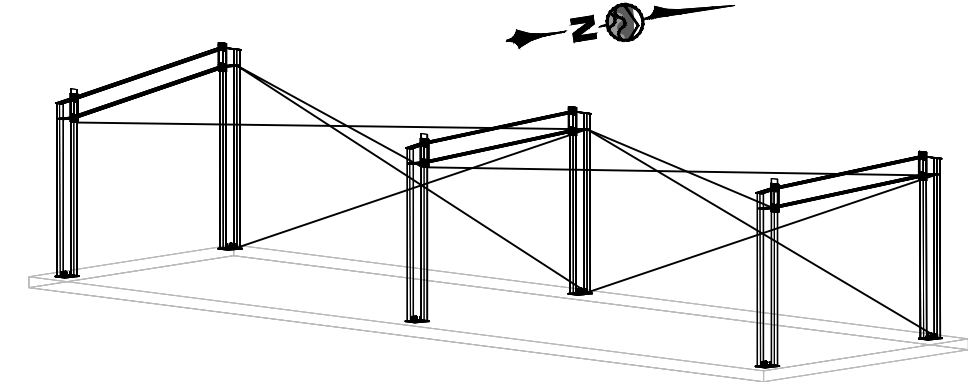
EASTSOUND SEWER AND WATER DISTRICT
WASHINGTON
SAN JUAN COUNTY
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

DATE	NOV 2, 2022
SCALE	AS SHOWN
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BLOWER BUILDING PLAN

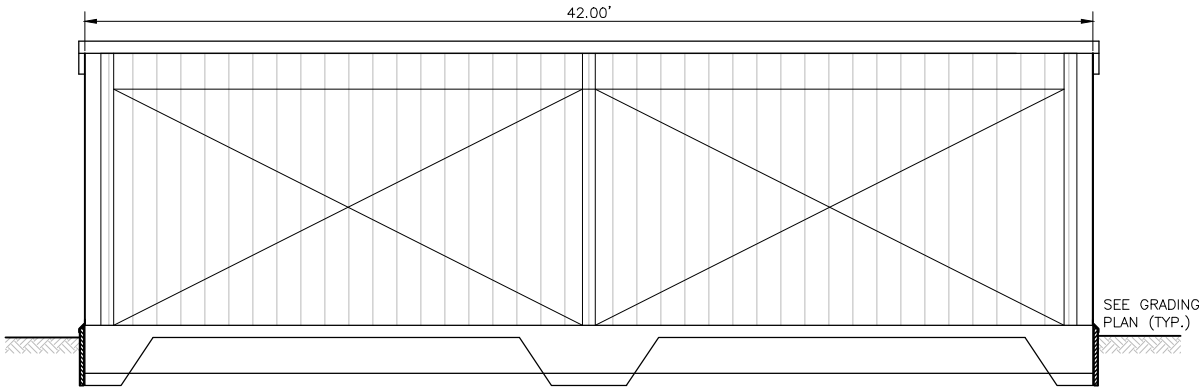
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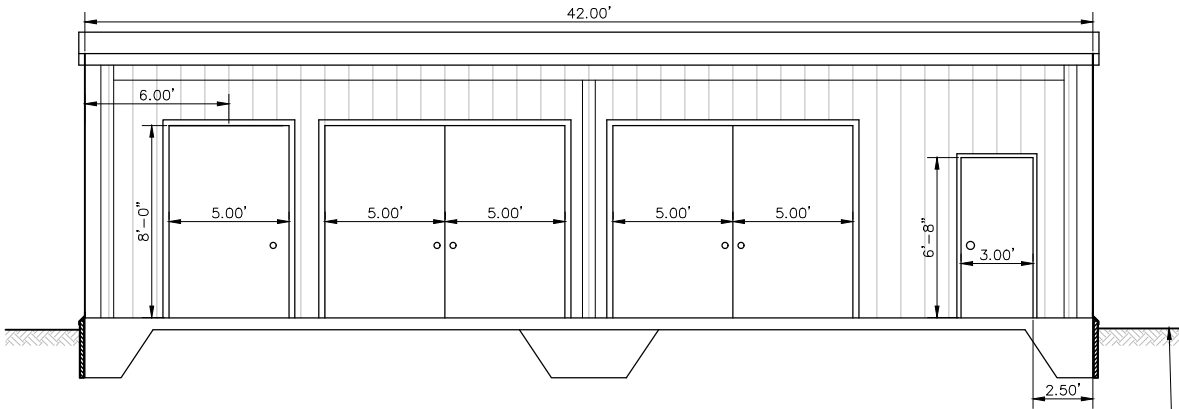
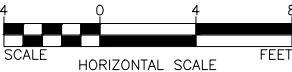
CONCEPTUAL FRAMING

NOT TO SCALE

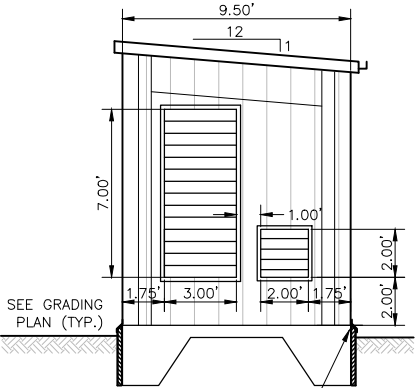
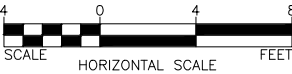
NOTE: CONTRACTOR SHALL PROVIDE
PRE-ENGINEERED METAL BUILDING WITH
SIMILAR COLUMN AND BRACING LAYOUT.



NORTH ELEVATION

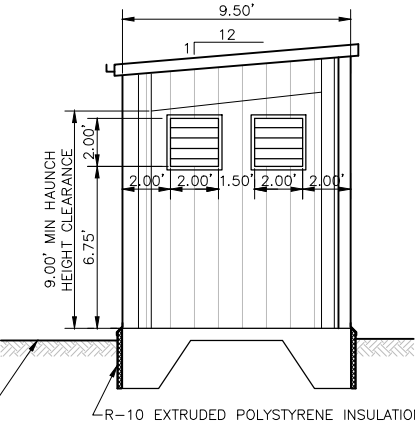
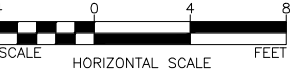


SOUTH ELEVATION

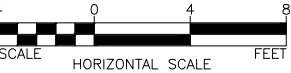


FLASHING TO COVER CHAMFERED
EDGE OF INSULATION BOARD.
MATCH METAL BUILDING TRIM
COLOR. LAP UNDER SIDING AND
OVER INSULATION BOARD A
MINIMUM OF 3-INCHES (TYP.).

WEST ELEVATION



EAST ELEVATION

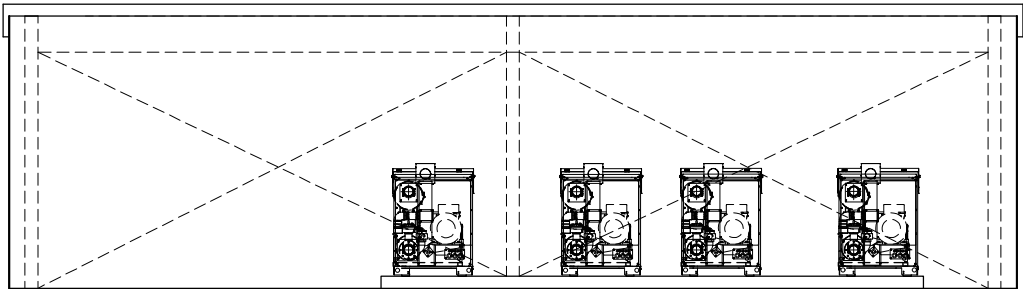


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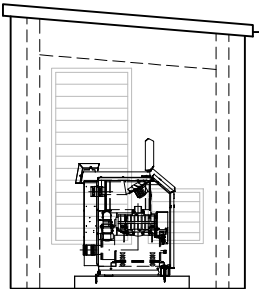
11-02-2022

SHEET		DATE		DESIGNED BY						WILSON ENGINEERING WILSONENGINEERING.COM	
A1.2		NOV 2, 2022		JCC							
PAGE		SCALE		DRAWN BY		SNR		CHECKED BY		AWL	
		AS SHOWN		SAN JUAN COUNTY		WASHINGTON					
		JOB NUMBER		WASTEWATER TREATMENT PLANT UPGRADE – PHASE I							
		2020–070		BLOWER BUILDING EXTERIOR ELEVATIONS 1							
54		OF		96							

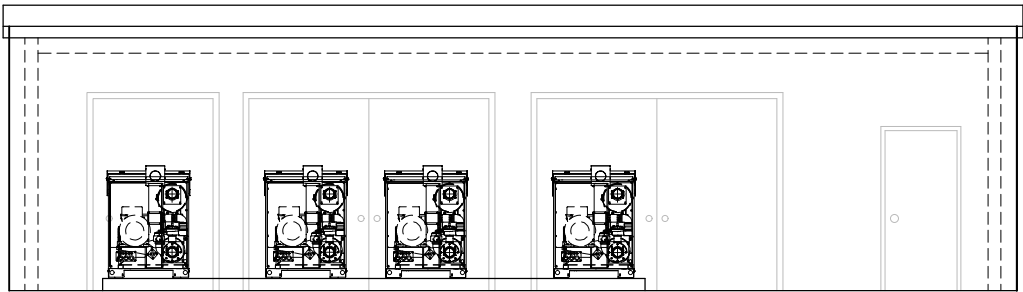
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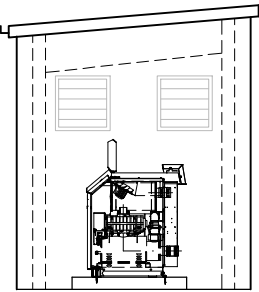
NORTH ELEVATION



WEST ELEVATION

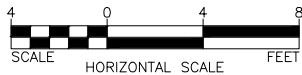


SOUTH ELEVATION

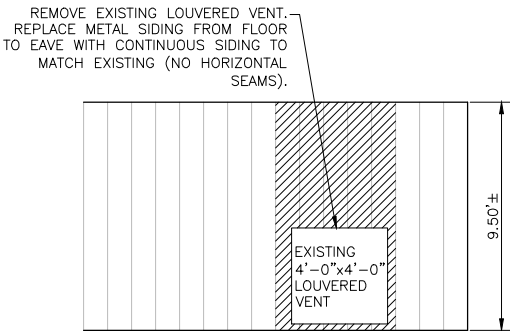


EAST ELEVATION

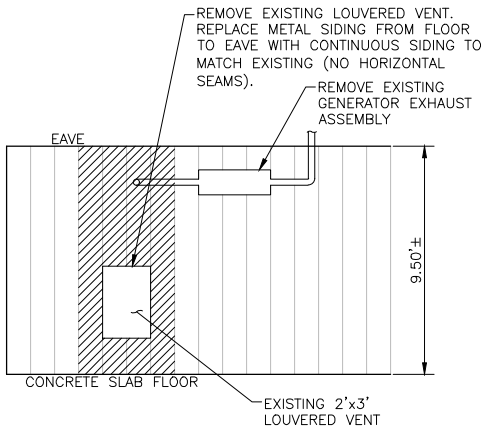
PROPOSED BLOWER BUILDING INTERIOR ELEVATIONS



GENERAL NOTES:
1) PROCESS AIR PIPING AND 2W PIPING NOT SHOWN FOR CLARITY.
SEE CIVIL SHEETS C1.1, C6.4, AND C6.5 FOR PIPING INFORMATION

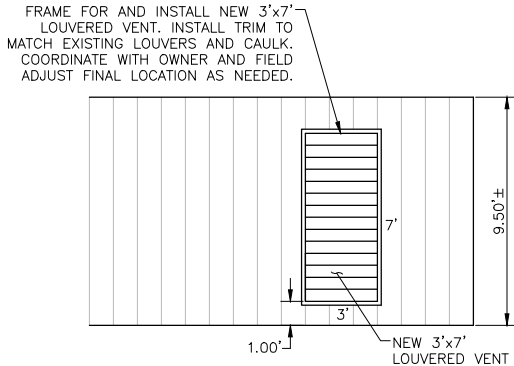
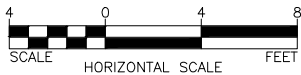


EXISTING OFFICE BUILDING
SOUTH ELEVATION @ 4'x4' LOUVER

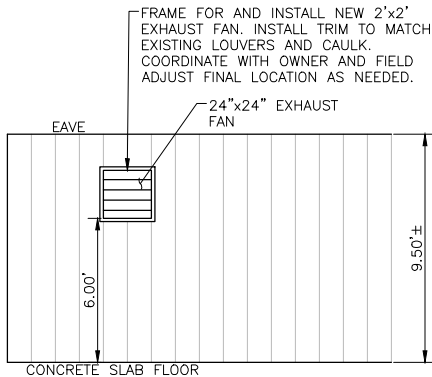


EXISTING OFFICE BUILDING
EAST ELEVATION @ 2'x3' LOUVER

EXISTING OFFICE
BUILDING ELEVATIONS

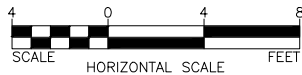


NEW LOUVER FOR
EXISTING WWTP BUILDING
(TYPICAL FOR ALL THREE LOCATIONS)



NEW EXHAUST FAN
FOR EXISTING BUILDING
(TYPICAL FOR ALL THREE LOCATIONS)

EXISTING WWTP
BUILDING ELEVATIONS



BID SET

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11-02-2022



DESIGNED BY	JCC
DRAWN BY	JCS
CHECKED BY	AWL

EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON
SAN JUAN COUNTY
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

DATE	NOV 2, 2022
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PAGE	55 OF 96



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-- GENERAL NOTES --

- 1 DESIGN BASIS: Designed in accordance with the 2018 International Building Code (IBC).
- 2 RISK CATEGORY: III (per IBC Table 1604.5)
- 3 DESIGN DEAD / LIVE LOADS
- Floor Live Load: 100 PSF Typical
- 4 DESIGN SNOW LOADS (ASCE 7-16)
- Ground Snow Load: Pg = 25 PSF
- Flat Roof Snow Load: Pf = 25 PSF
- Minimum Uniform Roof Snow Load = 25 PSF (unreducible)
- Thermal Factor: Ct = 1.1 / Exposure Factor: Ce = 1.1 / Snow Load Importance Factor: Is = 1.1
- 5 DESIGN WIND LOADS (ASCE 7-16)
- V(ult) = 105 MPH / V(asd) = 81 MPH
- Exposure: C / Internal Pressure Coefficient: +/- 0.18
- Components & Cladding: Components and cladding-wind pressures to be used for the design of exterior components and cladding (by others) shall be determined in accordance with ASCE 7 and the IBC.
- 6 DESIGN SEISMIC LOADS (ASCE 7-16)
- Site Class = D (default)
- Seismic Design Category = D / Importance Ie = 1.25
- Ss = 1.13g / S1 = 0.40g
- Sds = 0.79g / Sd1 = 0.51g
- 7 QUALITY: Contractor shall ensure high standards of workmanship throughout, with strict adherence to the contract documents and all governing codes and standards.
- 8 DESIGN RESPONSIBILITY: Kingworks is responsible only for the design of the primary structural system as shown in the contract documents. Design of all secondary structure or non-structural elements are by others.
- 9 DISCREPANCIES: Notify Wilson Engineering immediately of any discrepancies between these notes, the contract drawings, the specification, or the governing code. Wilson Engineering shall reply in writing. Any related work performed by the Contractor prior to receiving a reply from Wilson Engineering is at the Contractor's sole risk. For purposes of bidding, the most stringent of the conflicting documents shall apply.
- 10 VERIFICATIONS: Verify all existing conditions; verify all dimensions in the field; verify architectural, mechanical and electrical openings for size, location and number; notify Wilson Engineering of any discrepancies, substandard existing conditions, or conditions not included in or contrary to the Contract Documents prior to shop drawing submittal or construction.
- 11 DRAWING COORDINATION: Coordinate the structural drawings with drawings from all other disciplines (including but not limited to Architectural, Civil, Mechanical, and Electrical).
- 12 COMPLETED FORM: The structure shown in these drawings is designed to be stable and to resist the loads above only in a fully completed form. Contractor shall ensure that the structure is adequately braced and shored during construction for all temporary loads until all elements are in place, and shall ensure that temporary loadings do not exceed the allowable capacity of any structural elements both before and after these elements are in place.
- 13 MEANS AND METHODS: Contractor is solely responsible for site safety, coordination, procedures, construction methodology, shoring, bracing, sequencing, and all other "means and methods" of construction except where specifically shown in the Contract Documents.
- 14 PROTECTION AND BRACING: Contractor is solely responsible for the protection of existing buildings, utilities, streets, equipment, etc. during construction. Provide temporary bracing and protection as required.
- 15 SCALING: Do not scale drawings. See civil drawings for dimensions, and notify Wilson Engineering of any discrepancies.
- 16 ALTERATIONS: Any holes or other alterations to the structure which are not specifically detailed on the Contract Drawings shall be submitted to the engineer for approval.
- 17 LOAD COORDINATION: The design of all pre-engineered components shall include provisions for precise locations and weights of all mechanical units and other concentrated loads. Concentrated loads are not necessarily indicated on the Structural Drawings, and must be coordinated by the General Contractor, who shall also inform Wilson Engineering of any significant loads not shown in the contract documents. The Contractor shall assume full responsibility for coordination of weights, locations, hanger spacings, methods of attachment and seismic bracing of all mechanical units, sprinklers, pipes, ductwork, and other miscellaneous MEP assemblies.
- 18 MEP CONNECTIONS TO STRUCTURE: Unless specifically shown otherwise in the Contract Documents, all anchorage, support, and seismic bracing of mechanical and electrical equipment, piping, ceilings, fixtures and other non-structural components shall be designed by a Washington State Licensed Professional Engineer and installed by the Contractor. The engineer shall be retained by the Contractor, and the Contractor is responsible for all associated engineering, component, and installation costs. Design shall be per ASCE 7 Chapter 13.
- 19 DELIVERY, STORAGE AND HANDLING: All products shall be delivered, stored, and handled according to the Manufacturer's recommendations and installation instructions. Protect all items from damage, moisture, corrosion, or other deterioration before, during and after installation.
- 20 COPYRIGHT: These drawings, and all designs shown within these drawings, are copyrighted by Kingworks Structural Engineers. Duplication is not permitted without written permission. The designs shown herein are intended for this project only and may not be used on any other project or for any other purpose.

-- SUBMITTALS --

- 1 GENERAL: Provide PDF of all submittals to Wilson Engineering. Allow two weeks for review. Submittals will be reviewed for general conformance to the contract documents. Responsibility for adherence to the contract documents lies solely with the Contractor, including but not limited to dimensions, sizes, connections, and quantities.
- 2 CONTRACTOR REVIEW: Contractor shall review, mark, and stamp all submittals before submittal to Wilson Engineering. Unreviewed or unstamped submittals will be returned to the Contractor without review.
- 3 RESUBMITTALS: Resubmittals shall have all revisions clearly identified with "drawing clouds" and revision dates. KW shall not be responsible for review of any unmarked revisions.

-- SUBMITTALS --

- 4 SHOP DRAWINGS: To include typical and unique conditions and all connections, shall be submitted to the Structural Engineer of Record for the following products prior to fabrication. Shop drawings shall clearly demonstrate the Contractor's understanding of the contract documents. The following shall be considered minimum structural submittals for this project:
- Concrete Reinforcing Steel (incl. wall elevations)
- Concrete Mix Designs (confirm f'c prior to construction)
- Slab-on-Grade Jointing Plan
- Structural Steel
- 5 DEFERRED SUBMITTALS: Shall be designed by a Washington State Licensed Professional Engineer (Structural Engineer where required by jurisdiction). Submittals shall clearly indicate all loads and reactions to be imposed on the primary structure. Submittals shall include all components and connections required to perform the work. Where the Contract Documents indicate member or connection qualities and/or quantities, these shall be considered a minimum and remain the design responsibility of the component engineer to increase as required to satisfy load and deflection requirements. Deferred submittals shall include calculation set. Both the calculation set and all related shop drawings of the deferred submittal shall be sealed prior to submission. Wilson Engineering shall forward deferred submittals to the Building Official where required. The following shall be considered Deferred Submittals on this project:
- Metal Stairs & Landings
- Pre-engineered Metal Building for Blower Building (See Specifications)
- 6 SUBMITTAL REVIEW COMMENTS: Engineer marks and comments on shop drawings and other submittals are a normal and expected part of the submittal process, and are not to be used as a basis for change orders except in cases where these marks result in or derive from substantial changes to the Contract Drawings. Time required to revise and resubmit any submittal shall be considered inherent to the submittal review process and shall not be deemed a change order. If discrepancies are discovered between the submittals and the Contract Documents (either before, during, or after submittal review), the Contract Documents shall govern and be implemented unless specifically directed otherwise.

-- SPECIAL INSPECTIONS, TESTING, AND STRUCTURAL OBSERVATION --

- 1 GENERAL: A special inspection agency shall be retained by the owner to perform inspections according to IBC Chapter 17. The following scheduled structural special inspection and testing regimen shall be cross-referenced with the IBC and its referenced standards for more specific requirements and exceptions. Special inspection and test reports shall be submitted to the Building Official, Civil Engineer, and Structural Engineer in accordance with IBC 1704.2.4.
- 2 NONSTRUCTURALCOMPONENTS: Nonstructural components are also subject to special inspection for conformance to the Seismic Design Requirements of ASCE 7-10 Chapter 13. The support and seismic bracing of nonstructural components shall be designed by a Washington State Licensed Engineer, who shall be retained by the Contractor. Nonstructural components subject to seismic design and periodic special inspection requirements of ASCE 7-10 and IBC Chapter 17 include (but are not limited to) the following:
- Architectural components (IBC 1705.12.5)
- Nonstructural components (IBC 1705.13.2)
- MEP Components (IBC 1705.12.6)
- Reference the architectural, electrical, and mechanical drawings and/or specifications for additional special inspection and testing requirements.
- 3 STRUCTURAL OBSERVATION: Kingworks will perform Structural Observations in accordance with IBC Section 1704.6 if/as required. These observations provide intermittent checks of general conformance to the design intent and are in addition to (not replacing) the third-party special inspection regimen. It shall be the Contractor's responsibility to keep the Structural Engineer apprised of the general schedule of construction, such that observations may be made at appropriate stages before significant structural components (such as reinforcing bars, framing members, or wall holdowns) are obscured.

-- STRUCTURAL SPECIAL INSPECTION SCHEDULE --

	STRUCTURAL ITEM	FREQUENCY (C=continuous, P=periodic)	REFERENCE (2018 IBC Section, uon)
1	STEEL		1705.2, 1705.12.1
	- Structural Steel - All	Varies	AISC 360-16 Chap N
	- Steel Deck	P	1705.2.2
	- Cold-Formed Steel	P	1705.11.2, 1705.12.3, AWS D1.3
	- SLRS Steel	Varies	AISC 360-16 Chap N, AISC 341-16 Appx Q, 1705.13
2	CONCRETE		1705.3, ACI 318-14 26.13
	- Reinf Placement	P (and prior to all pours)	Table 1705.3, ACI 318-14 26.6.2
	- Anchor Placement	P (and prior to all pours)	Table 1705.3
	- Concrete Placement	C	Table 1705.3
	- Concrete Testing	Per ACI 318-14 26.12	Table 1705.3
	- Formwork & Curing	P	Table 1705.3
3	POST-INSTALLED ANCHORS		Table 1705.3
	- Epoxy or Adhesive Anchors Used in Horizontal or Overhead Position	C	See ICC-ES report
	- All Other Anchors Installed in Hardened Concrete or Masonry	P (except where C req'd by ICC-ES report)	See ICC-ES report
4	SOILS & FOUNDATIONS		1705.6, Table 1705.6, Geotech Report
	- Subgrade Adequacy	P (beneath fill and/or foundations)	Table 1705.6
	- Excavation Depth	P	Table 1705.6
	- Fill Materials	P	Table 1705.6
	- Fill Placement & Compaction	C	Table 1705.6

-- FOUNDATIONS & SUBGRADE --

- 1 SOIL ALLOWABLE BEARING PRESSURE: 2500 PSF (x 1.33 when subjected to wind or seismic loadings).
- 2 GEOTECHNICAL REPORT: Reference GeoEngineers geotechnical report dated September 7, 2021 for subgrade preparation, backfill, compaction, wet weather work, and other recommendations. All provisions of the geotechnical report shall be enacted by the Contractor.
- 3 GEOTECHNICAL VERIFICATION: The Owner shall retain a geotechnical inspection agency to inspect the excavations and confirm that the provisions of the Geotechnical Report are followed. All recommendations of the geotechnical inspector shall be followed and communicated with the design team via inspection report. Contractor shall coordinate this inspection effort with the Owner.
- 4 UTILITIES: Utilities are not to pass through or beneath footings, stemwalls, and other concrete work on grade except as shown in specific details.
- 5 MISCELLANEOUS VERIFICATIONS: Verify sizes, slopes and locations of tunnels, electrical cells, pits, pipes, floor drains, trenches and floor recesses with architectural, mechanical and electrical contractors.
- 6 ALIGNMENT: All footings shall be centered below columns and walls, unless dimensioned otherwise.
- 7 BACKFILL: Do not backfill against below-grade walls until strength has been achieved and floor framing is in place.
- 8 FROST PROTECTION: Maintain minimum 1'-6" soil cover, measured from finished grade to the bottom of the footing, for perimeter wall foundations and isolated exterior foundations.
- 9 EXCAVATION SLOPE: Excavation slope shall not exceed that permitted by local regulation, except as specifically approved by the geotechnical engineer.

-- REINFORCED CONCRETE --

- 1 REFERENCED STANDARDS: Reinforced concrete shall conform to the requirements of the following documents, except as modified in these drawings.
- ACI 301-16 "Standards Specifications for Structural Concrete"
- ACI 318-14 "Building Code Requirements for Structural Concrete"
- 2 MATERIALS
- Conform to the following concrete material schedule requirements:

-- CONCRETE MATERIAL SCHEDULE --

	f'c (PSI)	W/CM (MAX)	COARSE AGG (MAX)	AIR CONTENT	MIN FLY ASH AND/OR SLAG
BLOWER BUILDING SOG & GRADE BEAMS, MISC EQUIPMENT PADS	4500	0.45	3/4"	6%	30%
HYDRAULIC STRUCTURES (SEE ADDTL REQ BELOW)	4500	0.45	1"	SEE NOTES	20%
DRAIN PUMP STATION LID	4500	0.45	3/4"	5%	20%

- Required concrete strength "f'c" shall be evaluated at 28 days (56 days OK for fly ash or slag concrete).
- Cementitious Materials: Provide Type II Blended Hydraulic Cement per ASTM C595 for hydraulic structures. Provide Type I or II Cement per ASTM C150 (Type III OK for Precast) for all else. Cementitious materials for use in concrete exposed to soil or weather must meet or exceed the S1 exposure class per ACI 318-14 Table 19.3.2.1.
- Fly Ash: Class F per ASTM C618.
- Slag: GGBF Slag per ASTM C989, Grade 100 minimum.
- Air Entrainment: Provide air entrainment per concrete mix table +/- 1.5%, all other locations exposed to weather shall have air entrainment of 6% (+/- 1.5%).
- Typical Reinforcing Steel: ASTM A615, Grade 60 Typ, Grade 75 for Train #3 (H) bars
- Weldable Reinforcing Steel: ASTM A706, Grade 60
- Welded Wire Fabric: ASTM A185
- Mixes for hydraulic structures shall also satisfy the following:
A. Shrinkage reducing admixture per ASTM C-494 shall be added so that test mixes shall demonstrate shrinkage of no more than 400 microstrain at 7 days.
B. Air percentage shall be per the manufacturer of the shrinkage reducing admixture recommendation.
C. Hydraulic structure concrete shall contain a minimum of 500 pounds of cementitious materials per cubic yard.
D. For hydraulic structures, the requirement for stone aggregate shall not be waived.
- 3 SPLICES: All reinforcing steel lap splices are to be per typical schedule unless noted otherwise. All welded wire reinforcement lap slices shall be the greater of one space plus 2 inches or 6 inches, unless otherwise noted.
- 4 CONTINUITY: Horizontal reinforcing steel in walls and wall footings shall be continuous around corners, same size and spacing. At intersections of walls or footings, extend all bars as far as possible into continuous element and terminate with standard hook.
- 5 CLEAR COVER: Provide clear cover from outermost reinforcing to surface of concrete in accordance with the following:
- Elements cast against and permanently exposed to earth: 3" dr
- Elements formed and exposed to earth or weather: 2" dr (#6-#18), 1 1/2" dr (#3-#5)
- 6 INTEGRAL COLUMN & WALL FOOTINGS: Where column footings are shown integral with wall strip footings, transverse wall footing bars need not continue through width of column footing. Longitudinal wall footing bars shall extend continuous through column footing, in addition to column footing bars per typical detail.
- 7 ACCESSORIES: Provide all accessories, chairs, spacer bars and supports necessary to secure steel in accordance with ACI Code of Standard Practice.
- 8 CHAMFER: Chamfer all exposed corners and edges above grade per Civil.
- 9 FORM STRIPPING: Do not strip forms until concrete has reached adequate strength.
- 10 SLEEVES / OPENINGS: Furnish and place all sleeves and openings as shown on the drawings or as specified.

-- REINFORCED CONCRETE --

- 1 REINFORCING FABRICATION: All reinforcing shall be shop fabricated. Exception: #3 or #4 bars may be field bent one time in any location, do not rebend or restraighthen.
- 2 HOOKS: All hooked bars shall be a standard shop fabricated hook with bend radii and length per ACI 318, UON.
- 3 TESTING: Test cylinders shall be taken by qualified personnel according to ACI 318-14 Section 26.12.
- 4 MEMBRANE: Provide vapor barrier (per arch/spec, 10-mil min where not otherwise specified) directly below slab-on-grade (above compacted fill), reinforce all seams and repair tears as necessary.
- 5 WEATHER PROVISIONS: Observe all ACI recommendations for hot or cold weather concreting cure slabs using an approved curing compound or wet cure system per ACI recommendations, with special consideration for slag and fly ash concrete as appropriate.
- 6 CONSTRUCTION JOINTS: All construction joint locations shall be submitted to Wilson Engineering for approval. Roughen joints to 1/4-inch minimum amplitude, remove all laitance. Soak joint continuously for 2 hours minimum, then remove all standing water, immediately prior to second pour. Unless noted otherwise, all reinforcing shall be spliced/continuous across the construction joints.
- 7 RIGID FOAM: Where rigid foam is indicated on plans to be placed below concrete slabs, it shall consist of cellular polystyrene conforming to ASTM D6817. Polystyrene shall have a minimum compressive resistance of 3.6 psi at 1 percent deformation unless otherwise noted. Secure foam in place per the manufacturer's recommendations. Offset joints between foam blocks 24" minimum between adjacent layers. At contractors option it is acceptable to substitute polystyrene conforming to ASTM C578 Type XIV rated for 40 psi compressive resistance at 10 percent deformation with a minimum thickness of 2" per layer.

-- ANCHORAGE TO CONCRETE --

- 1 MATERIALS (unless noted otherwise in the drawings)
- Concrete or grout must cure for a minimum of 21 days prior to drilling any holes or placing post-installed anchors.
- Anchor type shall be according to the drawings. All post-installed anchors installed in concrete shall have ICC-ES reports demonstrating IBC compliance for use in cracked concrete and for seismic loading. Substitutions not permitted without written permission by KW.
- Pre-approved Epoxy for post-installed threaded rod or reinforcing in concrete base material: HILTI HIT-RE 500 V3 or Simpson SET-3G.
- Pre-approved "Expansion Anchors" in concrete base material: HILTI Kwik-Bolt TZ or Simpson Strong-Bolt 2, or DEWALT/Powers Power-Stud+ SD2.
- Pre-approved "Screw Anchors" in concrete base material: HILTI HUS-EZ or Simpson Titen HD or DEWALT/Powers Screw-Bolt+.
- Post-installed or Cast-in-Place Threaded Rod (Anchor): ASTM A36
- Post-installed Reinforcing: ASTM A615 Grade 60
- 2 EMBEDMENT: Anchor embedment in base material shall be per the drawings, which shall govern over the typical values shown below. Where not otherwise indicated, provide embedment as follows:
- Epoxy to Concrete: Min Embed = 12 x (Rod/Bar Ø) for 5/8"Ø (#5) or smaller, 14 x (Rod/Bar Ø) for 3/4"Ø (#6) or larger
- Expansion Anchor to Concrete: Min Embed = 8 x (Anchor Ø)
- Screw Anchor to Concrete: Min Embed = 9 x (Anchor Ø)
- Cast-In-Place Anchor: Min Embed = 7" to top of embedded washer or hook
- 3 INSTALLATION: Post-installed anchor hole diameter, drilling depth, cleaning and installation procedure shall be in accordance with the current Manufacturer's Printed Installation Instructions (MPII) provided in the ICC/ES report. Holes shall be drilled with rotohammer equipment. Core-drilled holes are not permitted unless specifically noted otherwise.
- 4 COLD-WEATHER INSTALLATION: Do not use epoxy or adhesive anchors outside of their rated temperature range. Contact the Structural Engineer for alternate if the base material temperature may be less than 40 degrees during installation or curing.
- 5 CAST-IN-PLACE ANCHORS: Cast-in-place anchors shall have nut and washer at embedded end, UON. Anchors shall be affixed to the form to prevent movement during pouring, vibration, or set-up and shall not be "stabbed" into wet concrete or grout. Verify adequate length of exposed thread to fully engage all attached work.
- 6 FINISHES: All anchors used at exterior, or where subject to moisture, or where in contact with pressure treated wood, shall be hot-dip galvanized per ASTM A153 or stainless steel, including matching washers and nuts.
- 7 REINFORCEMENT LOCATIONS: All post-installed anchors shall be located to avoid drilling into reinforcement, unless specifically approved by the Engineer. Reinforcement shall be placed with consideration for locations of post-installed anchors. Do not damage reinforcing during drilling operations.

-- STRUCTURAL STEEL --

- 1 REFERENCED STANDARDS: Fabrication, erection and dimensional tolerances of structural steel members shall be in accordance with AISC 303-16 "Code of Standard Practice for Steel Buildings and Bridges" and ASTM A6.
- 2 MATERIALS (unless otherwise noted on drawings)
- Wide Flange and WT shapes: ASTM A992, fy=50 ksi.
- Hollow Structural Sections (HSS): A500 Grade B, Rectangular fy=46 ksi, Round fy=42 ksi.
- Pipe Sections: A53 Grade B, fy=35 ksi.
- Bolts: A325(-X or -N or -SC per details) Type 1 high strength bolts with A563 nuts and F436 washers unless noted otherwise.
- Welding Electrodes: E70XX
- Anchor Rods: ASTM F1554 Grade 36
- Angles / Channels / Plates / Rods / Miscellaneous: ASTM A36
- 3 WELDING: All welding to be performed by qualified welders and shall conform to A.W.S. D1.1.
- 4 GALVANIZING: All steel elements and fasteners exposed to the exterior shall be hot-dip galvanized in accordance with ASTM A153/A123.
- 5 GALVANIZING REPAIR: Repair all damage to hot-dip galvanized coating by hot-stick or high-zinc paint method in accordance with ASTM A780. Where welding of galvanized steel is required, all galvanizing on the area of steel to be welded, plus 3/4-inch each side of the welded area, shall be ground off prior to welding; affected area shall be repaired per ASTM A780.

NO.	REVISIONS	BY	DATE
2	Bid Set	TH	9/6/22



-- STRUCTURAL STEEL --

- 6 ERECTION: Contractor shall comply with all current OSHA/WISHA requirements concerning steel erection. This may require additional bolts, shims, guys, bracing, and other elements that are not detailed in the Contract Documents but remain the Contractor's responsibility to implement.
- 7 BOLT SPECIFICATION: All bolted connections shall conform to the current "RCSC Specification for Structural Joints Using ASTM A325 or A490 Bolts". "-SC" bolts shall be approved twist-off tension-control type per ASTM F1852 or shall utilize compressible-washer-type direct tension indicators per ASTM F959.
- 8 BOLT INSTALLATION: Bolts shall be snug-tightened (-ST) unless noted as "-SC". "-SC" bolts shall be pretensioned with Class A faying surfaces per the RCSC Spec.
- 9 BOLT HOLES: Bolt holes in steel shall be 1/16" larger than bolt diameter, except where specifically detailed otherwise.
- 10 GROUT: Grout for baseplates, sleeves, and embedded steel shall be an approved nonshrink cementitious grout containing natural aggregates delivered to the job site in factory prepackaged containers requiring only the addition of water. The minimum 26-day compressive strength shall be at least 1000 psi higher than the supporting concrete strength, UON. Grout shall be mixed, applied, and cured strictly in accordance with the manufacturer's printed instructions. For grouting under base plates, grout shall be proportioned as a flowable mix. When a flowable mix does not provide the required strength, an epoxy grout shall be used. Grout thickness shall be per the typical details.



Verify Scale:

Enclosing box measures 1/2 inch tall x 2 inch wide when drawings are printed at full scale

DESIGNED BY	THaidersdorf	DRAWN BY	CKinney	CHECKED BY	THaidersdorf
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EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON

SAN JUAN COUNTY

WASTEWATER TREATMENT PLANT UPGRADE - PHASE 1

STRUCTURAL NOTES

SHEET

S1.1

PAGE

56 OF 96

DATE

10-15-2021

SCALE

(on 22"x34")

JOB NUMBER

Wilson: 2020-070

KW: 211.31



BID SET

ELEVATION

- NOTES**
1. CENTER OF PEDESTAL = CENTER OF TRAIN #3 SLAB, REFER TO 1/S2.1
 2. TOP BARS SHOWN, BOTTOM BARS SIM (3" CLR FROM BOT).

1
S4.1

CL

GUARDRAIL, MECHANICAL EQUIPMENT, AND LADDERS PER CIVIL

(2) #4 (H) AT TOP

CONCRETE WALL PER PLAN

#5 (H) @ 12" OC EF, GRADE 75 BARS, STAGGER LAPS PER 5/S2.1

#5 (V) @ 12" OC EF, 3" CLR EF AND FROM BOT

GROUT SURFACE PER SPEC

HYDROPHILIC WATER STOP AA, SUBMIT FOR APPROVAL

MAT FOUNDATION & REINFORCING PER PLAN

SUBGRADE PREP PER STRUCTURAL NOTES & GEOTECH. REPORT

PEDESTAL REINFORCING PER 2/S2.1

#5 (V) @ 16" OC AA, STD HOOK AT BOT.

#5 (V) @ 16" OC AA, TYP LAP AT TOP, STD HOOK AT BOT.

1'-0" AA

NOTE
TOP BARS SHOWN, BOTTOM
BARS SIM (3" CLR FROM BOT).

TRUE NORTH

Labels in the diagram include:
 - 12" CONCRETE WALL AA
 - 12" MAT SLAB FOUNDATION
 - PERIMETER OF RAISED PEDESTAL, SEE 2/S2.1
 - CONT #5 AA @ 12" OC
 - #5 RADIAL BAR @ 12" OC AT OUTER EDGE OF SLAB, 2" CLR FROM TOP, TERMINATE EVERY OTHER BAR W/ STD HOOK AGAINST (V) BAR AT PEDESTAL PERIMETER
 - 3/S2.1
 - 4/S2.1
 - 5/S2.1
 - 1'-0" TYP AA
 - cl1
 - clA

NOTE
TOP BARS SHOWN, BOTTOM
BARS SIM (3" CLR FROM BOT).



NO.	REVISIONS	BY	DATE
2	Bid Set	TH	9/6/22



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360.714.8260
www.king-works.com

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Theidersdorf	
DRAWN BY	
GKinney	
CHECKED BY	
Theidersdorf	

EASTSOUND SEWER AND WATER DISTRICT

SAN JUAN COUNTY

WASHINGTON

WASTEWATER TREATMENT PLANT UPGRADE - PHASE 1

TRAIN 3 PLANS & SECTION

DATE
15-2021

SCALE
As indicated
(on 22"x34")

57 OF 96

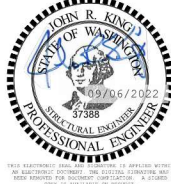
SHEET
S2.1

57 OF 96

811 Call 811
two business days
before you dig

BID SET

NO.	REVISIONS	BY	DATE
1	UV Disinfection	TH	5/4/22
2	Bid Set	TH	9/6/22



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when drawings are printed at full scale.

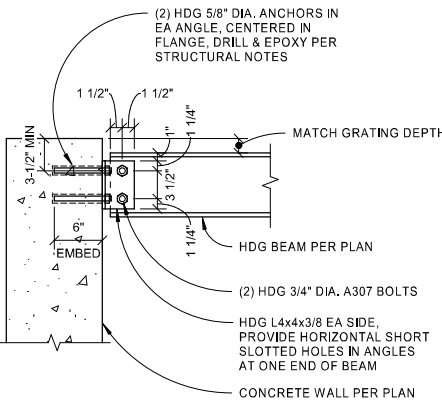
DESIGNED BY	THEILDERSDORF	DRAWN BY	CKINNEY	CHECKED BY	THEILDERSDORF
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EASTSOUND SEWER AND WATER DISTRICT
WASHINGTON
SAN JUAN COUNTY
WASTEWATER TREATMENT PLANT UPGRADE - PHASE 1
UV DISINFECTION CHAMBER & DIGESTER PLAN & SECTIONS

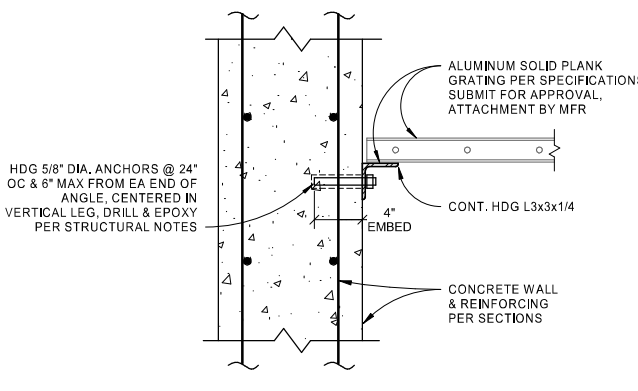
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SHEET	S2.2	PAGE	58 OF 96



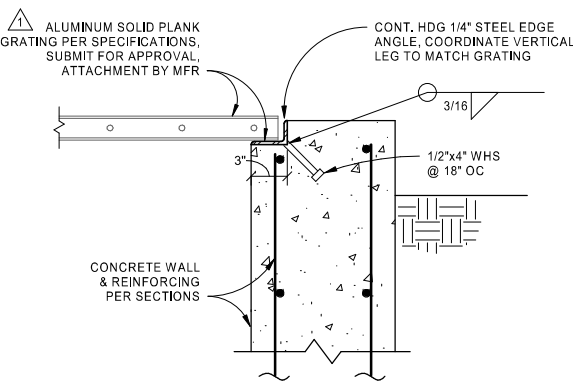
BID SET



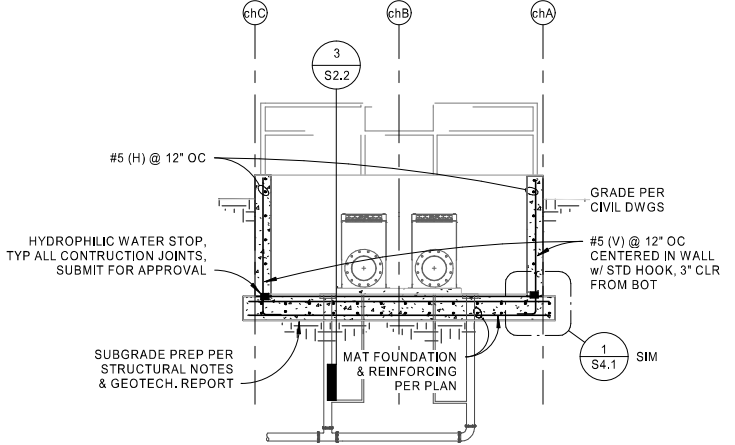
7 STEEL BEAM TO CONCRETE WALL
1" = 1'-0"



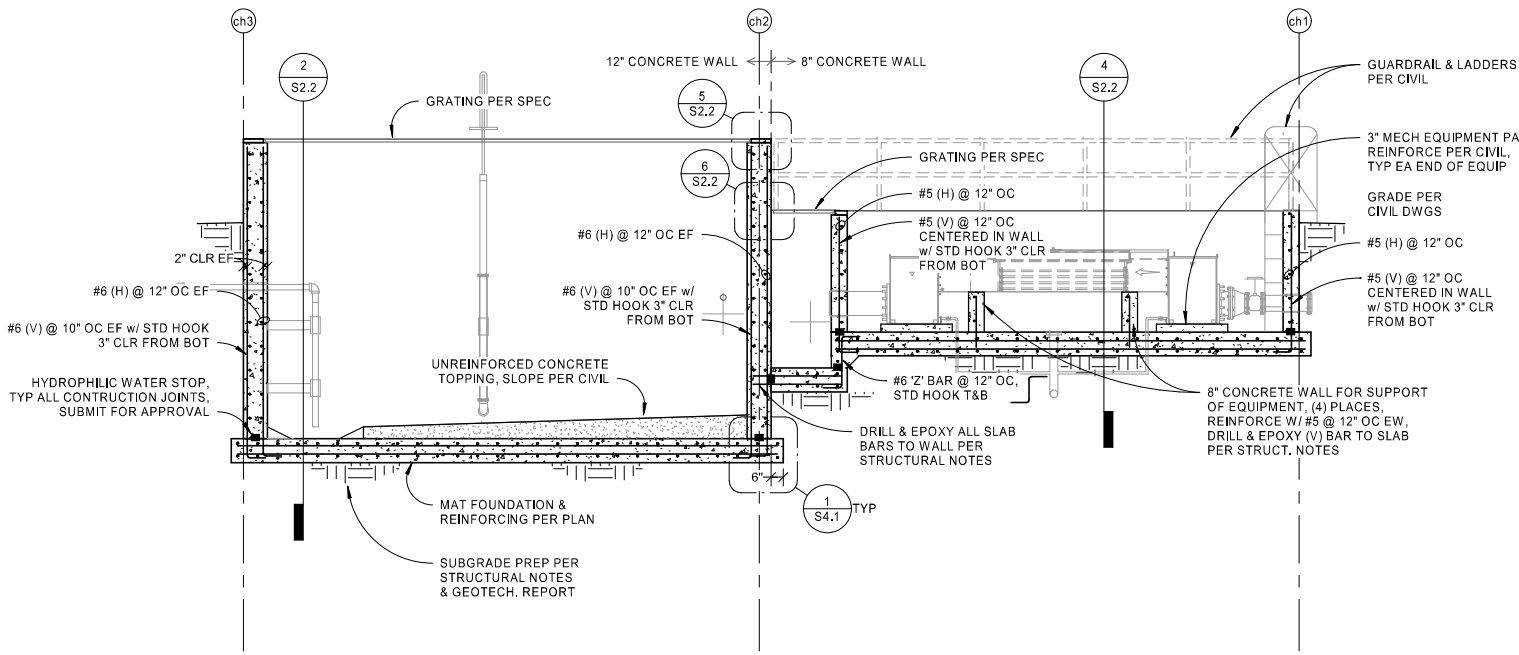
6 TYP GRATING SUPPORT AT SIDE OF CONCRETE WALL
1 1/2" = 1'-0"



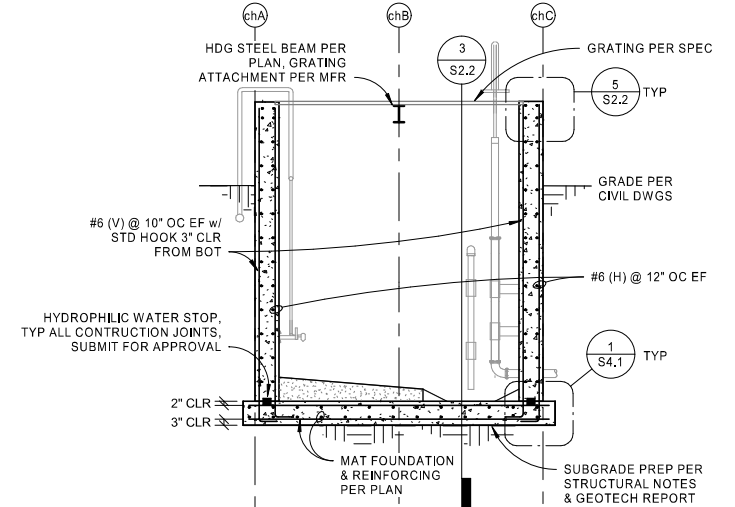
5 TYP GRATING SUPPORT AT TOP OF CONCRETE WALL
1 1/2" = 1'-0"



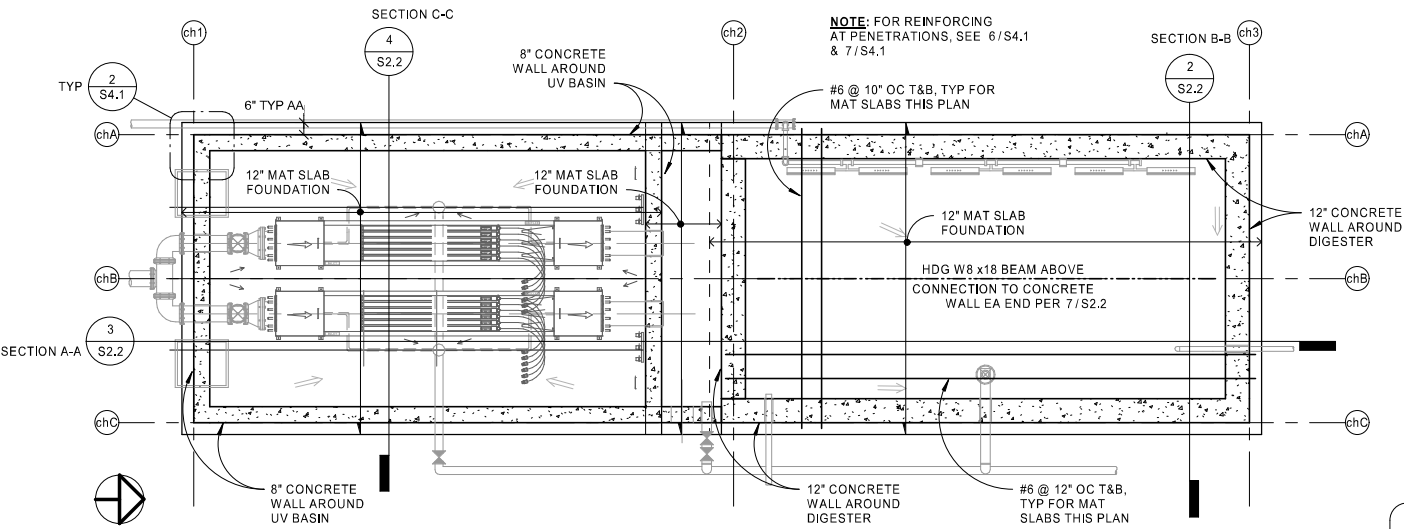
4 UV DISINFECTION CHAMBER & DIGESTER SECTION C-C
1/4" = 1'-0"



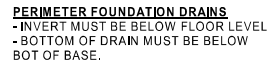
3 UV DISINFECTION CHAMBER & DIGESTER SECTION A-A
1/4" = 1'-0"



2 UV DISINFECTION CHAMBER & DIGESTER SECTION B-B
1/4" = 1'-0"



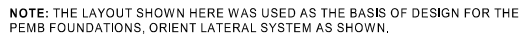
1 UV DISINFECTION CHAMBER & DIGESTER FOUNDATION PLAN
1/4" = 1'-0"



1. DRAIN PIPES MUST DISCHARGE BY GRAVITY OR MECHANICAL MEANS TO AN APPROVED DRAINAGE SYSTEM THAT COMPLIES WITH THE INTERNATIONAL PLUMBING CODE.
2. MINIMUM SLOPE FOR DRAIN PIPE RUNS IS 0.25%.
3. DO NOT CONNECT FOUNDATION DRAIN SYSTEMS TO RAIN LEADERS OR DOWN SPOUTS.
4. PROVIDE APPROVED CLEANOUTS FOR ALL DRAIN PIPE AT ENDS, CORNERS, AND NO MORE THAN 120'-0" ON CENTER.
5. SEE CIVIL DRAWINGS FOR GRADES, DISCHARGE POINTS, AND OTHER REQMTS. ALSO SEE GEOTECH REPORT FOR ADDITIONAL REQMTS.
6. SUBMIT A FOUNDATION DRAINAGE PLAN TO CIVIL FOR REVIEW.
7. DRAIN ROCK/BASE MATERIAL SHALL CONSIST OF GRAVEL OR STONE WITH LESS THAN 10% OF MATERIAL PASSING THROUGH A NUMBER 4 SIEVE.

$$1'' = 1' - 0'$$


1. PLACE CONTROL JOINTS AND CONST JOINTS AT NO MORE THAN 12'-0" O/C. E.W. ALIGN JOINTS WITH COLS AND SLAB CORNERS WHERE POSSIBLE.
2. CUT CJS AFTER CONC HAS SUFFICIENTLY HARDENED TO PREVENT SPALLING OR TEARING BUT NO MORE THAN 12 HOURS AFTER POUR.
3. SUBMIT A SLAB PLAN SHOWING CONTROL & CONSTRUCTION JOINT LAYOUT AND POUR SEQUENCE FOR APPROVAL.
4. THE SUBGRADE SHOWN ABOVE SHALL BE THE MINIMUM REQUIRED AT ANY LOCATION, AND SHALL BE INCREASED OR MODIFIED AS RECOMMENDED IN THE GEOTECH REPORT.
5. SEE SPEC AND ARCH/MECH/CIVIL DRAWINGS FOR SLAB EXTENTS, SLOPES, STEPS, DRAINS, ELEVATIONS, CHAMFERS, FINISHES AND OTHER RELATED INFORMATION. MAINTAIN MIN THICKNESS PER PLAN THROUGHOUT.
6. WHERE SINGLE PENETRATION (OR COMBINED GROUPING OF MULTIPLE PENETRATIONS, SUCH AS AT GROUPS OF HPDES OR GROUPS LARGER THAN 12" IN DIAMETER) EXCEEDS 4" IN EITHER HORIZONTAL OR VERTICAL EXTENSION, THE PENETRATION EXTENTS.
7. AT INT CORNERS, PROVIDE (2) ADD'L #3x36" @ 2' O/C, PLACED AT 45-DEG AND 2' CLR TO THE INT CORNER OF SOG.
8. SIM AT EXT SOG's/SIDEWALKS AROUND BLDG, EXCEPT OMIT VAPOR BARRIER, REDUCE C/J SPACING TO 10'-0" O/C MAX, PROVIDE AIR ENTRAPMENT PER SN.

$$1'' = 1' - 0'$$

$$1/4'' = 1'-0''$$

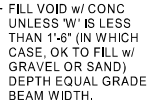
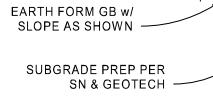
1. PROVIDE SLEEVES w/ ID 2.5" LARGER THAN PIPE OD. CENTER PIPE AND WRAP w/ COMPRESSIBLE MATERIAL.
2. ALL PROPOSED GB PENETRATIONS SHALL BE COORDINATED BY THE GC. PROVIDE 6" MIN CONC BETWEEN ADJACENT PENETRATIONS. CLEARLY INDICATE ALL PENETRATIONS ON THE SUBMITTED REINF SHOP DRAWINGS; THE ENGINEER SHALL REVIEW THE SHOP DRAWING FOR ADEQUACY AND ADD REINF AS REQUIRED. FOR BIDDING PURPOSES ONLY, ASSUME THAT ALL CONT LONG REINF IS LAP SPICED WITH EQUIVALENT BAR AREA.
3. SECTIONS SHOWS ACCEPTABLE PENETRATION LOCATIONS THROUGH AND BENEATH GB'S.
4. DO NOT INTERRUPT REINF IN ANY CASE. MAINTAIN COVER PER STRUCTURAL NOTES AT ALL REINFORCEMENT.



WHERE MIN TOP OR BOT EDGE DISTANCE IS NOT POSSIBLE, LOCALLY THICKEN GB AND INTERRUPT REINF AS SHOWN BELOW:

WHERE TYP CONT LONG REINF IS INTERRUPTED,
ADD EXTRA LONG BARS w/ SAME REINF AREA AS
INTERRUPTED BARS. 'LD' SHALL BE PER SCHED
ACCORDING TO THE SIZE OF ADDED BARS.

⑤ $1\frac{1}{2}'' = 1'-0''$


$$1'' = 1' - 0''$$


1. SEE CIVIL FOR INFORMATION NOT SHOWN.
2. PRE ENGINEERED METAL BUILDING (PEMB) IS DESIGNED BY OTHERS, SEE DEFERRED SUBMITTAL SECTION OF STRUCTURAL NOTES AND SPECIFICATION. FOUNDATION DESIGN IS SUBJECT TO CHANGE AFTER PEMB DESIGN IS FINALIZED.

○ $1/4'' = 1'-0''$

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when drawings are printed at full scale.

100

GKinney

19

EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON

WASTEWATER TREATMENT PLANT UPGRADE - PHASE 1

BLOWER BUILDING FOUNDATION PLAN & DETAILS

DATE _____

S2.3 SHEET

SCALE

JOB NUMBER
Wilson: 2020-070

9 OF

811 Call 811
two business days
before you dig

BID SET

NO.	REVISIONS	BY	DATE
2	Bid Set	TH	9/6/22



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when drawings are printed at full scale.

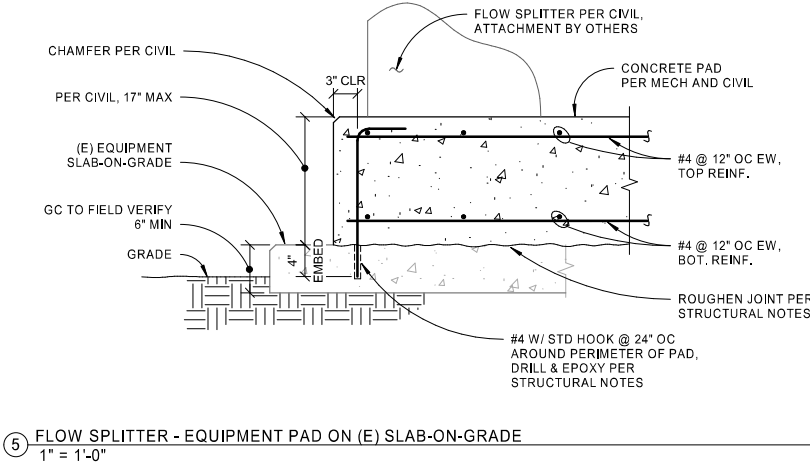
DESIGNED BY	Blomson	DRAWN BY	Theidersdorf	CHECKED BY	Blomson
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EASTSOUND SEWER AND WATER DISTRICT
WASHINGTON
SAN JUAN COUNTY
WASTEWATER TREATMENT PLANT UPGRADE - PHASE 1
PUMP STATION LID, GENERATOR PAD, & FLOW SPLITTER PAD

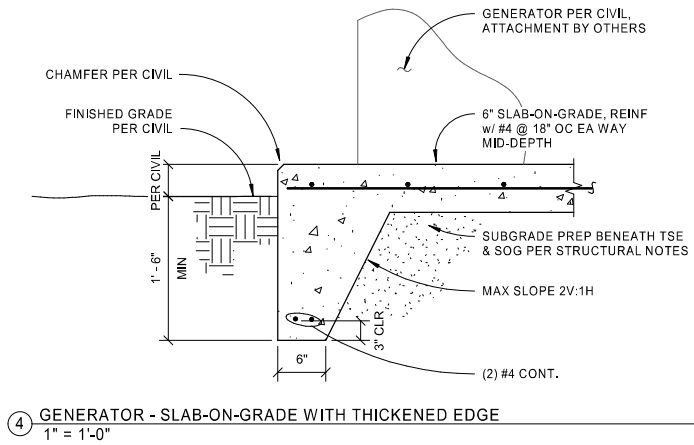
DATE	10-15-2021	SCALE	As indicated (on 22"x34")	JOB NUMBER	Wilson: 2020-070 KW: 21131
SHEET	S2.4	PAGE	60	OF	96



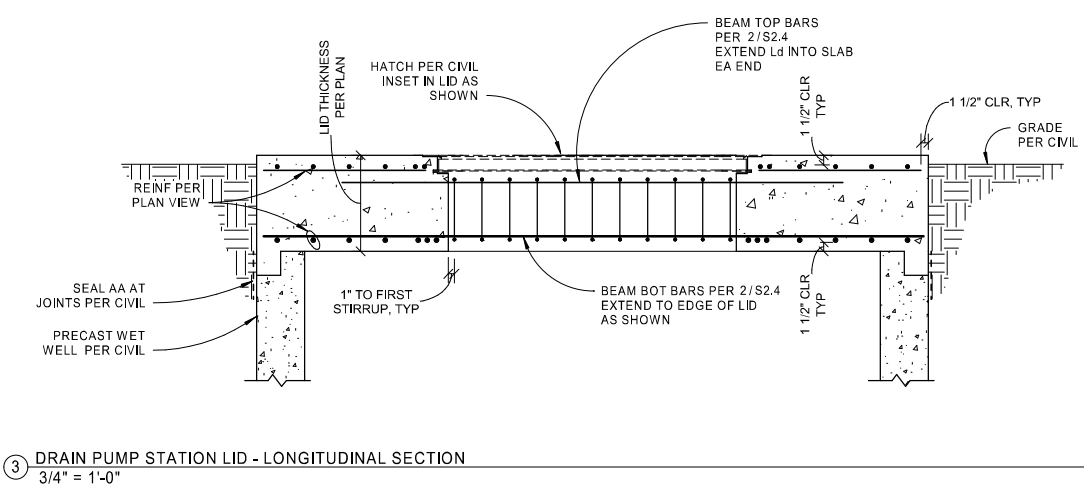
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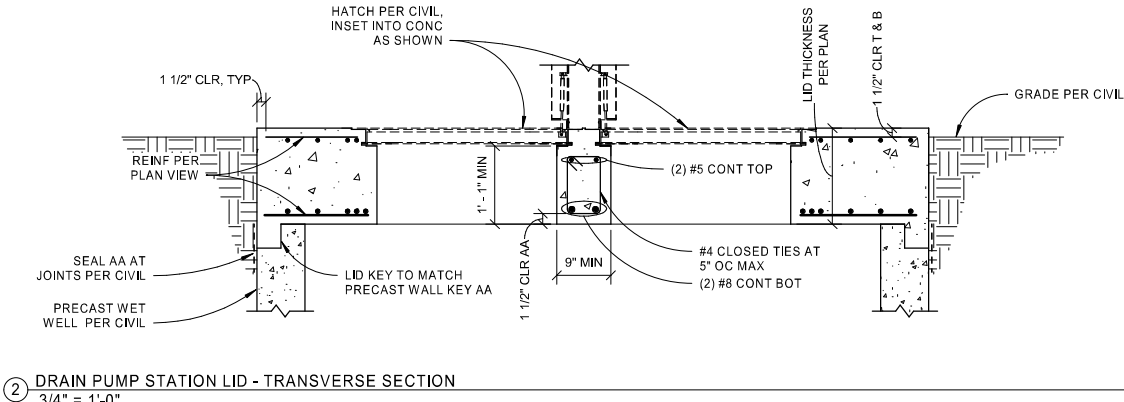
5 FLOW SPLITTER - EQUIPMENT PAD ON (E) SLAB-ON-GRADE
1" = 1'-0"



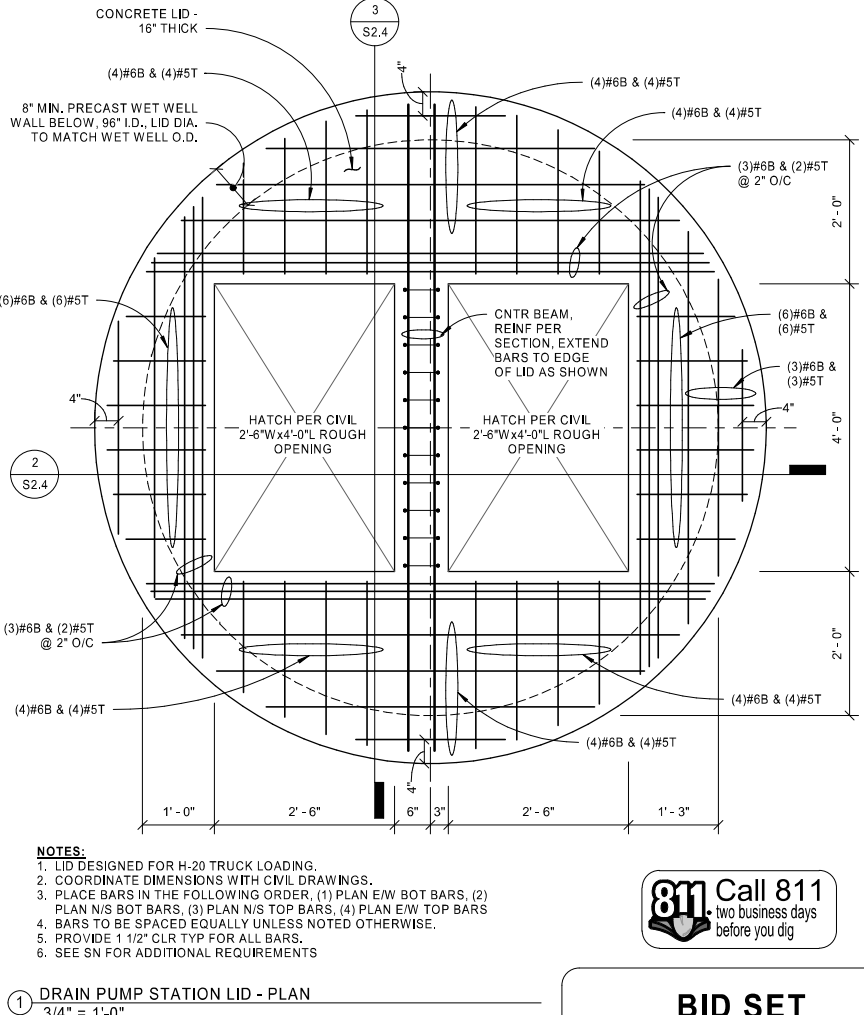
4 GENERATOR - SLAB-ON-GRADE WITH THICKENED EDGE
1" = 1'-0"



3 DRAIN PUMP STATION LID - LONGITUDINAL SECTION
3/4" = 1'-0"



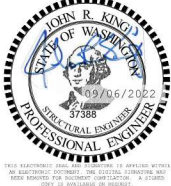
2 DRAIN PUMP STATION LID - TRANSVERSE SECTION
3/4" = 1'-0"



- NOTES:
- LID DESIGNED FOR H-20 TRUCK LOADING.
 - COORDINATE DIMENSIONS WITH CIVIL DRAWINGS.
 - PLACE BARS IN THE FOLLOWING ORDER: (1) PLAN E/W BOT BARS, (2) PLAN N/S BOT BARS, (3) PLAN N/S TOP BARS, (4) PLAN E/W TOP BARS
 - BARS TO BE SPACED EQUALLY UNLESS NOTED OTHERWISE.
 - PROVIDE 1 1/2" CLR TYP FOR ALL BARS.
 - SEE SN FOR ADDITIONAL REQUIREMENTS

1 DRAIN PUMP STATION LID - PLAN
3/4" = 1'-0"

NO.	REVISIONS	BY	DATE
1	UV Disinfection	TH	5/4/22
2	Bid Set	TH	9/6/22



Verify Scale:
Enclosing box measures 1/2 inch tall x 2 inch wide
when drawings are printed at full scale.

DESIGNED BY
Theidersdorf

DRAWN BY
Ckinney

CHECKED BY
Theidersdorf

EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON

SAN JUAN COUNTY

WASTEWATER TREATMENT PLANT UPGRADE - PHASE 1

TYP STRUCTURAL DETAILS

DATE
10-15-2021

SCALE
As indicated
(on 22"x34")

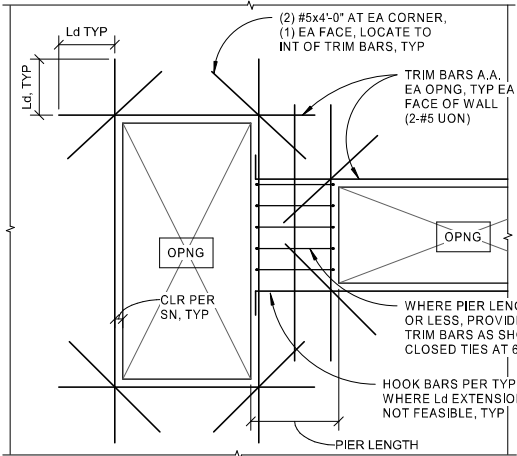
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96

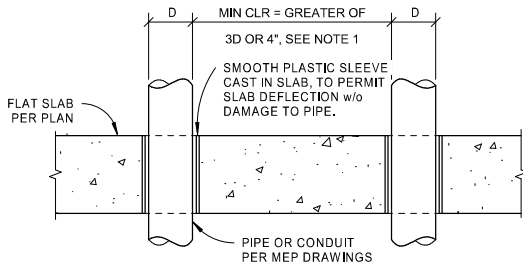


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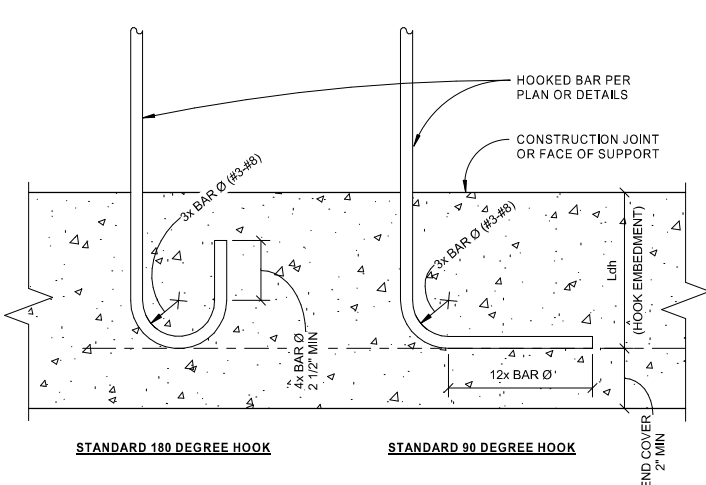
- NOTES**
1. DETAIL APPLIES TO ALL OPENINGS WITH EITHER HEIGHT OR WIDTH GREATER THAN 2'-0".
 2. Ld IS DEVELOPMENT LENGTH PER TYPICAL DETAILS.
 3. REINF SHOWN ABOVE IS TYP MINIMUM. OMIT ADDED REINFORCEMENT ABOVE WHEN SPECIAL REINF INDICATED ON PLANS OR DETAILS EXCEEDS THIS REINF.
 4. CONTRACTOR SHALL VERIFY ALL OPENINGS NOT SHOWN IN THE STRUCTURAL DRAWINGS WITH THE STRUCTURAL ENGINEER PRIOR TO SUBMITTAL DEVELOPMENT AND INSTALLATION.

7 TYP LARGE PENETRATIONS IN CONCRETE
1/2" = 1'-0"



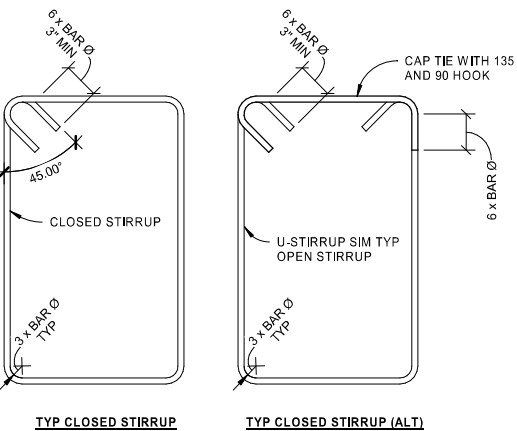
- NOTES**
1. IF MINIMUM CLR SPACING CANNOT BE ACHIEVED, REINF AROUND COMBINED EXTENTS PER TYP LARGE OPENING DETAIL.
 2. DETAIL APPLIES TO D= 9" OR LESS.

6 TYP SMALL PENETRATIONS IN CONCRETE
3/4" = 1'-0"



- NOTES**
1. FOR CONC STRENGTHS NOT INDICATED, USE NEXT LOWEST TABULATED f_c VALUE.
 2. PROVIDE 2 1/2" SIDE COVER AND 2" END COVER FOR HOOKS.
 3. TABULATED VALUES ARE APPLICABLE FOR GRADE 60 UNCOATED REINF BARS.

STANDARD HOOK EMBEDMENT (Ldh)	
BAR SIZE	CONC DESIGN STRENGTH (f_c)
#3	4,500 PSI
#4	6"
#5	7"
#6	9"
#7	10"
#8	12"
#8	14"

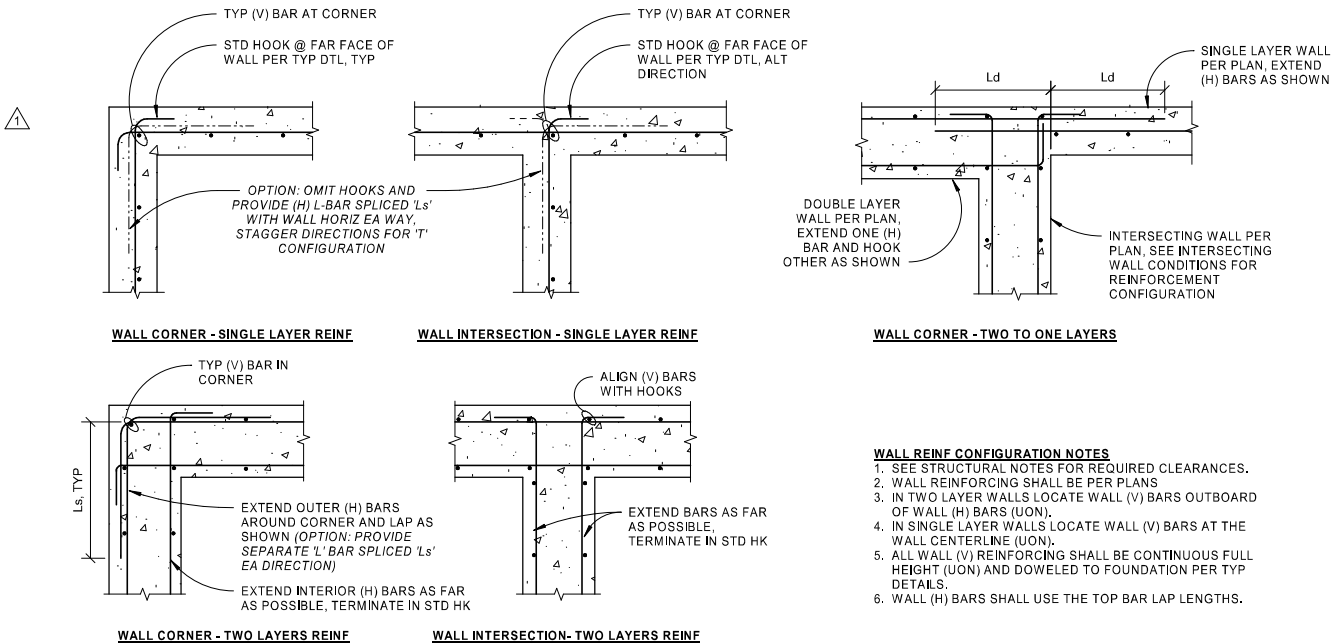


4 TYP CONCRETE BEAM STIRRUPS
1 1/2" = 1'-0"

4,500 PSI CONCRETE				
BAR SIZE	DEVELOPMENT LENGTHS (Ld)		LAP SPICE LENGTHS (Ls)	
	TOP	OTHER	TOP	OTHER
#3	16 in	12 in	18 in	16 in
#4	20 in	15 in	24 in	20 in
#5	28 in	22 in	36 in	28 in
#6	38 in	30 in	50 in	38 in
#7	60 in	46 in	78 in	60 in
#8	74 in	57 in	96 in	74 in
#9	90 in	69 in	116 in	90 in
#10	108 in	83 in	140 in	108 in
#11	126 in	97 in	164 in	126 in

- NOTES**
1. TABULATED VALUES ARE APPLICABLE FOR GRADE 60 UNCOATED REINF BARS.
 2. BAR C/C SPACING MUST BE GREATER THAN TWICE THE BAR DIAMETER AND COVER GREATER THAN ONE BAR DIAMETER.
 3. "TOP" BARS ARE DEFINED AS HORIZONTAL BARS WITH MORE THAN 12 INCHES OF FRESH CONCRETE CAST BELOW THE BARS, INCLUDING WALL HORIZONTALS.
 4. AT LOCATIONS WHERE Ld IS CALLED OUT BUT CANNOT BE ACHIEVED, EXTEND BAR AS FAR AS POSSIBLE AND PROVIDE STANDARD HOOK PER TYP DETAIL.

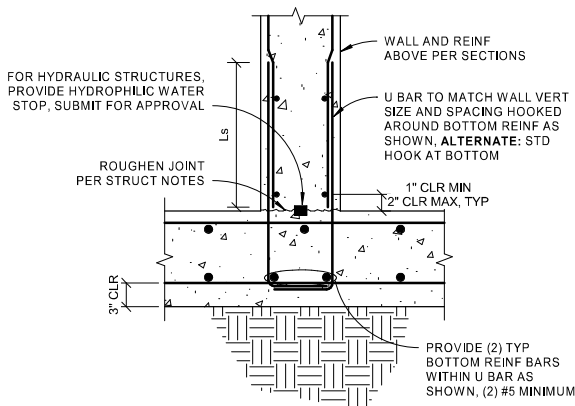
3 TYP CONCRETE REINFORCING DEVELOPMENT / SPICE LENGTHS
3/4" = 1'-0"



- WALL REINF CONFIGURATION NOTES**
1. SEE STRUCTURAL NOTES FOR REQUIRED CLEARANCES.
 2. WALL REINFORCING SHALL BE PER PLANS
 3. IN TWO LAYER WALLS LOCATE WALL (V) BARS OUTBOARD OF WALL (H) BARS (UON).
 4. IN SINGLE LAYER WALLS LOCATE WALL (V) BARS AT THE WALL CENTERLINE (UON).
 5. ALL WALL (V) REINFORCING SHALL BE CONTINUOUS FULL HEIGHT (UON) AND DOWELED TO FOUNDATION PER TYP DETAILS.
 6. WALL (H) BARS SHALL USE THE TOP BAR LAP LENGTHS.

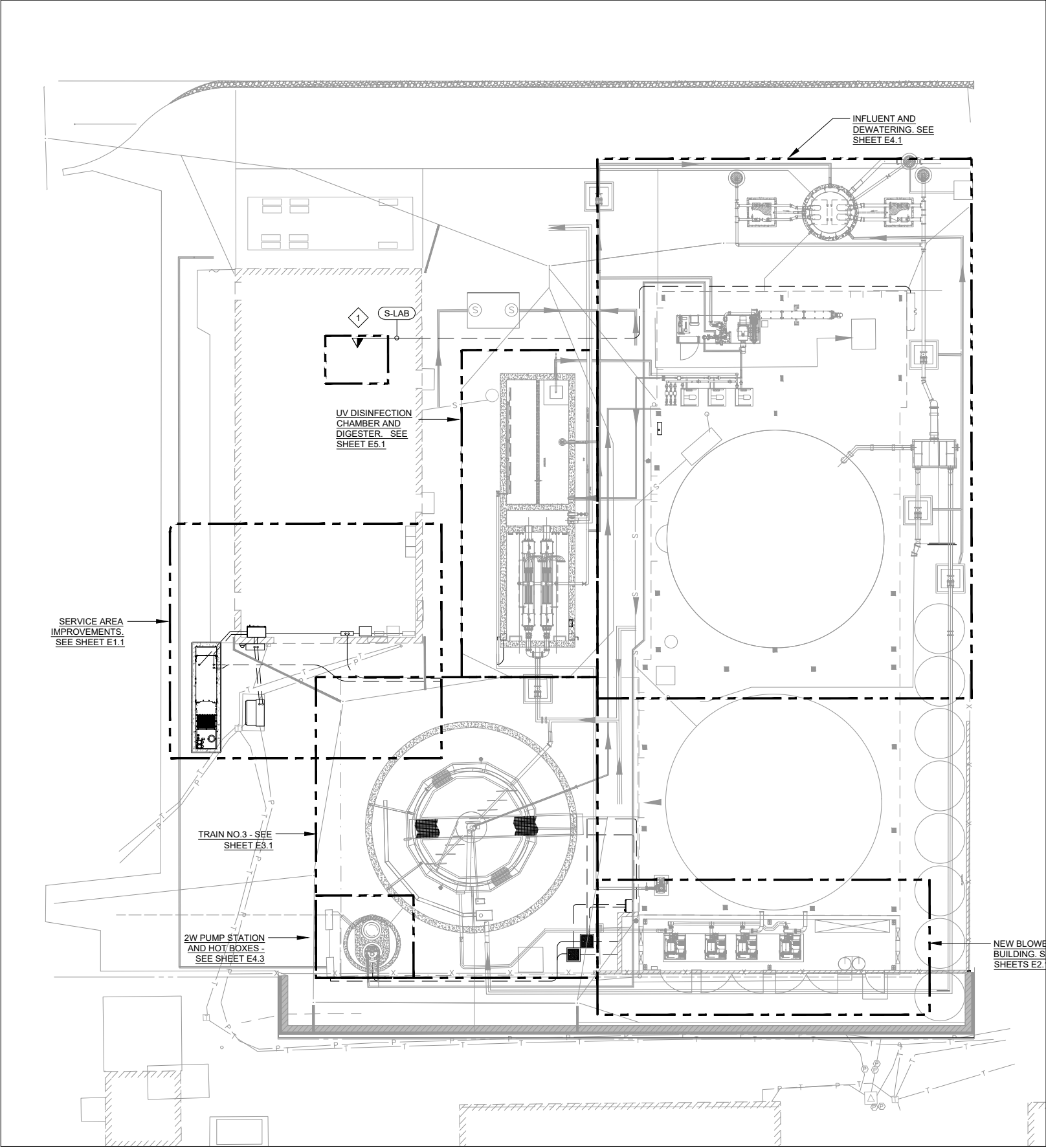
2 TYP CONCRETE WALL REINFORCING AT CORNERS & INTERSECTIONS
3/4" = 1'-0"

NOTE SEE PLANS AND STRUCTURAL NOTES FOR REQUIRED CLEARANCES OF REINF IN SLAB AND WALL.



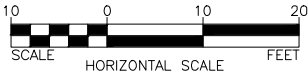
1 TYP CONCRETE WALL REINFORCING TO MAT SLAB
1" = 1'-0"

RACEWAYS AND CONDUCTORS		CALLOUTS AND DESIGNATIONS		CONTROLS AND INSTRUMENTATION		STANDARD ABBREVIATIONS		PROJECT GENERAL NOTES:	
MANUFACTURERS CORD/CABLE HEAT TAPE ON PIPING FLEXIBLE CONDUIT TWISTED SHIELDED PAIR SEWER LINE OVERHEAD ELECTRICAL EXISTING CONDUIT UNDERGROUND CONDUIT EXPOSED CONDUIT BELOW GRADE OR CONCEALED CONDUIT CAPPED CONDUIT BENT UP OR TOWARD CONDUIT BENT DOWN OR AWAY GROUNDING CAD WELD CONNECTION CONDUCTORS NOT CONNECTED CONDUCTORS CONNECTED CONDUIT SEALS CLASS 1, DIV. 1/2 EXPLOSION PROOF NEW EQUIPMENT (STANDARD LINEWEIGHT) EXISTING EQUIPMENT (E) (LIGHT LINEWEIGHT) EQUIPMENT TO BE REMOVED		CONDUIT CALLOUT TRENCH CALLOUT EQUIPMENT CALLOUT LIGHTING FIXTURE CALLOUT: SEE SCHEDULE DRAWING KEY NOTE CALLOUT DETAIL NUMBER DETAIL IDENTIFIER REFERENCE DRAWING NUMBER PANEL AND CIRCUIT (EXAMPLE: PANEL LPA, CIRCUITS 1 AND 3) PHASE/SWITCHLEG CONDUCTOR HOMERUN/CONDUIT GROUND CONDUCTOR NEUTRAL CONDUCTOR		NORMALLY CLOSED NORMALLY OPEN TEMPERATURE SWITCH - TS LEVEL SWITCH - LS PRESSURE SWITCH - PS LIMIT SWITCH - LS CONTACT - CR = CONTROL RELAY, MS-MOTOR STARTER, OR AS INDICATED SWITCH - SW FLOW SWITCH - FS PUSHBUTTON - PB SELECTOR SWITCH. HAND-OFF-AUTO SHOWN. X'S INDICATE CONTACT SWITCHING CONVENTION. AMMETER VOLTMETER GENERATOR MOTOR STARTER PHASE FAIL RELAY ELAPSED TIME METER STARTS COUNTER CONTROL RELAY TIME DELAY RELAY SV-SOLENOID VALVE INSTRUMENT (L=LEVEL, F=FLOW P=PRESSURE) INDICATING LIGHT, LETTER INDICATES: R-RED, G-GREEN, A-AMBER, W-WHITE, B-BLUE D.C. TERMINAL A.C. TERMINAL FIELD INSTRUMENT HORN SPEED POTENTIOMETER		A. AMP AC AFF AI AIC AL ALT AO ATS BAT BC BH BP C CAP CB CKT CNT CPT CR CT CV DEM DI DO DWG E OR (E) EF FE FS FT FU FVNR G, GND GEN GFCI/GFI H HH HID HOA HTR IC IS ISR KW KWH KWD LC LCP LE LS LT LTG M MCC MCP MFR MOV MS MTS N NC NO OI OIT OL OT P PB PE PFR PLC PNL POT PS PT PVC RGS RTM RV S SA SE SHT SS SSS SV T TC TDR TST TYP UH UPS VS VT VFD VSD W WHM WP XFMR XP XMTR ZS AMPERE AIR COMPRESSOR ABOVE FINISHED FLOOR ANALOG INPUT POINT (PLC) AMPERES INTERRUPTING CAPACITY ALARM ALTERNATOR ANALOG OUTPUT POINT (PLC) AUTOMATIC TRANSFER SWITCH BATTERY BATTERY CHARGER BLOCK HEATER BYPASS CONTACTOR CONDUIT (RGS) CAPACITOR CIRCUIT BREAKER CIRCUIT COUNTER CONTROL PANEL CONTROL POWER TRANSFORMER CONTROL RELAY CURRENT TRANSFORMER CHECK VALVE DEMAND DIGITAL INPUT POINT (PLC) DIGITAL OUTPUT POINT (PLC) DRAWING EXISTING DEVICE EXHAUST FAN FLOW ELEMENT FLOW SWITCH FLOW TRANSMITTER FUSED FULL VOLTAGE NON-REVERSING GROUND GENERATOR GROUND FAULT CIRCUIT INTERRUPTER HOT, HIGH HAND HOLE HIGH INTENSITY DISCHARGE HAND-OFF-AUTO HEATER ISOLATION CONTACTOR INTRINSICALLY SAFE INTRINSICALLY SAFE RELAY KILOWATT KILOWATT HOUR KILOWATT DEMAND LIGHTING CONTACTOR LOCAL CONTROL PANEL LEVEL ELEMENT LIMIT SWITCH LEVEL TRANSMITTER LIGHTING METER MOTOR CONTROL CENTER MAIN CONTROL PANEL MANUFACTURER MOTOR OPERATED VALVE MOTOR STARTER MANUAL TRANSFER SWITCH NEUTRAL NORMALLY CLOSED NORMALLY OPEN OPERATOR INTERFACE OPERATOR IN TROUBLE OVERLOAD RELAY OVER TEMP POWER PUSH BUTTON PHOTO ELECTRIC RELAY PHASE FAILURE RELAY PROGRAMMABLE LOGIC CONTROLLER PANEL POTENTIOMETER PRESSURE SWITCH POTENTIAL TRANSFORMER POLY VINYL CHLORIDE (CONDUIT) RIGID GALVANIZED STEEL (CONDUIT) RUN TIME METER REDUCED VOLTAGE SIGNAL SURGE ARRESTOR SERVICE ENTRANCE SHEET STAINLESS STEEL SOLID STATE STARTER SOLENOID VALVE THERMOSTAT TIME CLOCK TIME DELAY TWISTED SHIELDED 3 CONDUCTOR (TRIAD) TYPICAL UNIT HEATER UNINTERRUPTABLE POWER SUPPLY VIBRATION SWITCH VIBRATION TRANSMITTER VARIABLE FREQUENCY DRIVE VARIABLE SPEED DRIVE WATT WATT HOUR METER WEATHER PROOF TRANSFORMER EXPLOSION PROOF TRANSMITTER LIMIT SWITCH		1. THE ELECTRICAL DRAWINGS AND SCHEDULES ARE FUNCTIONAL IN NATURE AND DO NOT SPECIFY EXACT LOCATIONS OF EQUIPMENT OR EQUIPMENT TERMINATIONS. IT IS THE INTENT OF THESE DRAWINGS TO DESCRIBE AND PROVIDE FOR THE FURNISHING, INSTALLING, TESTING AND PLACING IN FULLY OPERATIONAL CONDITION ALL EQUIPMENT, MATERIALS, DEVICES AND NECESSARY APPURTENANCES TO PROVIDE A COMPLETE ELECTRICAL SYSTEM, TOGETHER WITH SUCH OTHER MISCELLANEOUS INSTALLATIONS AND EQUIPMENT SHOWN ON THE DRAWINGS. THE WORK SHALL INCLUDE ALL MATERIALS, APPLIANCES AND APPARATUS NOT SPECIFICALLY MENTIONED HEREIN OR SHOWN ON THE DRAWINGS, BUT WHICH ARE NECESSARY TO MAKE A COMPLETE, FULLY OPERATIONAL INSTALLATION OF ALL ELECTRICAL SYSTEMS SHOWN ON THE DRAWINGS. 2. THIS PROJECT INCLUDES THE INSTALLATION OF PACKAGED EQUIPMENT SYSTEM(S) OR SUB-SYSTEM(S) THAT WILL REQUIRE COORDINATION BETWEEN THE CONTRACTOR AND THE MANUFACTURER TO DETERMINE THE DETAILED INSTALLATION REQUIREMENTS. THE ENGINEER HAS SHOWN GENERAL INSTALLATION INFORMATION FOR THESE SYSTEMS BASED ON THE BEST INFORMATION AVAILABLE AT THE TIME OF DESIGN. WHERE INDICATED ON THE DRAWINGS TO "PROVIDE A COMPLETE AND OPERATIONAL SYSTEM" THE CONTRACTOR SHALL PROVIDE ALL MATERIALS, INSTALLATION, AND COORDINATION WITH THE MANUFACTURER REQUIRED SO THE EQUIPMENT IS INSTALLED AND OPERATES IN A SATISFACTORY MANNER. MINOR CHANGES IN EQUIPMENT LOCATIONS, QUANTITY OF TERMINATIONS OR WIRES, JUNCTION BOXES, CONDUIT, ETC SHALL BE INCLUDED IN THE CONTRACT PRICE. 3. CONTRACTOR SHALL COORDINATE WITH OWNER FOR REMOVAL OF EXISTING EQUIPMENT AND ANY REQUIRED PHASING TO MAINTAIN FACILITY OPERATIONS . 4. DISPOSE OF ALL DEMO MATERIALS NOT WANTED BY OWNER. 5. ALL CONDUIT IN NEW WALLS OR CEILINGS SHALL BE CONCEALED WHERE POSSIBLE. 6. THE NUMBER OF CONDUCTORS AND CONDUIT ROUTING WILL VARY BASED ON HOW THE CONTRACTOR ELECTS TO ROUTE AND COMBINE CIRCUITING. THE CONTRACTOR SHALL PROVIDE DETAILED REDLINE MARKUPS ON A DEDICATED SET OF CONSTRUCTION DRAWINGS TO THE ENGINEER UPON COMPLETION OF THE PROJECT FOR PREPARATION OF RECORD DRAWINGS. THIS INCLUDES ACTUAL RACEWAY ROUTING, CONDUCTOR QUANTITIES, PANEL SCHEDULES, RECEPTACLE CONFIGURATIONS AND MOUNTING ELEVATIONS, ETC. 7. ALL MATERIALS SHALL CONFORM TO THE NATIONAL ELECTRICAL CODE ARTICLE 110-14C. WIRING AND CIRCUIT BREAKERS ON THIS PROJECT ARE DESIGNED FOR 75 DEG C OPERATION ABOVE 100 AMPERES; 60 DEG C FOR 100 AMPERES AND BELOW. ALL PRODUCTS FURNISHED ON THIS PROJECT SHALL HAVE ELECTRICAL TERMINATIONS RATED FOR 60 DEG C FOR AMPACITIES OF 100 AMPERES AND BELOW, AND RATED FOR 75 DEG C FOR AMPACITIES ABOVE 100 AMPERES. ALL CONDUCTORS SHALL BE COPPER.	
LIGHTING AND RECEPTACLES		ELECTRICAL AND POWER DISTRIBUTION						ELECTRICAL SHEET INDEX:	
FLUORESCENT LIGHTING FIXTURE. FIXTURE IDENTIFIER AND SWITCHED CIRCUIT INDICATED. REFER TO LIGHTING SCHEDULE FOR FIXTURE AND LAMP TYPE. FLUORESCENT LIGHTING FIXTURE WITH EMERGENCY BATTERY PACK LIGHTING FIXTURE, EMERGENCY DUAL HEAD WITH INTEGRAL BATTERY PACK. EXIT SIGN WHERE INDICATED. LED EXIT SIGN INCANDESCENT, COMPACT FLUORESCENT OR H.I.D. LIGHTING FIXTURE, CEILING MOUNTED PHOTOELECTRIC CONTROL UNIT. WALL MOUNTED LIGHTING FIXTURE, POLE MOUNT LIGHTING FIXTURE, WALL MOUNT DUPLEX RECEPTACLE, NUMBER INDICATES CIRCUIT. GFCI WHERE INDICATED FOURPLEX RECEPTACLE, NUMBER INDICATES CIRCUIT. DUPLEX RECEPTACLE MOUNTED 6" ABOVE COUNTER NUMBER INDICATES CIRCUIT. DUPLEX RECEPTACLE FLOOR MOUNTED, NUMBER INDICATES CIRCUIT. SPECIAL PURPOSE RECEPTACLE OR DEDICATED EQUIPMENT CONNECTION, AS NOTED. TELEPHONE OUTLET DATA OUTLET SPLIT TELEPHONE DATA OUTLET INTERCOM SWITCH, NUMBERS REFER TO SWITCH TYPE AND SWITCHED CIRCUIT. JUNCTION BOX JUNCTION BOX, EXPLOSION PROOF THERMOSTAT		PANELBOARD 208Y/120V OR 120/240V PANELBOARD 480Y/277V UTILITY METER MOTOR CONNECTION NUMBER INDICATES HORSEPOWER THERMAL OVERLOAD RELAY FULL VOLTAGE NON REVERSING MOTOR STARTER NUMBER INDICATES NEMA SIZE REDUCED VOLTAGE SOLID STATE STARTER VARIABLE FREQUENCY DRIVE LINE REACTOR BUS CONNECTION (N=NEUTRAL, G=GROUND) HEATER, NUMBER INDICATES KW DISCONNECT SWITCH - HP RATED, AS INDICATED DISCONNECT SWITCH (FUSED) TRANSFORMER CARTRIDGE FUSE AND FUSEHOLDER ATS - AUTOMATIC TRANSFER SWITCH MTS - MANUAL TRANSFER SWITCH THERMAL MAG CIRCUIT BREAKER, RATING/NO. POLES MOTOR CIRCUIT PROTECTOR, RATING/NO. POLES GROUND ROD AND WELL GROUNDING ELECTRODE PULL OUT PLUG-RECEPTACLE/MCC CONNECTION						# SHEET TITLE 62 E0.1 ELECTRICAL SYMBOLS AND ABBREVIATIONS 63 E0.2 ELECTRICAL DEMOLITION SITE PLAN AND ONE LINE DIAGRAM 64 E0.3 OVERALL ELECTRICAL SITE PLAN 65 E0.4 AREA CLASSIFICATION PLAN 66 E1.1 ELECTRICAL SERVICE AREA PLAN 67 E2.1 BLOWER BUILDING POWER, CONTROL AND GROUNDING PLAN 68 E2.2 BLOWER BUILDING LIGHTING AND HVAC PLAN 69 E3.1 TRAIN NO.3 ELECTRICAL PLAN AND ELEVATION 70 E4.1 INFLUENT PUMP STATION, WAS, DEWATERING ELECTRICAL PLAN 71 E4.2 INFLUENT PUMP STATION ELECTRICAL PLAN AND ELEVATION 72 E4.3 2W PUMP STATION ELECTRICAL PLAN AND ELEVATION 73 E5.1 UV DISINFECTION CHAMBER & DIGESTER ELECTRICAL PLAN AND ELEVATION 74 E6.1 ELECTRICAL ONE LINE DIAGRAM AND LOAD CALCULATIONS 75 E6.2 ELECTRICAL PANEL SCHEDULES 76 E6.3 CONDUIT AND CONDUCTOR SCHEDULES 77 E7.0 CONTROL SYSTEM OVERVIEW 78 E7.1 MAIN PLC CONTROL PANEL ELEVATION 79 E7.2 MAIN PLC CONTROL PANEL - POWER WIRING DIAGRAM 80 E7.3 MAIN PLC CONTROL PANEL - DIGITAL INPUTS 81 E7.4 MAIN PLC CONTROL PANEL - DIGITAL OUTPUTS 82 E7.5 MAIN PLC CONTROL PANEL - ANALOG INPUTS - SHEET 1 83 E7.6 MAIN PLC CONTROL PANEL - ANALOG INPUTS - SHEET 2 84 E7.7 MAIN PLC CONTROL PANEL - ANALOG OUTPUTS 85 E8.1 INFLUENT PUMP STATION CONTROL PANEL ELEVATIONS 86 E8.2 INFLUENT PUMP STATION CONTROL PANEL - POWER WIRING DIAGRAMS 87 E8.3 INFLUENT PUMP STATION CONTROL PANEL - DIGITAL INPUTS 88 E8.4 INFLUENT PUMP STATION CONTROL PANEL - DIGITAL OUTPUTS 89 E8.5 INFLUENT PUMP STATION CONTROL PANEL - ANALOG INPUTS 90 E8.6 INFLUENT PUMP STATION CONTROL PANEL - ANALOG OUTPUTS 91 E9.1 BLOWER BUILDING MOTOR CONTROL CENTER ELEVATION 92 E9.2 BLOWER BUILDING MOTOR CONTROL CENTER - VFD WIRING DIAGRAMS 93 E9.3 VENTILATION CONTROL PANEL ELEVATION AND WIRING DIAGRAM 94 E10.1 ELECTRICAL DETAILS - SHEET 1 95 E10.2 ELECTRICAL DETAILS - SHEET 2 96 E10.3 STANDBY DIESEL GENERATOR ELEVATIONS	
								BID SET	
E0.1		EASTSOUND SEWER AND WATER DISTRICT WASHINGTON SAN JUAN COUNTY		DATE 09-12-2022		SCALE AS SHOWN		JOB NUMBER 2020-070	
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OVERALL ELECTRICAL SITE PLAN

SCALE: 1" = 10' AT FULL SCALE



GENERAL NOTES:

1. ALL CONDUIT ROUTING IS NOT SHOWN, AND WHERE SHOWN, IS SCHEMATIC. ELECTRICAL CONTRACTOR SHALL DETERMINE THE BEST ROUTING PATH AND CIRCUIT COMBINATIONS BASED ON FIELD CONDITIONS AND ELECTRICAL CODES.
2. EQUIPMENT LOCATIONS AND ARRANGEMENT ARE SCHEMATIC. CONTRACTOR SHALL COORDINATE WITH EQUIPMENT MANUFACTURER FOR DETAILED CONNECTION REQUIREMENTS AND PROVIDE MATERIALS AND INSTALLATION FOR A COMPLETE AND OPERATIONAL SYSTEM.
3. ELECTRIC HAND HOLES ARE NOT SHOWN ON THESE PLANS BUT SHALL BE REQUIRED TO MEET CODE AND AS SPECIFIED IN DIVISION 26. COORDINATE ALL FINAL LOCATIONS WITH ENGINEER.

KEY NOTES:

1. PROVIDE NEW SCADA COMPUTER SYSTEM PER SPECIFICATIONS. COORDINATE WITH OWNER FOR FINAL LOCATION IN LAB. INSTALL NEW CAT 6 NETWORK JUNCTION BOX WITH COMBO NETWORK/TELEPHONE OUTLET. EXTEND CAT6 CABLING TO NETWORK SWITCH LOCATED IN INFLUENT PUMP STATION CONTROL PANEL. WIRE EXISTING TELEPHONE CIRCUIT TO JBOX FOR SCADA ALARM DIALER.

NO.	REVISIONS	BY	DATE



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WILSONENGINEERING.COM



09-12-2022

DESIGNED BY	BZ
DRAWN BY	BZ
CHECKED BY	BZ

EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON
SAN JUAN COUNTY
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

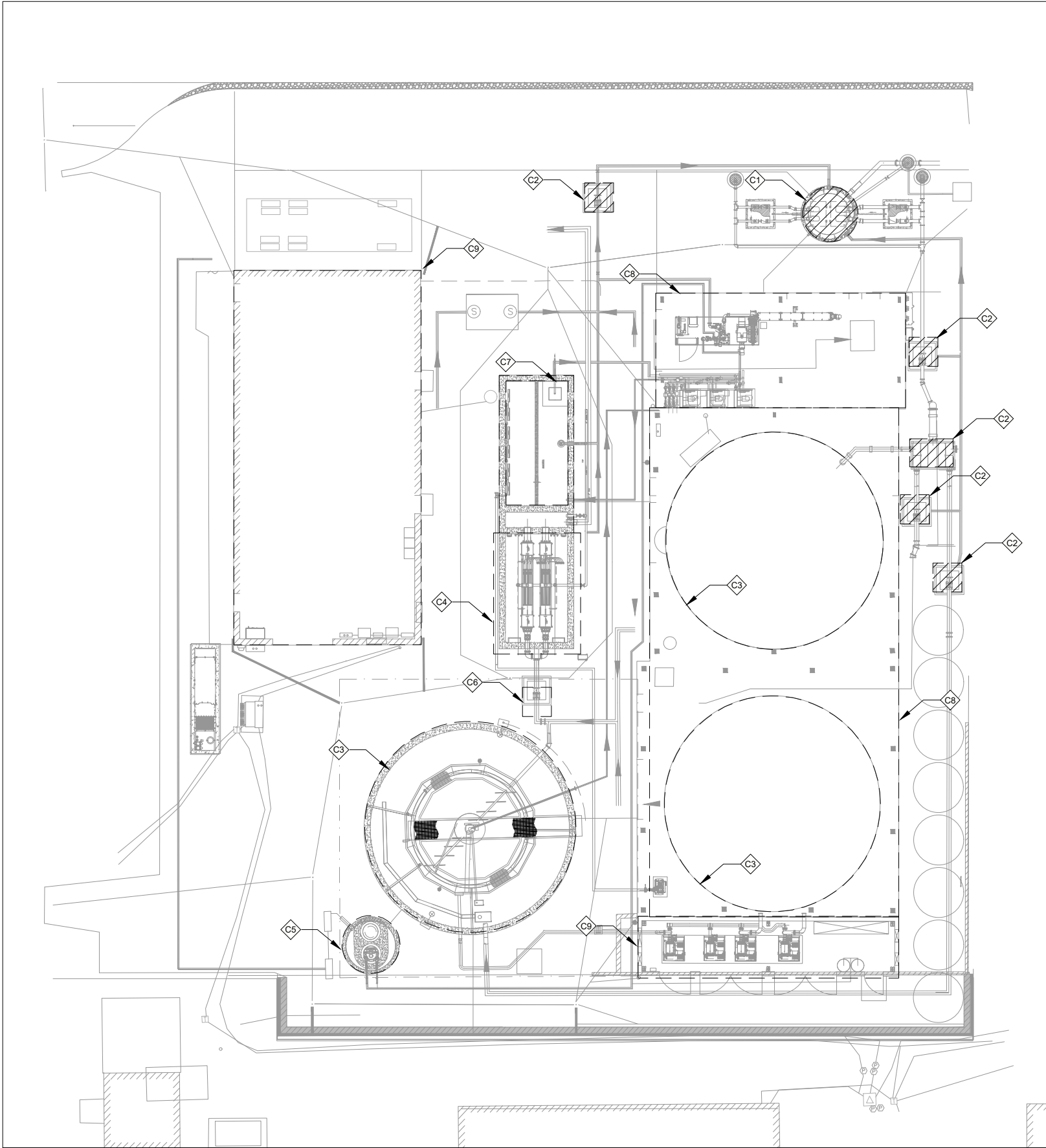
OVERALL ELECTRICAL SITE PLAN

DATE	09-12-2022
SCALE	AS SHOWN
JOB NUMBER	2020-070

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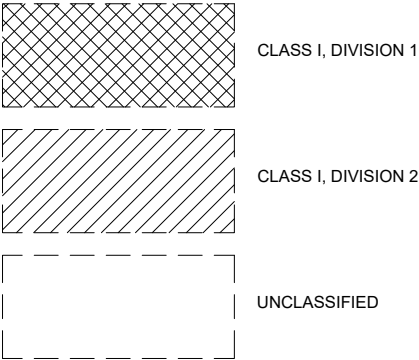


BID SET



AREA CLASSIFICATION SITE PLAN
SCALE: 1" = 10' AT FULL SCALE
10 0 10 20
SCALE HORIZONTAL SCALE FEET

AREA CLASSIFICATION LEGEND:



GENERAL NOTES:

- 1. AREA CLASSIFICATIONS BASED ON NFPA 820, STANDARD FOR FIRE PROTECTION IN WASTEWATER TREATMENT AND COLLECTION FACILITIES, 2020 EDITION.
- 2. PROCESS BASED ON TREATMENT OF WASTEWATER FROM RESIDENTIAL STEP STATIONS PER NFPA 820-2020, TABLE 4.2.2, ROW 9.
- 3. ELECTRICAL INSTALLATION IN CLASSIFIED AREAS SHALL COMPLY WITH THE REQUIREMENTS OF NFPA 70, NATIONAL ELECTRICAL CODE, AND OTHER STATE AND AGENCY REQUIREMENTS AS REQUIRED BY THE AUTHORITY HAVING JURISDICTION.

CLASSIFIED AREA NOTES:

- C1 INFLUENT PUMP STATION: PRIMARILY RESIDENTIAL SEWER - CLASS I, DIV 2 PER NFPA 820-2020 TABLE 4.2.2, ROW 10a. EXTENT OF CLASSIFIED LOCATION: WITHIN ENCLOSED SPACE.
- C2 BELOW GRADE METERING VAULT WITH CLOSED PIPING SYSTEM: CLASS I, DIV 2 PER NFPA 820-2020 TABLE 4.2.2, ROW 10a AND 34a. EXTENT OF CLASSIFIED LOCATION: WITHIN ENCLOSED SPACE.
- C3 AEROBIC TREATMENT OF WASTEWATER PRECEDED BY PRIMARY TREATMENT: UNCLASSIFIED PER NFPA 820-2020 TABLE 5.2.2.11.
- C4 UV DISINFECTION UNITS: UNCLASSIFIED PER NFPA 820-2020 TABLE 5.2.2.26.
- C5 2W PUMP STATION: UNCLASSIFIED PER NFPA 820-2020 TABLE 5.2.2.21.
- C6 EFFLUENT FLOW METER VAULT: UNCLASSIFIED PER NFPA 820-2020 TABLE 5.2.2.27.
- C7 AERATED DIGESTER OPEN TO ATMOSPHERE: AEROBIC TREATMENT OF WASTEWATER PRECEDED BY PRIMARY TREATMENT WITH SKIMMING: UNCLASSIFIED PER NFPA 820-2020 TABLE 5.2.2.11.
- C8 DEWATERING BUILDING WITH DEWATERING PRESS: UNCLASSIFIED PER NFPA 820-2020 TABLE 6.2.2, ROW 12a WITH TYPE 'C' VENTILATION, CONTINUOUSLY VENTILATED AT 6 AIR CHANGES PER HOUR AND IN ACCORDANCE WITH NFPA 820-2020 CHAPTER 9.
- C9 OFFICE BUILDING AND BLOWER BUILDING NOT CLASSIFIED.

NO.	REVISIONS	BY	DATE



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DRAWN BY	BZ
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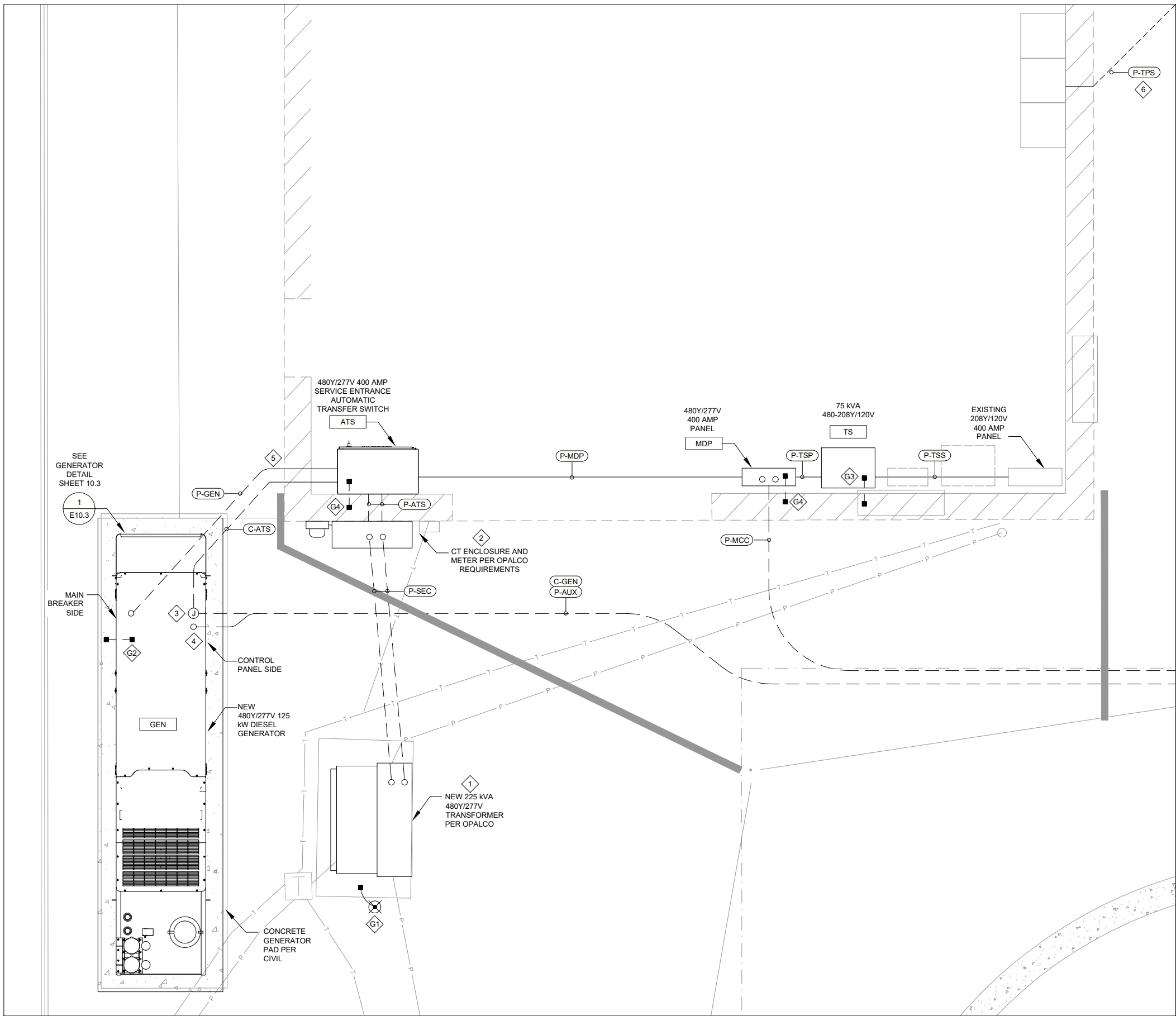
EASTSOUND SEWER AND WATER DISTRICT	WASHINGTON
SAN JUAN COUNTY	WASTEWATER TREATMENT PLANT UPGRADE – PHASE I
AREA CLASSIFICATION SITE PLAN	

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ELECTRICAL SERVICE AREA PLAN

SCALE: 1" = 2' AT FULL SCALE



NO.	REVISIONS	BY	DATE



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09-12-2022

GENERAL NOTES:

1. ALL CONDUIT ROUTING IS NOT SHOWN. ELECTRICAL CONTRACTOR SHALL DETERMINE THE BEST ROUTING PATH AND CIRCUIT COMBINATIONS BASED ON FIELD CONDITIONS AND ELECTRICAL CODES.
2. CONDUCTOR AND CONDUIT SIZING SHALL BE AS REQUIRED BY NEC.
3. EQUIPMENT LOCATIONS AND ARRANGEMENT ARE SCHEMATIC. CONTRACTOR SHALL COORDINATE WITH EQUIPMENT MANUFACTURER FOR DETAILED CONNECTION REQUIREMENTS AND PROVIDE MATERIALS AND INSTALLATION FOR A COMPLETE AND OPERATIONAL SYSTEM.

KEY NOTES:

- 1 COORDINATE WITH OPALCO FOR REMOVAL OF EXISTING 208Y/120V UTILITY TRANSFORMER AND INSTALLATION OF NEW 480Y/277V TRANSFORMER. CONTRACTOR SHALL MODIFY EXISTING PAD AND PREP FOR INSTALLATION PER OPALCO REQUIREMENTS. SECONDARY CONDUIT AND WIRE BY CONTRACTOR. OPALCO SHALL FURNISH TRANSFORMER AND INSTALL ON MODIFIED PAD. PRIMARY WIRE AND CONNECTIONS BY OPALCO.
- 2 CONTRACTOR TO PROVIDE NEW CT ENCLOSURE, LANDING PADS AND METER BASE PER OPALCO REQUIREMENTS. METER AND CTS BY UTILITY.
- 3 PROVIDE J-BOX TO COMBINE ATS AND GENERATOR STATUS SIGNALS AND WIRE TO PLC CONTROL PANEL IN BLOWER BUILDING.
- 4 PROVIDE 120V 20 AMP CIRCUIT FROM LOAD CENTER IN MCC TO GENERATOR AUX POWER CONNECTIONS TO POWER BATTERY CHARGER AND COOLANT HEATER.
- 5 REPAIR CONCRETE AND BUILDING WALLS AND STRUCTURE AS NEEDED TO ROUTE NEW CONDUITS INTO EXISTING BUILDING.
- 6 REPLACE EXISTING 200 AMP 208Y/120V POWER FEEDER FROM EXISTING SWITCHGEAR TO EXISTING TPS PANEL IN TREATMENT BUILDING. SEE SITE PLAN AND SCHEDULES.

GROUNDING NOTES:

- G1 PROVIDE GROUNDING FOR UTILITY TRANSFORMER PER NEC AND OPALCO REQUIREMENTS.
- G2 GROUND CONCRETE ENCASED GENERATOR PAD REBAR TO GENERATOR AND CONNECT TO GROUNDING ELECTRODE SYSTEM.
- G3 PROVIDE GROUNDING FOR TRANSFORMER AS SEPARATELY DERIVED SYSTEM.
- G4 INSTALL NEW GROUNDING ELECTRODE SYSTEMS AND TIE INTO EXISTING BUILDING GROUNDING ELECTRODE SYSTEM.



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EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON
SAN JUAN COUNTY
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

ELECTRIC SERVICE AREA PLAN

DATE
09-12-2022

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GENERAL NOTES:

1. ALL CONDUIT ROUTING IS NOT SHOWN, AND WHERE SHOWN, IS SCHEMATIC. ELECTRICAL CONTRACTOR SHALL DETERMINE THE BEST ROUTING PATH AND CIRCUIT COMBINATIONS BASED ON FIELD CONDITIONS AND ELECTRICAL CODES.
2. EQUIPMENT LOCATIONS AND ARRANGEMENT ARE SCHEMATIC. CONTRACTOR SHALL COORDINATE WITH EQUIPMENT MANUFACTURER FOR DETAILED CONNECTION REQUIREMENTS AND PROVIDE MATERIALS AND INSTALLATION FOR A COMPLETE AND OPERATIONAL SYSTEM.
3. ELECTRIC HANDHOLE LOCATIONS SHOWN ARE SCHEMATIC. COORDINATE ALL FINAL LOCATIONS WITH ENGINEER.

KEY NOTES:

- 1 PROVIDE GROUNDING IN ACCORDANCE WITH NEC ART.250. GROUNDING ELECTRODE SYSTEM TO INCLUDE GROUND RODS, CONCRETE ENCASED ELECTRODE, STRUCTURAL STEEL, METAL WATER PIPES AND OTHERS WHERE AVAILABLE.
- 2 CONDUIT ROUTING IS SCHEMATIC. COORDINATE STUBUP LOCATIONS WITH EQUIPMENT MANUFACTURER SO POWER AND CONTROL CONDUITS ENTER EQUIPMENT ENCLOSURES AT DESIGNATED AREAS AND ROUTE TO APPROPRIATE MCC SECTION OR CONTROL PANEL.
- 3 PROVIDE ALL CONNECTIONS TO BLOWERS AS REQUIRED BY MANUFACTURER. INCLUDES MOTOR CONNECTIONS AND SENSORS. GROUND SKIDS PER NEC.
- 4 PROVIDE CEILING MOUNTED SMOKE/HEAT DETECTOR, COORDINATE FINAL LOCATION WITH BUILDING CONSTRUCTION. SYSTEM SENSOR MODEL 4WTAR-B FORM C RELAY, SOUNDER, 4-WIRE. WIRE TO PLC CONTROL PANEL.
- 5 PROVIDE DOOR SWITCH FOR INTRUSION MONITORING. MAGNETIC REED STYLE, SPDT, INTERLOGIX SENTROL MODEL 2807T OR EQUAL. WIRE TO PLC CONTROL PANEL.
- 6 PROVIDE NEMA 4X STAINLESS STEEL LOCAL DISCONNECT.
- 7 ROUTE ALL FIELD CONDUITS FROM BLOWER BUILDING THROUGH HANDHOLES. MAINTAIN SEPARATION OF POWER AND SIGNAL/CONTROL RACEWAYS. PROVIDE A MINIMUM OF 2 SPARES FROM EACH POWER HANDHOLE TO MCC AND EACH CONTROL HANDHOLE TO PLC PANEL. SEE PLANS AND SCHEDULES FOR DETAILS.
- 8 PROVIDE SEPARATE RACEWAY FROM PLC CONTROL PANEL TO MCC AS REQUIRED TO MAINTAIN SEPARATION OF POWER, SIGNAL, AND DATA CABLING. COORDINATE STUB UP LOCATIONS WITH EQUIPMENT REQUIREMENTS. WIRE SIZE AND QUANTITY PER SCHEDULES AND AS REQUIRED FOR ALL CONNECTIONS BETWEEN MCC AND PLC CONTROL PANEL.



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EASTSOUND SEWER AND WATER DISTRICT

11

COUNTY
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

BLOWER BUILDING POWER, CONTROL AND GROUNDING PLAN

DATE 09-12-2022

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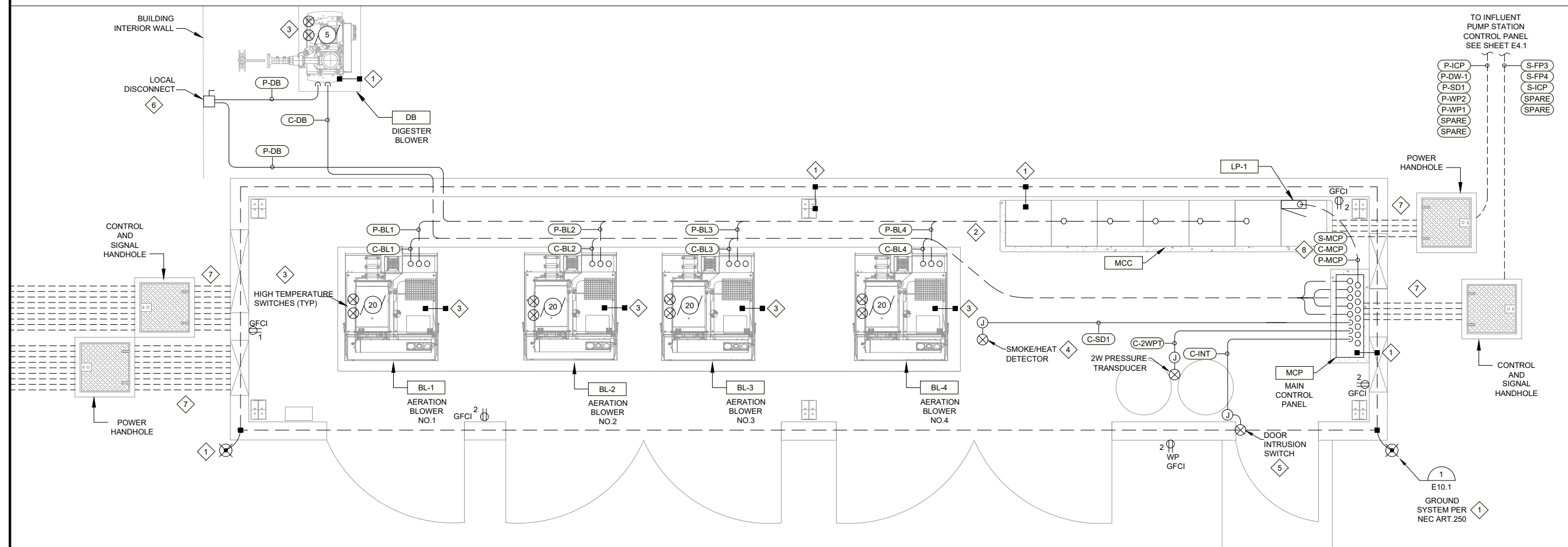
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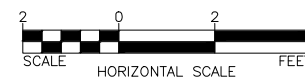


811 Call 811
two business days
before you dig

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BLOWER BUILDING - POWER, CONTROL AND GROUNDING PLAN

SCALE: 1" = 2' AT FULL SCALE



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09-12-2022

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EASTSOUND SEWER AND WATER DISTRICT
SAN JUAN COUNTY
WASHINGTON
WASTEWATER TREATMENT PLANT UPGRADE – PHASE I
BLOWER BUILDING LIGHTING AND HVAC PLAN

DATE	09-12-2022	SCALE	AS SHOWN	JOB NUMBER	2020-070
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SHEET	E2.2	PAGE	68	OF	96
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GENERAL NOTES:

- ROUTE ALL BRANCH CIRCUITS TO LIGHTING PANEL(S) CIRCUIT SHOWN ON THE PLANS. ALL CONDUIT ROUTING IS NOT SHOWN. ELECTRICAL CONTRACTOR SHALL DETERMINE THE BEST ROUTING PATH AND CIRCUIT COMBINATIONS BASED ON FIELD CONDITIONS AND ELECTRICAL CODE.
- BRANCH CIRCUITS NOT SHOWN ON CONDUIT SCHEDULE SHALL BE SIZED AS REQUIRED BY NEC.
- ALL ELECTRICAL CONDUIT AND BOXES SHALL BE UNDERGROUND OR CONCEALED IN WALLS. COORDINATE INSTALLATION WITH CMU INSTALLER WHERE NECESSARY.

KEY NOTES:

- PROVIDE PHOTOCELL ON EXTERIOR FIXTURE. FIXTURE SHALL BE ON DEDICATED SWITCH AND DARY SKY COMPLAINT, SEE SCHEDULE.
- WIRE EMERGENCY LIGHTS TO INTERIOR LIGHTING CIRCUIT.
- PROVIDE OCCUPANCY SENSOR FOR INTERIOR LIGHTING. COMPLY WITH WASHINGTON STATE ENERGY CODE REQUIREMENTS.
- CONTRACTOR SHALL PROVIDE COMPLETE VENTILATION CONTROL PANEL, INCLUDING RELAY CONTROLS AND THERMOSTATS FOR EXHAUST FAN AND LOUVER CONTROL. ONE THERMOSTAT FOR EACH EXHAUST FAN. MOTORIZED LOUVER SHALL OPERATE WHEN EITHER FAN IS RUNNING.

PANEL: LP1				PANEL SCHEDULE							PROJECT: Eastsound WWTP			
120/240V, 1Ph, 3W.			100A Bus				100A M.C.B.				SURFACE MOUNTED			
CKT NO	DESCRIPTION / LOCATION	LOAD (VA)	LOAD TYPE	C.B. AMP	C.B. POLE	PHASE	C.B. POLE	C.B. AMP	LOAD TYPE	LOAD (VA)	DESCRIPTION / LOCATION	CKT NO		
1	BLOWER ROOM RECEPTACLES	900	R	20	1	A	1	20	C	1,000	MAIN CONTROL PANEL	2		
3	LOUVER AND HVAC CONTROL	500	M	20	1	B	1	20	L	239	BLOWER BUILDING LIGHTING	4		
5	EXHAUST FAN EF-1 (BLOWER BUILDING)	1,127	M	20	1	A	1				SPARE	6		
7	EXHAUST FAN EF-2 (BLOWER BUILDING)	1,127	M	20	1	B	1	20	H	1,300	GENERATOR - HEATER AND BATTERY CHGR	8		
9	UV1 POWER	2,520	C	30	1	A	1	20	H	1,000	HOT BOX 1 - GFEP	10		
11	UV2 POWER	2,520	C	30	1	B	1	20	H	1,000	HOT BOX 2 - GFEP	12		
13	UV1 CONTROLLER	600	C	20	1	A	1	20	M	500	EFFLUENT SAMPLER	14		
15	UV2 CONTROLLER	600	C	20	1	B	1	20	M	500	HYPOCHLORITE FEED PUMP	16		
17	SPARE			20	1	A	1	20			SPARE	18		
19	SPACE					B					SPACE	20		
21	SPACE					A					SPACE	22		
23	SPACE					B					SPACE	24		
TOTAL CONNECTED LOAD:		PH A	7,647 VA	63.7 AMPS				DATE: September 15, 2022						
TOTAL CONNECTED LOAD:		PH B	7,786 VA	64.9 AMPS										
MAX PHASE CONNECTED LOAD:		PH B	7,786 VA					PANEL RATING:		10,000 AIC				
TOTAL CONNECTED LOAD (2 x MAX):			15.6 kVA	64.9 AMPS				TOTAL DEMAND LOAD:		17.3 kVA		72.1 AMPS		

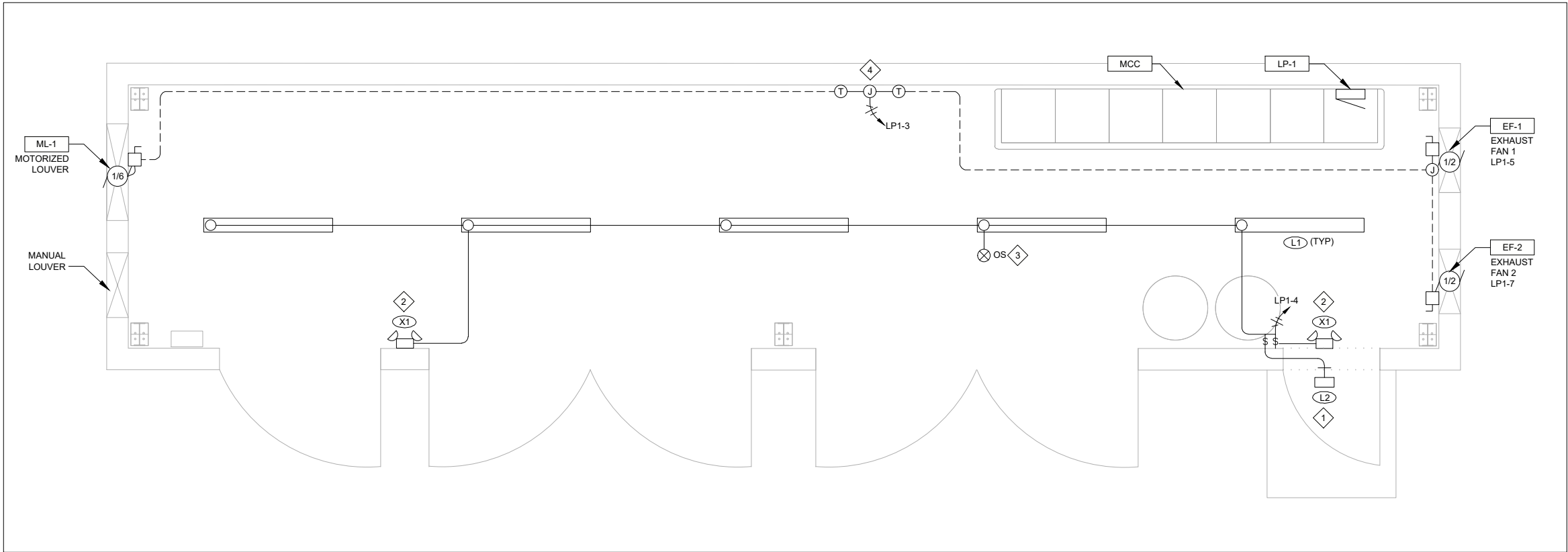
PANEL LP1 (MCC) SCHEDULE

SCALE: NONE

LIGHTING FIXTURE SCHEDULE						
ID	DESCRIPTION	LAMPS	VOLTAGE	VA	MANUFACTURER	CATALOG NUMBER
L1	NARROW LED 4' FIXTURE, VAPOR TIGHT	LED	120V	42	ACUITY	CSVT-L48-AL03-MVOLT-40K-80CRI-STSL
L2	LED WALL PACK, WET LOCATION WITH PHOTOCELL	LED	120V	17	ACUITY	ARC1LED-P2-30K-MVOLT-FAO-PE-DDBXD
X1	EMERGENCY LIGHTING W/BATTERY PACK	LED	120V	6	LITHONIA	ECC EMERGENCY COMBO UNIT

BLOWER BUILDING LIGHTING FIXTURE SCHEDULE

SCALE: NONE

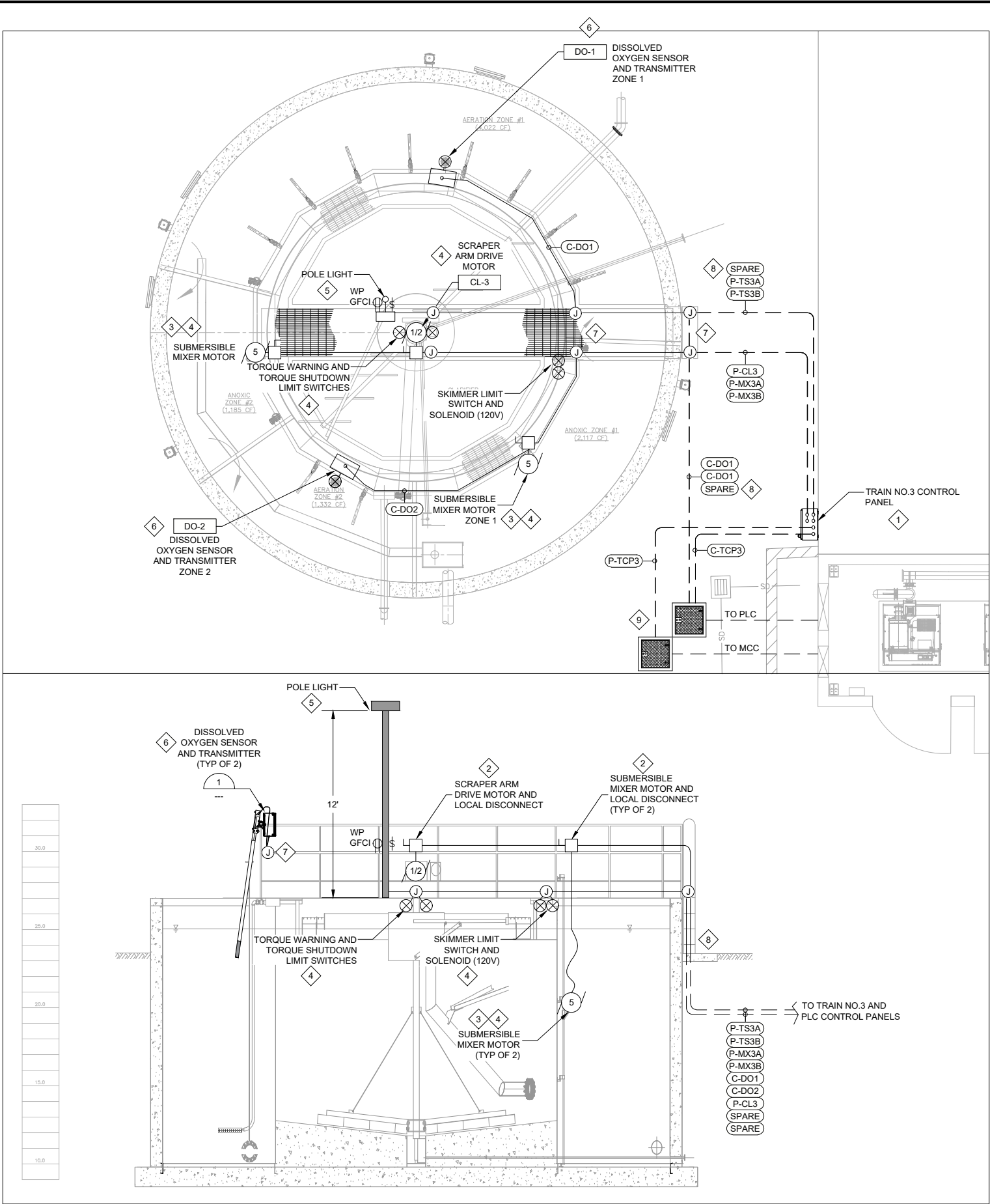


BLOWER BUILDING - LIGHTING AND HVAC PLAN

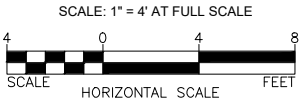
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TRAIN NO.3 ELECTRICAL PLAN AND ELEVATION

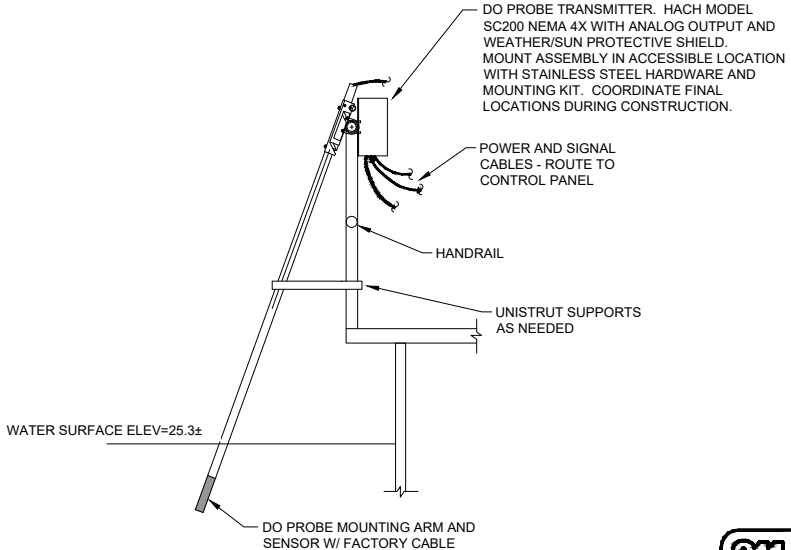


GENERAL NOTES:

- ALL CONDUIT ROUTING IS NOT SHOWN. ELECTRICAL CONTRACTOR SHALL DETERMINE THE BEST ROUTING PATH AND CIRCUIT COMBINATIONS BASED ON FIELD CONDITIONS AND ELECTRICAL CODES.
- CONDUCTOR AND CONDUIT SIZING SHALL BE AS REQUIRED BY NEC.
- EQUIPMENT LOCATIONS AND ARRANGEMENT ARE SCHEMATIC. CONTRACTOR SHALL COORDINATE WITH EQUIPMENT MANUFACTURER FOR DETAILED CONNECTION REQUIREMENTS AND PROVIDE MATERIALS AND INSTALLATION FOR A COMPLETE AND OPERATIONAL SYSTEM.
- ALL OUTDOOR CONTROL PANEL AND DISCONNECT ENCLOSURES SHALL BE TYPE 316 STAINLESS STEEL.

KEY NOTES:

- TREATMENT SYSTEM PROVIDER SHALL PROVIDE COMPLETE TREATMENT SYSTEM PER SPECIFICATIONS. CONTROL PANEL SHALL ACCEPT 480V 3 PHASE INPUT AND INCLUDE ALL WIRING AND CONTROLS FOR THE COMPLETE TREATMENT SYSTEM. FEATURES SHALL INCLUDE BUT NOT BE LIMITED TO:
 - NEMA 4X TYPE 316 STAINLESS STEEL ENCLOSURE SIZED FOR THE APPLICATION. INCLUDE DRIP SHIELD FOR OUTDOOR PANELS.
 - NEMA FULL VOLTAGE MOTOR STARTERS FOR BOTH CLARIFIER DRIVE MOTOR AND TWO SUBMERSIBLE MIXERS.
 - SEAL FAIL AND OVERTEMP SENSOR FOR SUBMERSIBLE MOTOR PROTECTION.
 - 480-120V STEP DOWN TRANSFORMER FOR CONTROL CIRCUITS.
 - HARD WIRE RELAY LOGIC CONTROL CIRCUITS FOR SYSTEM CONTROL AND PROTECTION.
 - ALARM LIGHT AND HORN WITH SILENCE PUSH BUTTON.
 - INDICATOR LIGHTS, PUSH BUTTONS, HAND-OFF-AUTO SWITCHES FOR MOTOR CONTROL.
 - WIRING TERMINALS FOR FIELD MOUNTED DEVICES INCLUDING TORQUE SWITCHES, LIMIT SWITCHES, SOLENOIDS AND ANY OTHER REQUIRED DEVICES.
 - DRY RELAY CONTACTS FOR CUSTOMER CONNECTION INCLUDING: RUN AND FAIL FOR EACH MOTOR, SEAL FAIL AND OVERTEMP FOR MIXER, ALARM AND SHUTDOWN FOR CLARIFIER.
 - PANEL HEATER WITH FAN TO PREVENT BUILDUP OF CONDENSATION IN PANEL.
 - 120V CIRCUIT FOR POLE LIGHT AND RECEPTACLE.
- ELECTRICAL CONTRACTOR SHALL PROVIDE LOCAL MOTOR DISCONNECTS FOR DRIVE MOTOR AND MIXER. NEMA 4X TYPE 316 STAINLESS STEEL.
- ROUTE SEAL FAIL AND OVERTEMP MOTOR WIRING BACK TO CONTROL PANEL SENSOR. COMBINE WIRING IN CONTROL RACEWAY AS REQUIRED.
- CONTRACTOR SHALL INSTALL ALL TREATMENT SYSTEM COMPONENTS PROVIDED BY MANUFACTURER AND PROVIDE WIRING AND RACEWAY BACK TO TREATMENT CONTROL PANEL. RATINGS, RACEWAY AND CONDUCTORS SHOWN ARE ESTIMATED; ADJUST DURING SUBMITTAL PERIOD BASED ON TREATMENT SYSTEM EQUIPMENT SUBMITTAL REQUIREMENTS.
- CONTRACTOR SHALL PROVIDE POLE LIGHT ON PLATFORM. SECURE TO HANDRAIL WITH STAINLESS STEEL HARDWARE. PROVIDE 4" SQUARE ALUMINUM POLE 12' TALL WITH RECEPTACLE, PHOTOCELL AND LOCAL ON/OFF SWITCH. LITHONIA LIGHTING #DXS0-LED-P3-30K-T3M-MVOL-T-SPA-DNAXD WITH PHOTOCELL, OR APPROVED EQUAL.
- PROVIDE (2) COMPLETE DISSOLVED OXYGEN MONITORS FOR EACH CELL. MOUNT TO TOP OF WALL WITH FACTORY HARDWARE TO ALLOW FOR TRANSMITTER ACCESS AND CLEANING. HACH MODEL LDO PROBE WITH SC200 TRANSMITTER AND ACCESSORIES. 24VDC POWERED. SEE SPECIFICATIONS.
- PROVIDE JUNCTION BOXES AS REQUIRED TO ROUTE ALL POWER AND SIGNAL CIRCUITS TO TRAIN NO.3 CONTROL PANEL AND PLC CONTROL PANEL.
- STUB SPARE CONDUITS ABOVE GROUND NEAR TREATMENT CELL WALLS. ONE FROM TREATMENT PANEL AND ONE TO PLC PANEL. CAP FOR FUTURE USE.
- NOTE: POWER AND SIGNAL CONDUITS TO UV AND DIGESTER AREA AND 2W PUMP STATION ROUTE THRU THIS AREA AND HANDHOLES BUT ARE NOT SPECIFICALLY INDICATED. FIELD COORDINATE ROUTING, SEE SHEETS E5.1 AND E4.3.

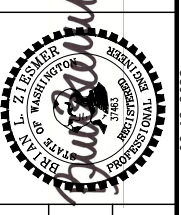


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SCALE: NONE

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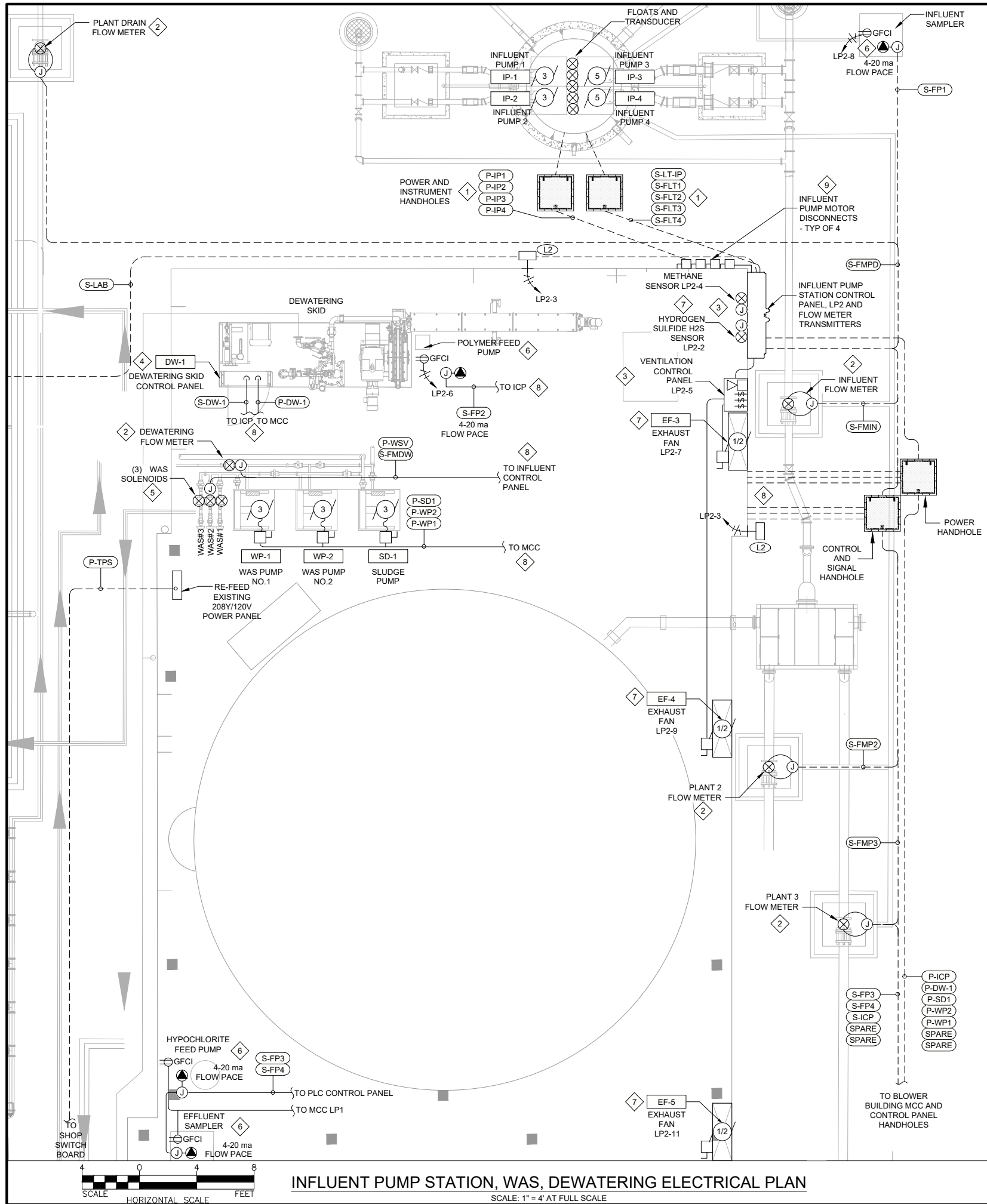
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EASTSOUND SEWER AND WATER DISTRICT
WASHINGTON
SAN JUAN COUNTY
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I
TRAIN NO.3 ELECTRICAL PLAN AND ELEVATION

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GENERAL NOTES:

- ALL CONDUIT ROUTING IS NOT SHOWN, AND WHERE SHOWN, IS SCHEMATIC. ELECTRICAL CONTRACTOR SHALL DETERMINE THE BEST ROUTING PATH AND CIRCUIT COMBINATIONS BASED ON FIELD CONDITIONS AND ELECTRICAL CODES.
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- UPSIZE CONDUITS FROM WET WELLS TO JUNCTION BOXES AS REQUIRED FOR PUMP AND FLOAT CABLES TO MEET NEC FILL REQUIREMENTS.
- ALL OUTDOOR CONTROL PANEL AND DISCONNECT ENCLOSURES SHALL BE TYPE 316 STAINLESS STEEL. ALL OUTDOOR ENCLOSURES SHALL BE LOCKABLE AND INCLUDE DRIP SHIELDS.
- POWER AND CONTROL HANDHOLE LOCATIONS SHOWN ARE SCHEMATIC IN NATURE. FIELD COORDINATE FINAL LOCATIONS TO ALIGN WITH RACEWAY ROUTING AND TO AVOID CONFLICTING UNDERGROUND PIPING AND UTILITIES.

KEY NOTES:

- PROVIDE INDIVIDUAL CONDUITS FOR EACH PUMP, FLOAT, AND TRANSDUCER CABLE. OVERSIZE CONDUITS AS REQUIRED BETWEEN WET WELL AND HANDHOLE FOR EASY REMOVAL AND REPLACEMENT OF FACTORY PUMP AND INSTRUMENT CABLES. SEAL ALL CONDUIT ENTRANCES WITH DUCT SEAL AT COMPLETION OF PROJECT.
- FLOW METER: SIZE PER CIVIL DRAWINGS. MAGNETIC FLOW METER, ENDRESS-HAUSER PROMAG W400, CLASS 1 DIV 2, CARBON STEEL FLANGE, 120-240VAC OR 24VAC/DC, ETHERNET/IP, POLYURETHANE LINER, 316L ELECTRODES, LOCAL DISPLAY ON FLOW METER (WAS), REMOTE METERS WITH SUBMERSIBLE KITS (VAULT METERS), NEMA 4X, NPT 1/2. PROVIDE (2) 316L GROUNDING RINGS FOR EACH FLOW TUBE.
- DEWATERING BUILDING WITH DEWATERING PRESS: UNCLASSIFIED PER NFPA 820-2020 TABLE 6.2.2, ROW 12a WITH TYPE 'C' VENTILATION, CONTINUOUSLY VENTILATED AT 6 AIR CHANGES PER HOUR AND IN ACCORDANCE WITH NFPA 820-2020 CHAPTER 9. PROVIDE VENTILATION FANS, GAS SENSORS, LOCAL ALARM LIGHT AND HORN, AND ACCESSORIES TO MEET NFPA 820 VENTILATION AND MONITORING REQUIREMENTS.
- DEWATERING SKID SHALL BE PROVIDED PRE WIRED AND TESTED BY THE MANUFACTURER. SKID ASSEMBLY SHALL BE UL OR ETL LABELED IN ACCORDANCE WITH WASHINGTON STATE L&I REQUIREMENTS. PROVIDE 480V THREE PHASE POWER FEEDER CONNECTION, AND CAT 6 SIGNAL CABLE FOR SYSTEM MONITORING.
- ROUTE SOLENOID WIRING TO INFLUENT PUMP STATION PANEL FOR SOLENOID VALVE CONTROL.
- PROVIDE DEDICATED 120V GFCI RECEPTACLE AND 4-20 mA FLOW PACE SIGNAL FROM PLC SYSTEM FOR 4 LOCATIONS: HYPO FEED PUMP, POLYMER FEED PUMP, INFLUENT SAMPLER, AND EFFLUENT SAMPLER. SAMPLERS PROVIDED BY OWNER FOR CONTRACTOR INSTALLATION.
- PROVIDE 120VAC POWER CIRCUIT FROM PANEL LP2 AND CONTROL SIGNALS AS SHOWN ON PLC I/O DRAWINGS. WIRE EACH VENTILATION FAN TO CONTROL PANEL - TYP 3 IN TREATMENT BUILDING.
- ROUTE RACEWAY FROM TREATMENT BUILDING EQUIPMENT TO HANDHOLES, AND TO ASSOCIATES CONTROL, MCC OR POWER PANEL. QUANTITY AND SIZE AS REQUIRED; PROVIDE MINIMUM TWO 1" SPARE CONDUITS FROM EACH POWER AND CONTROL HANDHOLE TO SIDE OF BUILDING AND STUB UP AND CAP FOR FUTURE USE.
- INSTALL INFLUENT PUMP MOTOR DISCONNECTS ON BUILDING WALL WITHIN SIGHT OF WET WELL. COORDINATE FINAL LOCATION DURING CONSTRUCTION.

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EASTSOUND SEWER AND WATER DISTRICT

SAN JUAN COUNTY WASHINGTON

WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

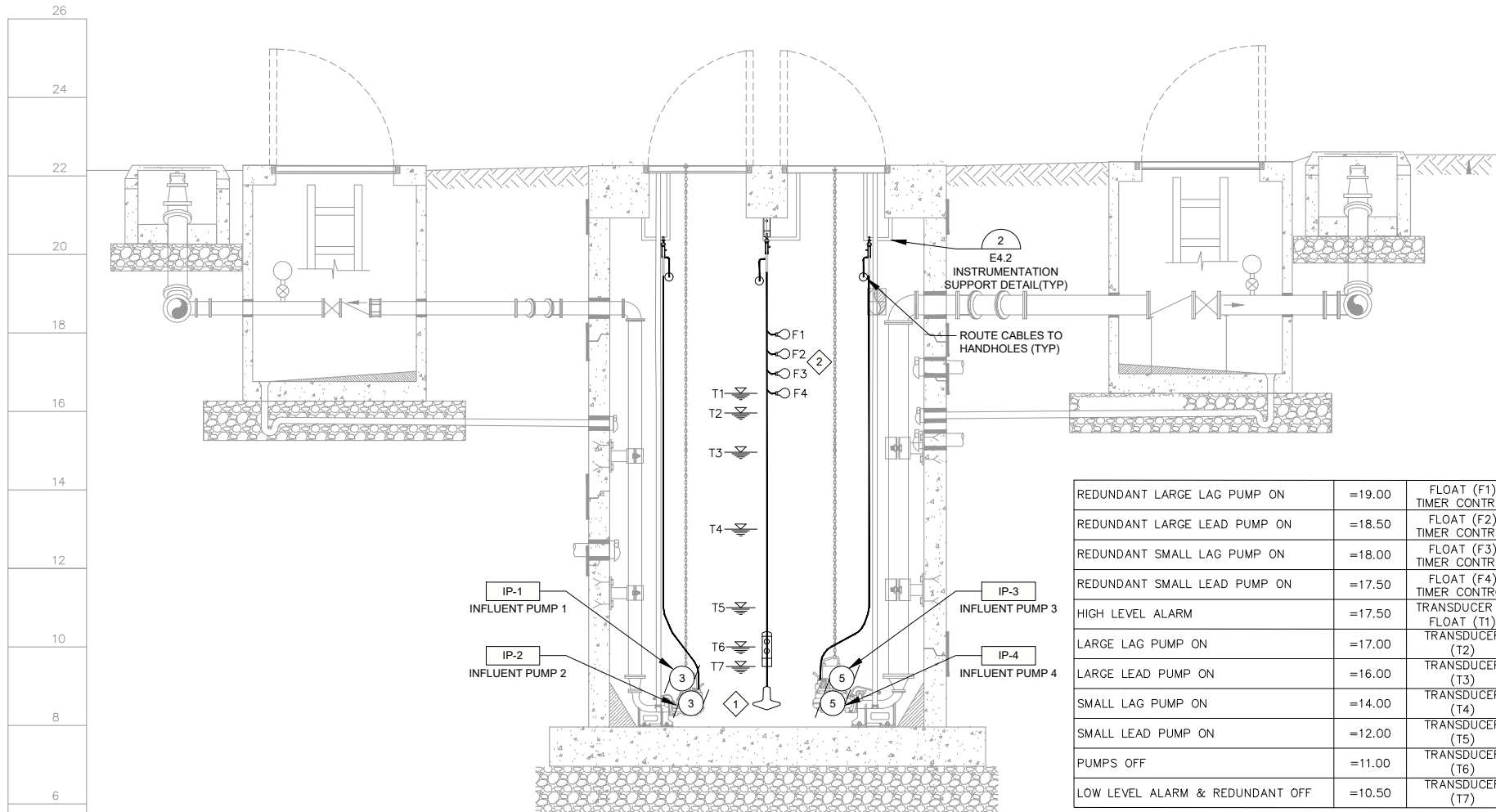
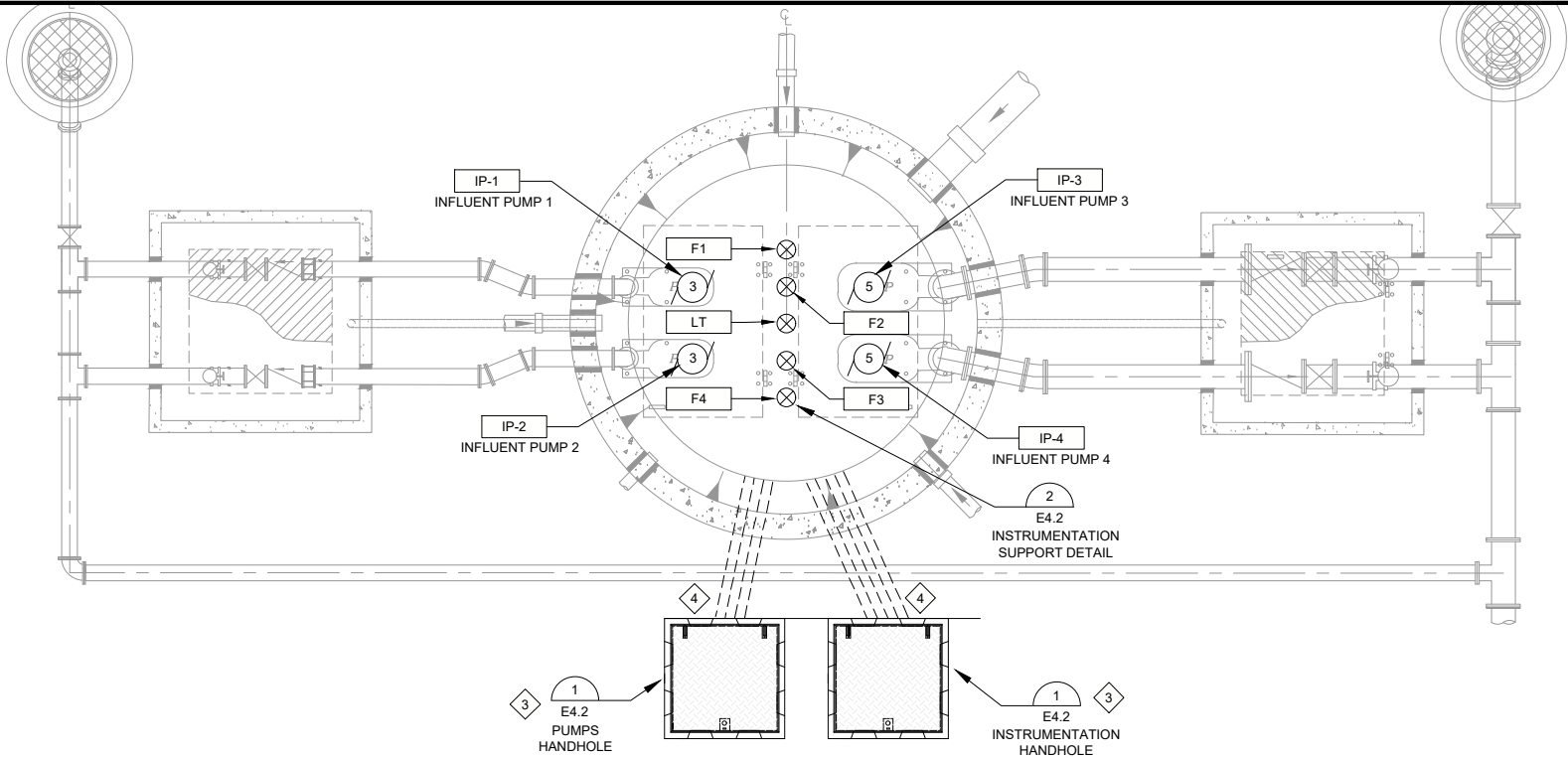
INFLUENT PUMP STATION, WAS, DEWATERING ELECTRICAL PLAN

DATE	09-12-2022
SCALE	AS SHOWN
JOB NUMBER	2020-070

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REDUNDANT LARGE LAG PUMP ON	=19.00	FLOAT (F1) TIMER CONTROL
REDUNDANT LARGE LEAD PUMP ON	=18.50	FLOAT (F2) TIMER CONTROL
REDUNDANT SMALL LAG PUMP ON	=18.00	FLOAT (F3) TIMER CONTROL
REDUNDANT SMALL LEAD PUMP ON	=17.50	FLOAT (F4) TIMER CONTROL
HIGH LEVEL ALARM	=17.50	TRANSDUCER & FLOAT (T1)
LARGE LAG PUMP ON	=17.00	TRANSDUCER (T2)
LARGE LEAD PUMP ON	=16.00	TRANSDUCER (T3)
SMALL LAG PUMP ON	=14.00	TRANSDUCER (T4)
SMALL LEAD PUMP ON	=12.00	TRANSDUCER (T5)
PUMPS OFF	=11.00	TRANSDUCER (T6)
LOW LEVEL ALARM & REDUNDANT OFF	=10.50	TRANSDUCER (T7)

INFLUENT PUMP STATION PLAN AND ELEVATION
SCALE: 1" = 2' AT FULL SCALE

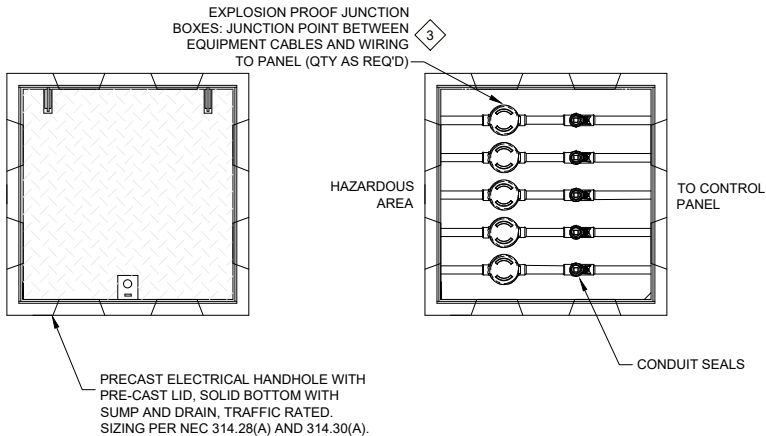
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GENERAL NOTES:

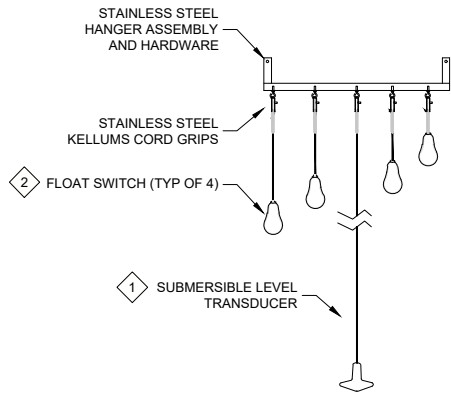
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- ELECTRIC HAND HOLES ARE NOT SHOWN ON THESE PLANS BUT SHALL BE REQUIRED TO MEET CODE AND AS SPECIFIED. COORDINATE ALL FINAL LOCATIONS WITH ENGINEER.
- UPSIZE CONDUITS FROM WET WELLS TO JUNCTION BOXES AS REQUIRED FOR PUMP AND FLOAT CABLES TO MEET NEC FILL REQUIREMENTS.
- ALL OUTDOOR CONTROL PANEL AND DISCONNECT ENCLOSURES SHALL BE TYPE 316 STAINLESS STEEL.

KEY NOTES:

- LEVEL TRANSDUCER: WIKA SUBMERSIBLE PRESSURE TRANSMITTER, 316SS BODY, 0-15 PSI RANGE WITH LEVELGUARD, POLYURETHANE CABLE, 4-20 mA LOOP POWERED, TERMINATION ENCLOSURE, FOR USE WITH VENTED PRESSURE TRANSMITTER, NEMA 4X, INTEGRALLY MOUNTED MOISTURE PROTECTION.
- HIGH LEVEL FLOATS: ECO-FLOAT MODEL GP60NONC, NON-MERCURY, NO/NC CONTACT, POLYPROPYLENE HOUSING AND CLAMP, PIPE MOUNT.
- PROVIDE SERVICE LOOPS FOR EXTRA PUMP AND INSTRUMENT CABLES INSIDE WET WELL. ROUTE TO HANDHOLES FOR CONNECTION POINT FOR CABLES TO PERMANENT WIRING. MAKE CONNECTIONS WITH APPROVED CONNECTORS. SEAL CONDUITS LEAVING HANDHOLES TO INFLUENT PUMP STATION PER NEC ART 500.
- PROVIDE INDIVIDUAL CONDUITS FOR EACH PUMP, FLOAT, AND TRANSDUCER CABLE. OVERSIZE CONDUITS AS REQUIRED BETWEEN WET WELL AND HANDHOLE FOR EASY REMOVAL AND REPLACEMENT OF FACTORY PUMP AND INSTRUMENT CABLES.



HAZARDOUS AREA CONNECTION DETAIL
SCALE: NONE



INSTRUMENTATION SUPPORT DETAIL
SCALE: NONE

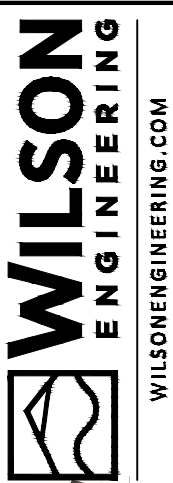


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EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON
SAN JUAN COUNTY
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I
INFLUENT PUMP STATION ELECTRICAL PLAN AND ELEVATION

DATE
09-12-2022

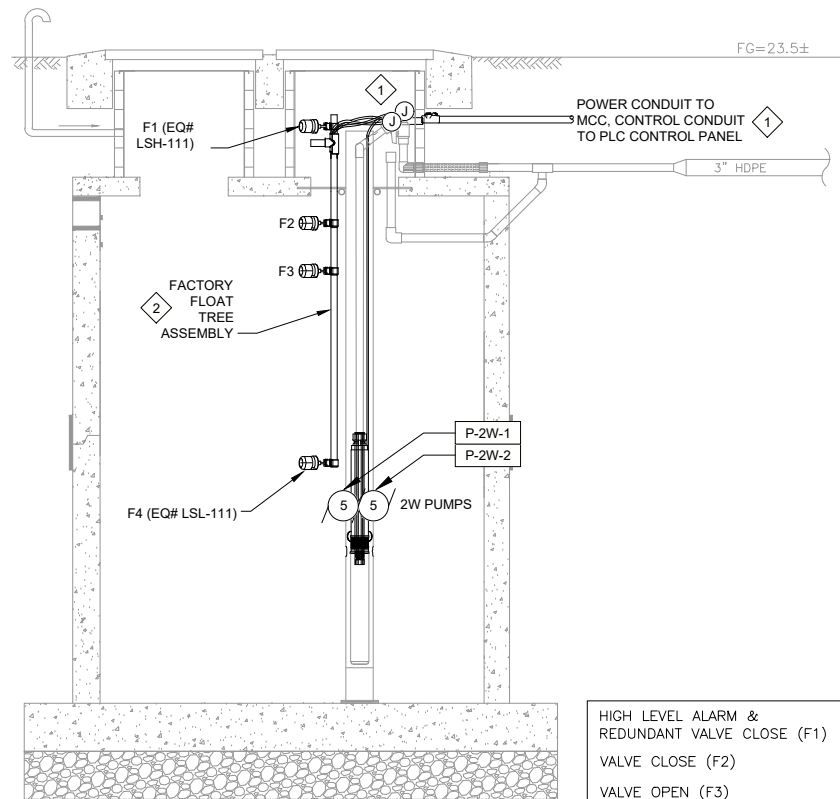
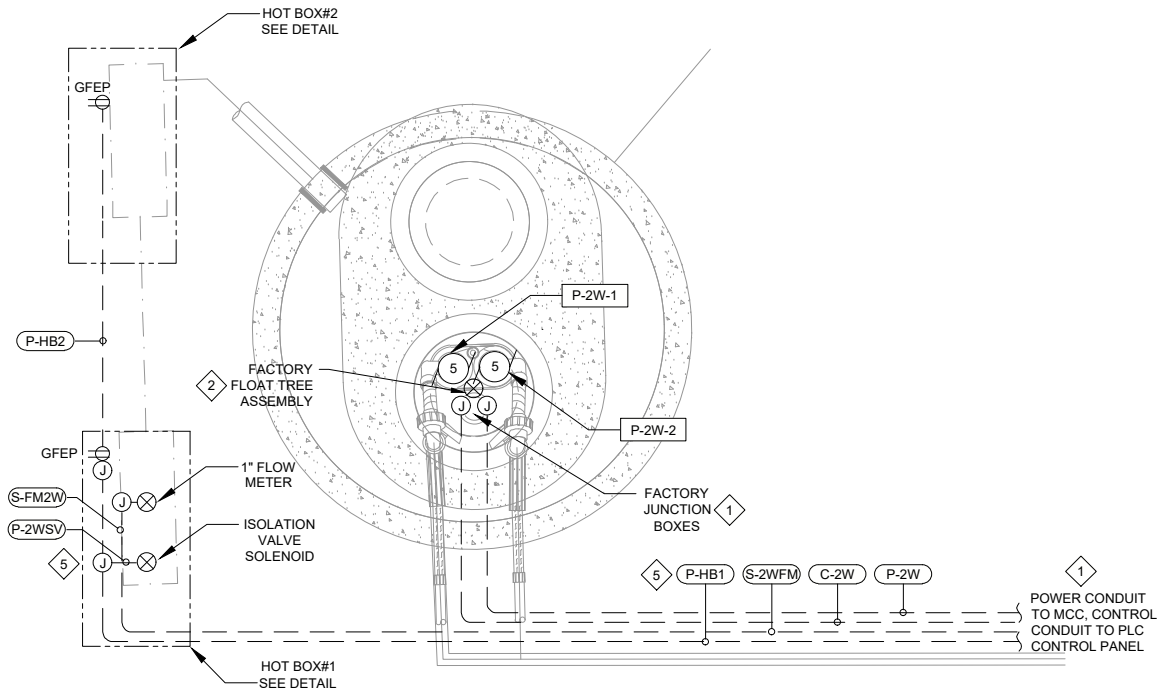
SCALE
AS SHOWN

JOB NUMBER
2020-070

SHEET
E4.2

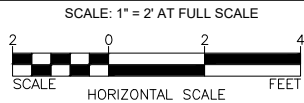
PAGE
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09-12-2022



HIGH LEVEL ALARM & REDUNDANT VALVE CLOSE (F1)	22.00
VALVE CLOSE (F2)	20.00
VALVE OPEN (F3)	19.00
LOW LEVEL ALARM (F4) (& PUMP(S) "OFF" FLOAT SWITCH)	15.00

2W PUMP STATION PLAN AND ELEVATION

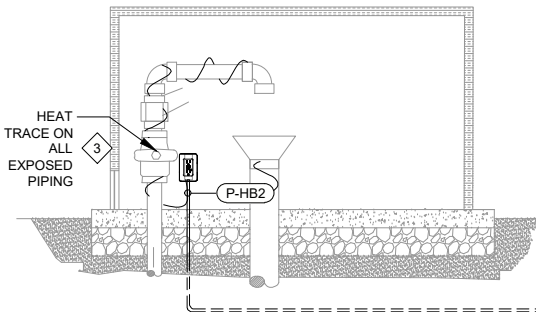


GENERAL NOTES:

1. ALL CONDUIT ROUTING IS NOT SHOWN, AND WHERE SHOWN, IS SCHEMATIC. ELECTRICAL CONTRACTOR SHALL DETERMINE THE BEST ROUTING PATH AND CIRCUIT COMBINATIONS BASED ON FIELD CONDITIONS AND ELECTRICAL CODES.
2. EQUIPMENT LOCATIONS AND ARRANGEMENT ARE SCHEMATIC. CONTRACTOR SHALL COORDINATE WITH EQUIPMENT MANUFACTURER FOR DETAILED CONNECTION REQUIREMENTS AND PROVIDE MATERIALS AND INSTALLATION FOR A COMPLETE AND OPERATIONAL SYSTEM.
3. ELECTRIC HAND HOLES ARE NOT SHOWN ON THESE PLANS BUT SHALL BE REQUIRED TO MEET CODE AND AS SPECIFIED. COORDINATE ALL FINAL LOCATIONS WITH ENGINEER.
4. UPSIZE CONDUITS FROM WET WELLS TO JUNCTION BOXES AS REQUIRED FOR PUMP AND FLOAT CABLES TO MEET NEC FILL REQUIREMENTS.
5. ALL OUTDOOR CONTROL PANEL AND DISCONNECT ENCLOSURES SHALL BE TYPE 316 STAINLESS STEEL.

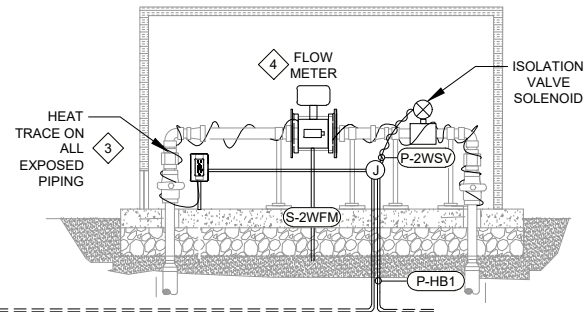
KEY NOTES:

1. 2W PUMP SYSTEM SHALL INCLUDE (2) FACTORY JUNCTION BOXES WITH CORD STRAIN RELIEF GRIPS FOR CABLES. PROVIDE ONE JBOX FOR THE MOTOR CABLES AND ONE FOR THE FLOAT SWITCHES. ROUTE POWER RACEWAY TO MCC BUCKETS, AND CONTROL WIRING TO PLC CONTROL PANEL.
2. PROVIDE FACTORY FLOAT TREE ASSEMBLY WITH 4 FLOATS.
3. PROVIDE HEAT TRACE ON ALL EXPOSED PIPING. RAY-CHEM BTV SELF REGULATING HEATING CABLE, OR EQUAL. SYSTEM SHALL INCLUDE GFEP BREAKER ON 120V 20AMP DEDICATED CIRCUIT. PROVIDE NEMA 4X THERMOSTAT AND END OF LINE TERMINATION KIT WITH INDICATOR LIGHT.
4. 2W FLOW METER: 1" MAGNETIC FLOW METER, ENDRESS-HAUSER PROMAG W400, CLASS I DIV 2, CARBON STEEL FLANGE, 24VAC/DC, ETHERNET/IP, POLYURETHANE LINER, 316L ELECTRODES, LOCAL DISPLAY, NEMA 4X, NPT 1/2". PROVIDE (2) 316L GROUNDING RINGS FOR EACH FLOW TUBE.
5. ROUTE POWER WIRING FOR SOLENOID AND HEAT TRACE CIRCUITS TO JBOX IN HB#1 AND EXTEND TO HB#2.



HOT BOX #2 ELECTRICAL DETAIL

NOT TO SCALE



HOT BOX #1 ELECTRICAL DETAIL

NOT TO SCALE

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WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

2W PUMP STATION ELECTRICAL PLAN AND ELEVATION

DATE
09-12-2022

SCALE
AS SHOWN

JOB NUMBER
2020-070

SHEET
E4.3

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1	ADD UV DISINFECTION	BZ	04/18/22



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WASTEWATER TREATMENT PLANT UPGRADE - PHASE I
UV DISINFECTION CHAMBER & DIGESTER ELECTRICAL PLAN AND ELEVATION

DATE 09-12-2022
SCALE AS SHOWN
JOB NUMBER 2020-070

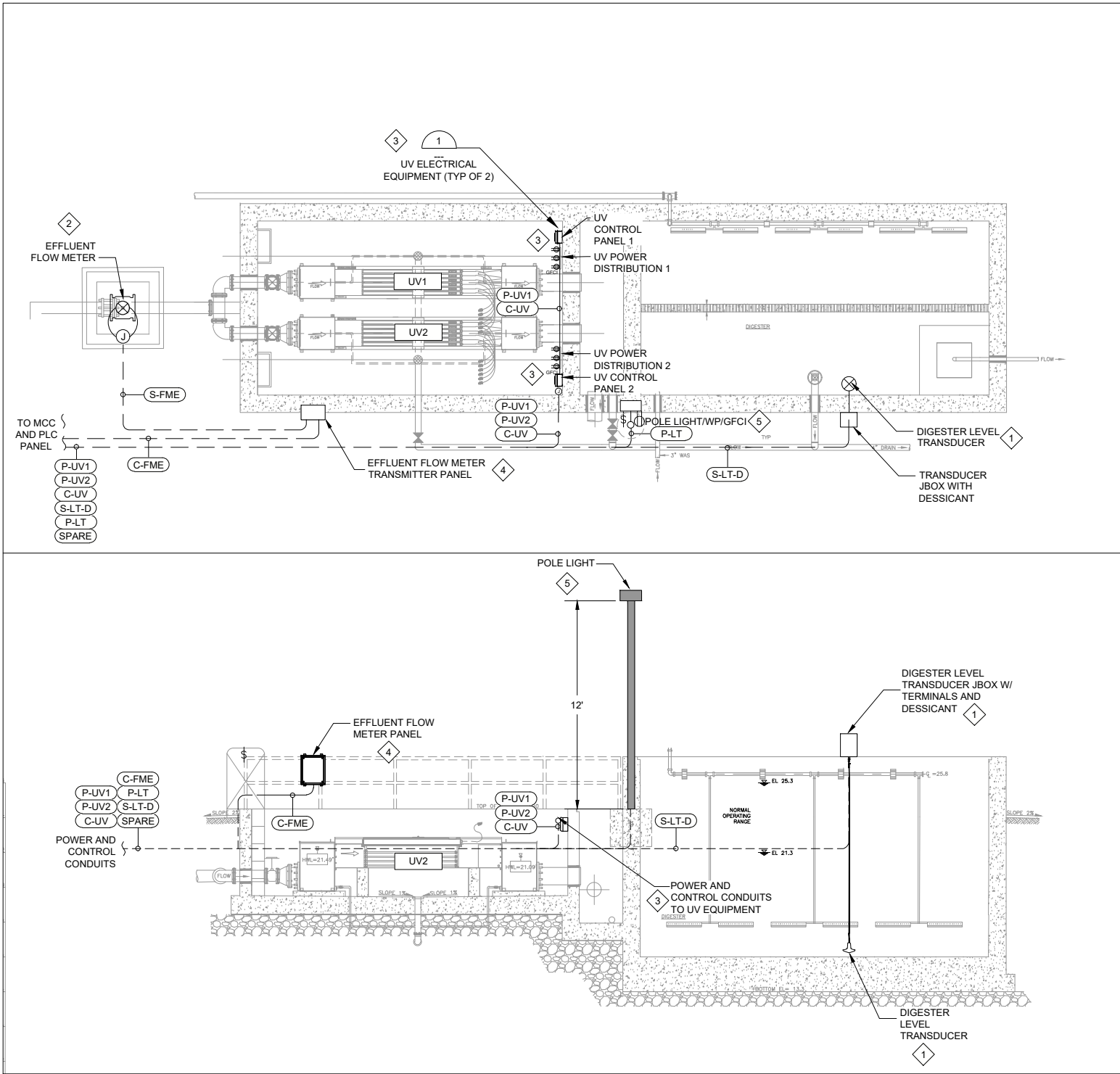
SHEET E5.1
PAGE 73 OF 96

GENERAL NOTES:

1. ALL CONDUIT ROUTING IS NOT SHOWN, AND WHERE SHOWN, IS SCHEMATIC. ELECTRICAL CONTRACTOR SHALL DETERMINE THE BEST ROUTING PATH AND CIRCUIT COMBINATIONS BASED ON FIELD CONDITIONS AND ELECTRICAL CODES.
2. EQUIPMENT LOCATIONS AND ARRANGEMENT ARE SCHEMATIC. CONTRACTOR SHALL COORDINATE WITH EQUIPMENT MANUFACTURER FOR DETAILED CONNECTION REQUIREMENTS AND PROVIDE MATERIALS AND INSTALLATION FOR A COMPLETE AND OPERATIONAL SYSTEM.
3. ELECTRIC HAND HOLES ARE NOT SHOWN ON THESE PLANS BUT SHALL BE REQUIRED TO MEET CODE AND AS SPECIFIED. COORDINATE ALL FINAL LOCATIONS WITH ENGINEER.
4. UPSIZE CONDUITS FROM WET WELLS TO JUNCTION BOXES AS REQUIRED FOR PUMP AND FLOAT CABLES TO MEET NEC FILL REQUIREMENTS.
4. ALL OUTDOOR CONTROL PANEL AND DISCONNECT ENCLOSURES SHALL BE TYPE 316 STAINLESS STEEL.

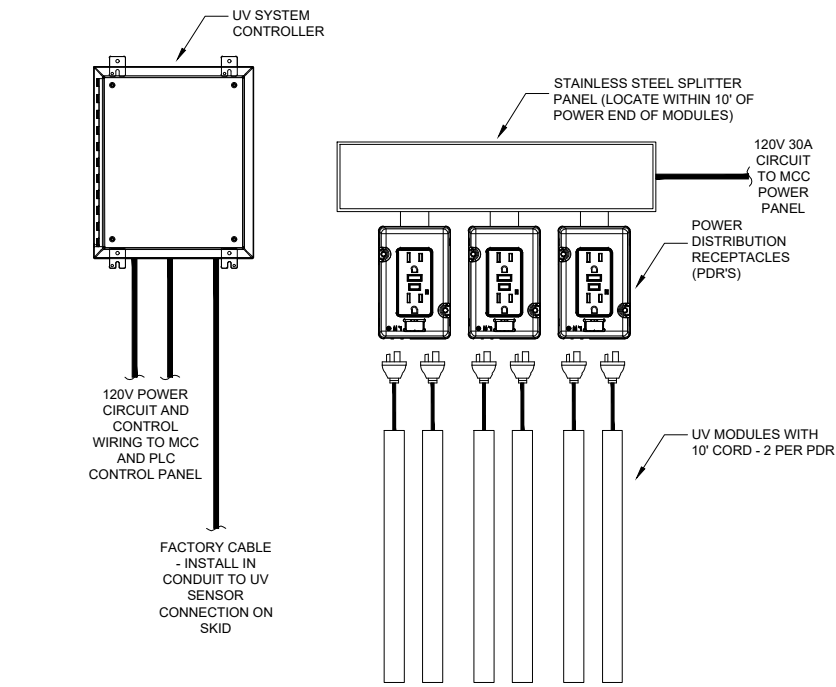
KEY NOTES:

1. LEVEL TRANSDUCER: WIKA SUBMERSIBLE PRESSURE TRANSMITTER, 316SS BODY, 0-15 PSI RANGE WITH LEVELGUARD, POLYURETHANE CABLE, 4-20 mA LOOP POWERED, TERMINATION ENCLOSURE, FOR USE WITH VENTED PRESSURE TRANSMITTER, DESSICANT, NEMA 4X, INTEGRALLY MOUNTED MOISTURE PROTECTION. MOUNT ENCLOSURE TO HANDRAIL WITH SS HARDWARE.
2. FLOW METERS: SIZE PER CIVIL DRAWINGS. MAGNETIC FLOW METER, ENDRESS-HAUSER PROMAG W400, CLASS I DIV 2, CARBON STEEL FLANGE, 120-240VAC OR 24VAC/DC, ETHERNET/IP, POLYURETHANE LINER, 316L ELECTRODES, REMOTE HOUSING, NPT 1/2" SENSOR IP68 TYPE 6, POTTED CORROSION PROTECTION AND SUBMERSIBLE. PROVIDE (2) 316L GROUNDING RINGS FOR EACH FLOW TUBE.
3. INSTALL LOCAL UV CONTROL PANELS AND POWER DISTRIBUTION FOR UV SYSTEMS. CONTRACTOR SHALL PROVIDE WIRING GUTTER, JUNCTION BOXES AND (3) DUPLEX GFCI RECEPTACLES (PDR'S) WITH IN-USE WET LOCATION COVERS FOR EACH UV UNIT. MOUNT TO BASIN WALL WITH STAINLESS STEEL UNITSTRUT AND HARDWARE. CONDUIT, FITTINGS AND JUNCTION BOXES SHALL BE STAINLESS STEEL. PROVIDE ALL WIRING AND CONNECTIONS PER MANUFACTURER REQUIREMENTS.
4. PROVIDE LOCAL CONTROL PANEL TO MOUNT EFFLUENT FLOW METER TRANSMITTER, AND TERMINALS FOR LOCAL INSTRUMENTATION SIGNALS. ROUTE BACK TO PLC CONTROL PANEL. ENCLOSURE SHALL BE STAINLESS STEEL, MOUNT TO HANDRAIL OR RACK WITH SS HARDWARE.
5. CONTRACTOR SHALL PROVIDE POLE LIGHT TO ILLUMINATE UV AREA. PROVIDE CONCRETE POLE BASE AND SECURE TO SIDE OF BASIN WITH STAINLESS STEEL HARDWARE; FIELD COORDINATE WITH CIVIL. PROVIDE 4" SQUARE ALUMINUM POLE 12' TALL WITH PHOTOCELL AND LOCAL ON/OFF SWITCH. LITHONIA LIGHTING #DXS0-LED-P3-30K-T3M-MVOLT-SPA-DNAXD WITH PHOTOCELL, OR APPROVED EQUAL.



1 UV DISINFECTION CHAMBER AND DIGESTER ELECTRICAL PLAN AND ELEVATION

SCALE: 1" = 4' AT FULL SCALE
SCALE HORIZONTAL SCALE FEET



1 UV SYSTEM INTERCONNECTION DIAGRAM - TYP OF 2
SCALE: NONE



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EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON
WASTEWATER TREATMENT PLANT UPGRADE – PHASE I

ELECTRICAL PANEL SCHEDULES

DATE	09-12-2022	SCALE	AS SHOWN	JOB NUMBER	2020-070
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SHEET
E6.2

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BID SET

PANEL: MDP				PANEL SCHEDULE						PROJECT: Eastsound WWTP			
480Y/277V, 3Ph, 4W.				400A Bus			400A		M.L.O.		SURFACE MOUNTED		
CKT NO	DESCRIPTION / LOCATION	LOAD (VA)	LOAD TYPE	C.B. AMP	C.B. POLE	PHASE	C.B. POLE	C.B. AMP	LOAD TYPE	LOAD (VA)	DESCRIPTION / LOCATION	CKT NO	
1	BLOWER BUILDING MCC	60,000	M	400	3	A	3	125	G	15,000	TRANSFORMER FOR EXISTING BUILDING	2	
3	---	60,000	M	---	---	B	---	---	G	15,000	480-208Y/120V	4	
5	---	60,000	M	---	---	C	---	---	G	15,000	---	6	
7	SPACE					A					SPACE	8	
9	---					B					---	10	
11	---					C					---	12	
13	SPACE					A					SPACE	14	
15	---					B					---	16	
17	---					C					---	18	
19	SPACE					A					SPACE	20	
21	---					B					---	22	
23	---					C					---	24	
25	SPACE					A					SPACE	26	
27	---					B					---	28	
29	---					C					---	30	
DATE: September 15, 2022													
TOTAL CONNECTED LOAD:		PH A	75,000 VA	270.6 AMPS									
TOTAL CONNECTED LOAD:		PH B	75,000 VA	270.6 AMPS									
TOTAL CONNECTED LOAD:		PH C	75,000 VA	270.6 AMPS									
MAX PHASE CONNECTED LOAD:		PH A	75,000 VA					PANEL RATING:		42,000 AIC			
TOTAL CONNECTED LOAD (3 x MAX):			225.0 kVA	270.6 AMPS				TOTAL DEMAND LOAD:		225.0 kVA	270.6 AMPS		

PANEL MDP
SCALE: NONE

PANEL: LP1				PANEL SCHEDULE						PROJECT: Eastsound WWTP			
120/240V, 1Ph, 3W.			100A Bus			100A M.C.B.			SURFACE MOUNTED				
CKT NO	DESCRIPTION / LOCATION	LOAD (VA)	LOAD TYPE	C.B. AMP	C.B. POLE	PHASE	C.B. POLE	C.B. AMP	LOAD TYPE	LOAD (VA)	DESCRIPTION / LOCATION	CKT NO	
1	BLOWER ROOM RECEPTACLES	900	R	20	1	A	1	20	C	1,000	MAIN CONTROL PANEL	2	
3	LOUVER AND HVAC CONTROL	500	M	20	1	B	1	20	L	239	BLOWER BUILDING LIGHTING	4	
5	EXHAUST FAN EF-1 (BLOWER BUILDING)	1,127	M	20	1	A	1				SPARE	6	
7	EXHAUST FAN EF-2 (BLOWER BUILDING)	1,127	M	20	1	B	1	20	H	1,300	GENERATOR - HEATER AND BATTERY CHGR	8	
9	UV1 POWER	2,520	C	30	1	A	1	20	H	1,000	HOT BOX 1 - GFEP	10	
11	UV2 POWER	2,520	C	30	1	B	1	20	H	1,000	HOT BOX 2 - GFEP	12	
13	UV1 CONTROLLER	600	C	20	1	A	1	20	M	500	EFFLUENT SAMPLER	14	
15	UV2 CONTROLLER	600	C	20	1	B	1	20	M	500	HYPOCHLORITE FEED PUMP	16	
17	SPARE			20	1	A	1	20			SPARE	18	
19	SPACE					B					SPACE	20	
21	SPACE					A					SPACE	22	
23	SPACE					B					SPACE	24	
DATE: September 15, 2022													
TOTAL CONNECTED LOAD:		PH A	7,647 VA	63.7 AMPS									
TOTAL CONNECTED LOAD:		PH B	7,786 VA	64.9 AMPS									
MAX PHASE CONNECTED LOAD:		PH B	7,786 VA					PANEL RATING:		10,000 AIC			
TOTAL CONNECTED LOAD (2 x MAX):			15.6 kVA	64.9 AMPS				TOTAL DEMAND LOAD:		17.3 kVA		72.1 AMPS	

PANEL LP1 (MCC)
SCALE: NONE

PANEL: LP2				PANEL SCHEDULE						PROJECT: Eastsound WWTP			
120/240V, 1Ph, 3W.				100A Bus			60A M.C.B.			SURFACE MOUNTED			
CKT NO	DESCRIPTION / LOCATION	LOAD (VA)	LOAD TYPE	C.B. AMP	C.B. POLE	PHASE	C.B. POLE	C.B. AMP	LOAD TYPE	LOAD (VA)	DESCRIPTION / LOCATION	CKT NO	
1	PUMP STATION CONTROL CIRCUITS	1,000	R	20	1	A	1	20	C	500	H2S SENSOR	2	
3	WALL PACK LIGHTING	500	L	20	1	B	1	20	C	500	METHANE SENSOR	4	
5	VENTILATION CONTROL PANEL	500	G	20	1	A	1	20	M	500	POLYMER FEED PUMP RECEPTACLE	6	
7	EX-HAUST FAN EF-3 (TREATMENT BUILDING)	1,127	M	20	1	B	1	20	M	500	INFLUENT SAMPLER RECEPTACLE	8	
9	EX-HAUST FAN EF-4 (TREATMENT BUILDING)	1,127	M	20	1	A	1	20			SPARE	10	
11	EX-HAUST FAN EF-5 (TREATMENT BUILDING)	1,127	M	20	1	B	1	20			SPARE	12	
13	SPARE			20	1	A	1	20			SPARE	14	
15	SPARE			20	1	B	1	20			SPARE	16	
DATE: September 15, 2022													
TOTAL CONNECTED LOAD:		PH A	3,627 VA	30.2 AMPS									
TOTAL CONNECTED LOAD:		PH B	3,754 VA	31.3 AMPS									
MAX PHASE CONNECTED LOAD:		PH B	3,754 VA			PANEL RATING:			10,000 AIC				
TOTAL CONNECTED LOAD (2 x MAX):			7.5 kVA	31.3 AMPS		TOTAL DEMAND LOAD:			7.8 kVA 32.3 AMPS				

PANEL LP2 (INFLUENT PUMP STATION PANEL)
SCALE: NONE



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CONTROL AND SIGNAL SCHEDULE

POWER SCHEDULE

SCALE: NONE



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EASTSOUND SEWER AND WATER DISTRICT

COUNTY WA
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

CONDUIT AND CONDUCTOR SCHEDULES

DATE 09-12-2022

SHEET

E6.3

SCALE

2.2

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Diagram illustrating the connection of a CompactLogix PLC rack to various components:

- Allen Bradley CompactLogix PLC Rack:** Shows slots 0 through 9. Slot 0 is labeled "SLOT 0" and contains a power supply. Slots 1 through 6 are labeled "SLOT 1" through "SLOT 6". Slot 7 is labeled "SLOT 7" and contains a network switch. Slots 8 and 9 are labeled "SLOT 8" and "SLOT 9".
- Allen Bradley Panelview Plus Operator Interface:** A large monitor unit connected to the PLC rack.
- MCC Power Monitor:** A device connected to the power supply in Slot 0.
- Network Switch:** A central hub connected to the PLC rack (Slot 7) and the MCC Power Monitor. It has multiple ports for connecting other devices.
- 2W Flow Meter:** A device connected to the Network Switch.
- Effluent Flow Meter:** A device connected to the Network Switch.
- VFDs in MCC (Typ of 10):** Variable Frequency Drives connected to the Network Switch.

The diagram illustrates the hardware configuration of an Allen Bradley CompactLogix PLC system. The PLC rack consists of six slots:

- SLOT 0:** 1768-ENBT (Ethernet Network Interface Module)
- SLOT 1:** 1768-PA1 (Power Supply)
- SLOT 2:** 1768-PA1 (Power Supply)
- SLOT 3:** 1768-L3 (CompactLogix Processor)
- SLOT 4:** 1768-L3 (CompactLogix Processor)
- SLOT 5:** 1768-PA1 (Power Supply)

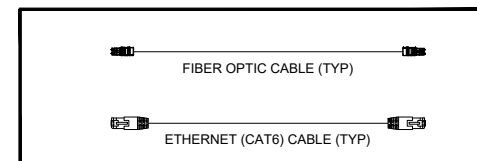
Below the rack, a **NETWORK SWITCH** is connected to the Ethernet module in Slot 0. The switch has several ports, with some connected to the PLC rack and others to external devices:

- TRAIN #1 FLOW METER (FUTURE):** Connected to the switch via a cable labeled "1-2".
- TRAIN #2 FLOW METER:** Connected to the switch via a cable labeled "1-2".
- TRAIN #3 FLOW METER:** Connected to the switch via a cable labeled "1-2".
- PLANT DRAIN FLOW METER:** Connected to the switch via a cable labeled "1-2".
- INFLUENT FLOW METER:** Connected to the switch via a cable labeled "1-2".
- WAS FLOW METER:** Connected to the switch via a cable labeled "1-2".

The diagram illustrates an office environment with a dashed border. Inside, there is a tall server rack on the left and a large computer monitor on the right. The monitor displays the text "SCADA COMPUTER WITH ALARM DIALER". A line connects the server rack to the monitor. Below the monitor, a network switch is shown with four ports and a power button. A cable connects the network switch to a telephone on the right, labeled "TELEPHONE (E)". Another cable connects the network switch to the internet, labeled "INTERNET (E)". A small icon of a person is shown at the bottom left, connected to the network switch.

The diagram illustrates a compact PLC system setup. At the top left, a 'NETWORK SWITCH' is shown with a patch panel and a rack of modules. A cable connects the switch to a 'COMPACT LOGIX PLC' at the bottom. The PLC is a multi-slot unit with various modules, including a power supply, CPU, and I/O modules. To the right of the PLC is an 'OPERATOR INTERFACE', which is a large rectangular screen or panel. The entire system is enclosed in a dashed-line box, indicating a single enclosure or a compact configuration.

CONTROL SYSTEM CABLE LEGEND



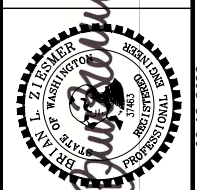
1. SYSTEM INTEGRATOR SHALL PROVIDE DETAILED CONTROL PANEL DESIGN AND DOCUMENTATION FOR A COMPLETE AND OPERATIONAL SYSTEM. ONLY MAJOR PANEL COMPONENTS ARE SPECIFIED.
2. ALL ETHERNET ENABLED EQUIPMENT SHALL BE CONNECTED TO CONTROL SYSTEM COMMUNICATION NETWORK. PROVIDE QUANTITY OF PORTS AS REQUIRED.



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COUNTY
WASTEWATER TREATMENT PLANT UPGRADE – PHASE I

CONTROL SYSTEM OVERVIEW

DATE	09-12-2022
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SHEET
E7.0

SCALE
AS SHOWN

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two business days
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GENERAL NOTES:

1. SYSTEM INTEGRATOR SHALL PROVIDE DETAILED CONTROL PANEL DESIGN AND DOCUMENTATION FOR A COMPLETE AND OPERATIONAL SYSTEM. ONLY MAJOR PANEL COMPONENTS ARE SPECIFIED.



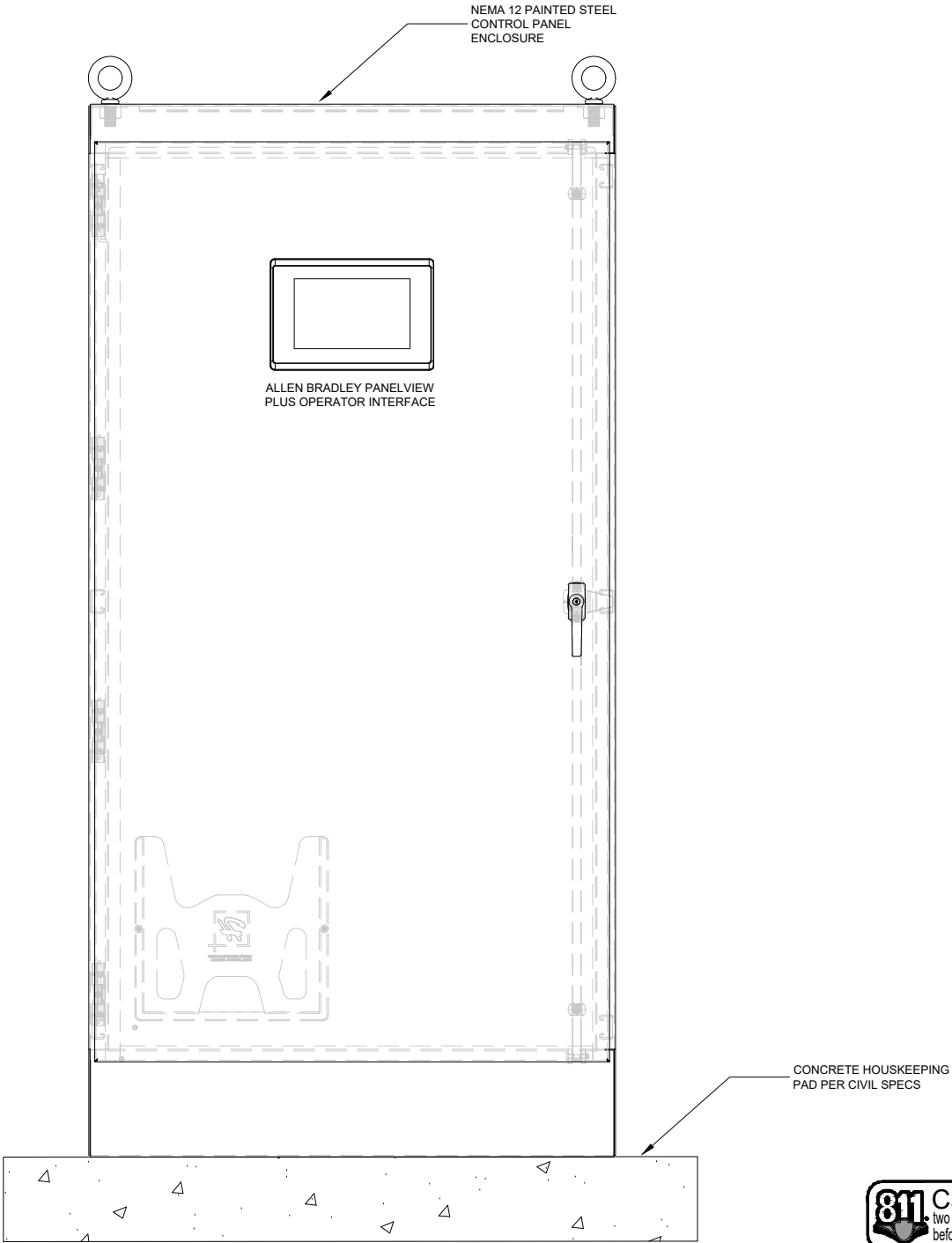
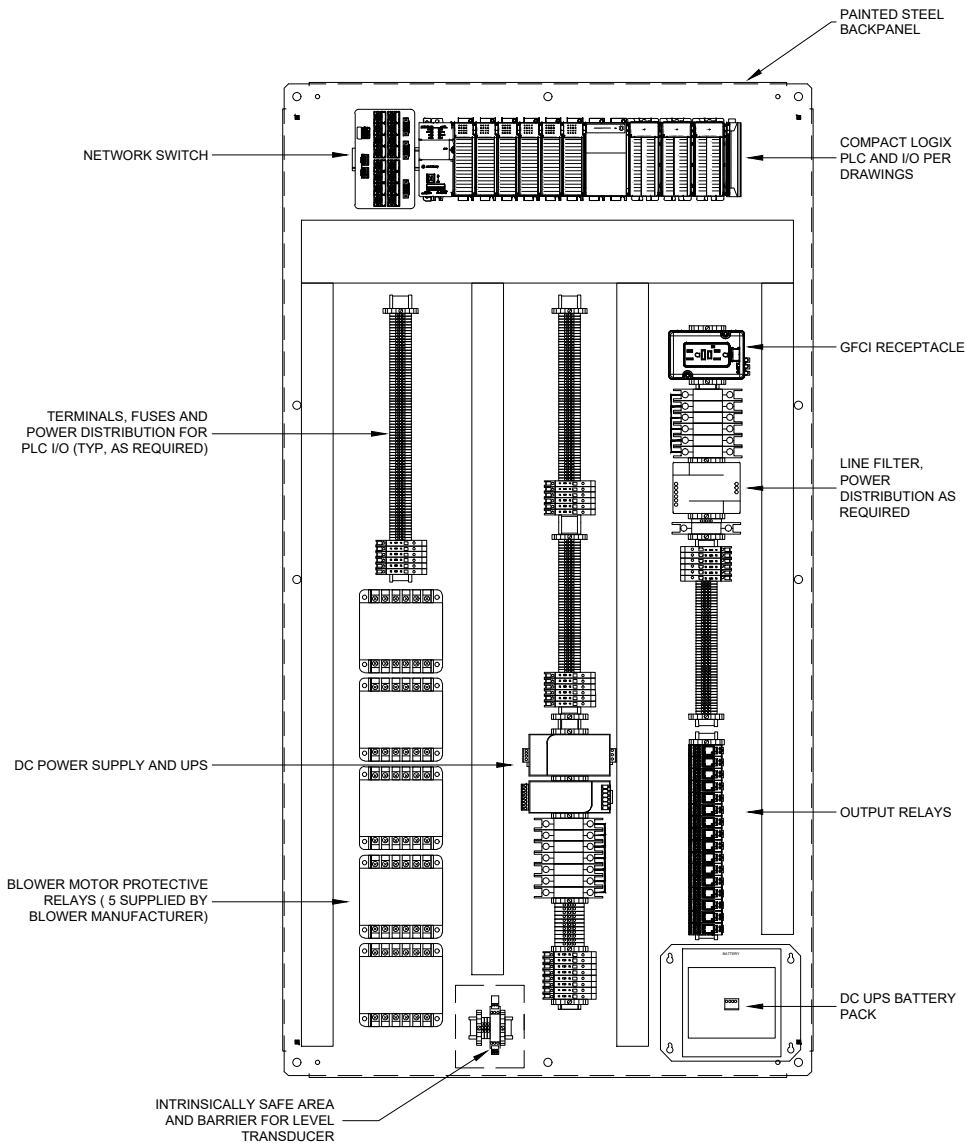
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MAIN PLC CONTROL PANEL ELEVATIONS
SCALE: NONE



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WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

MAIN PLC CONTROL PANEL ELEVATION

DATE
09-12-2022

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JOB NUMBER
2020-070

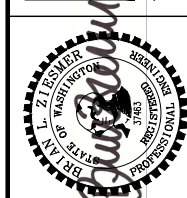
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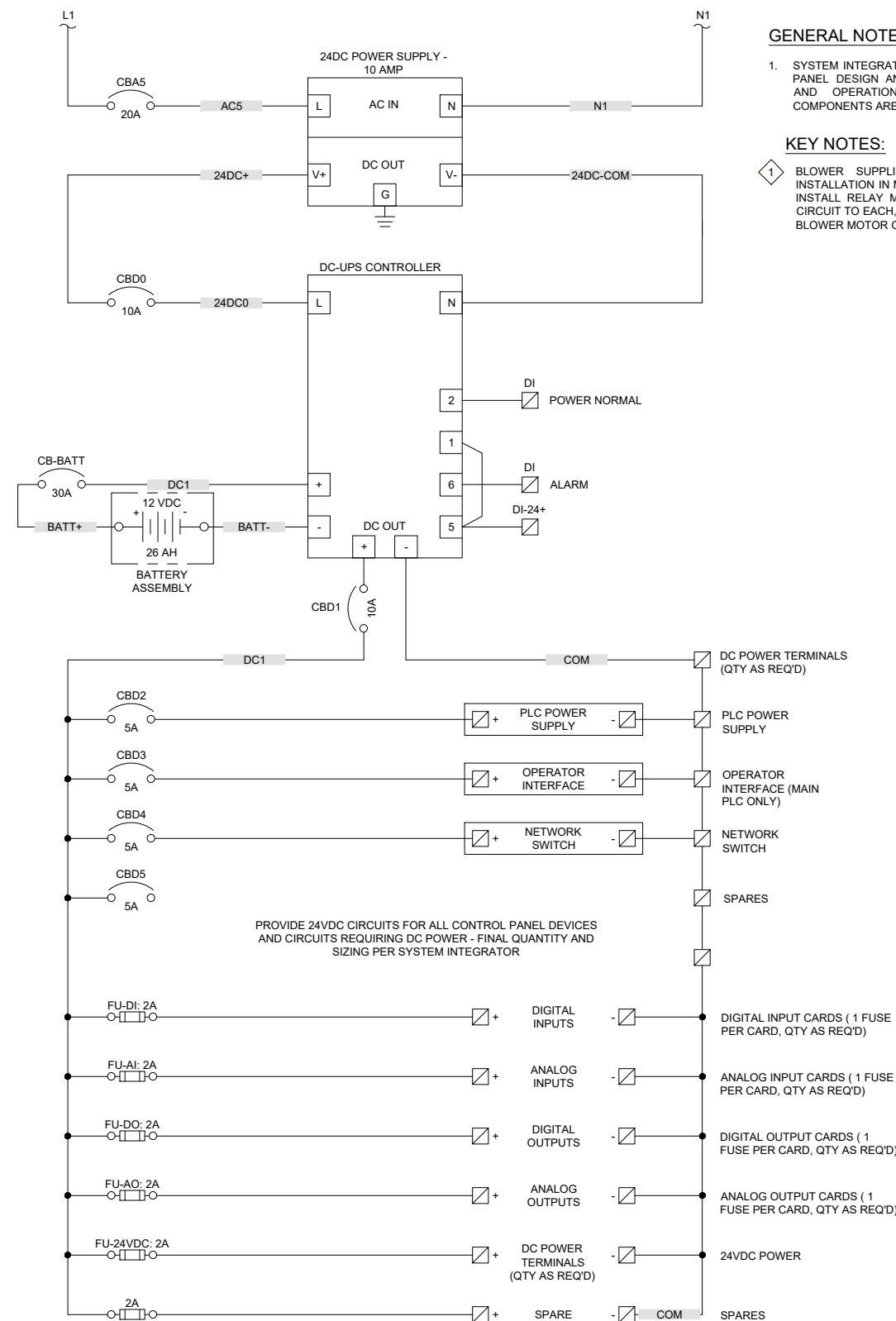
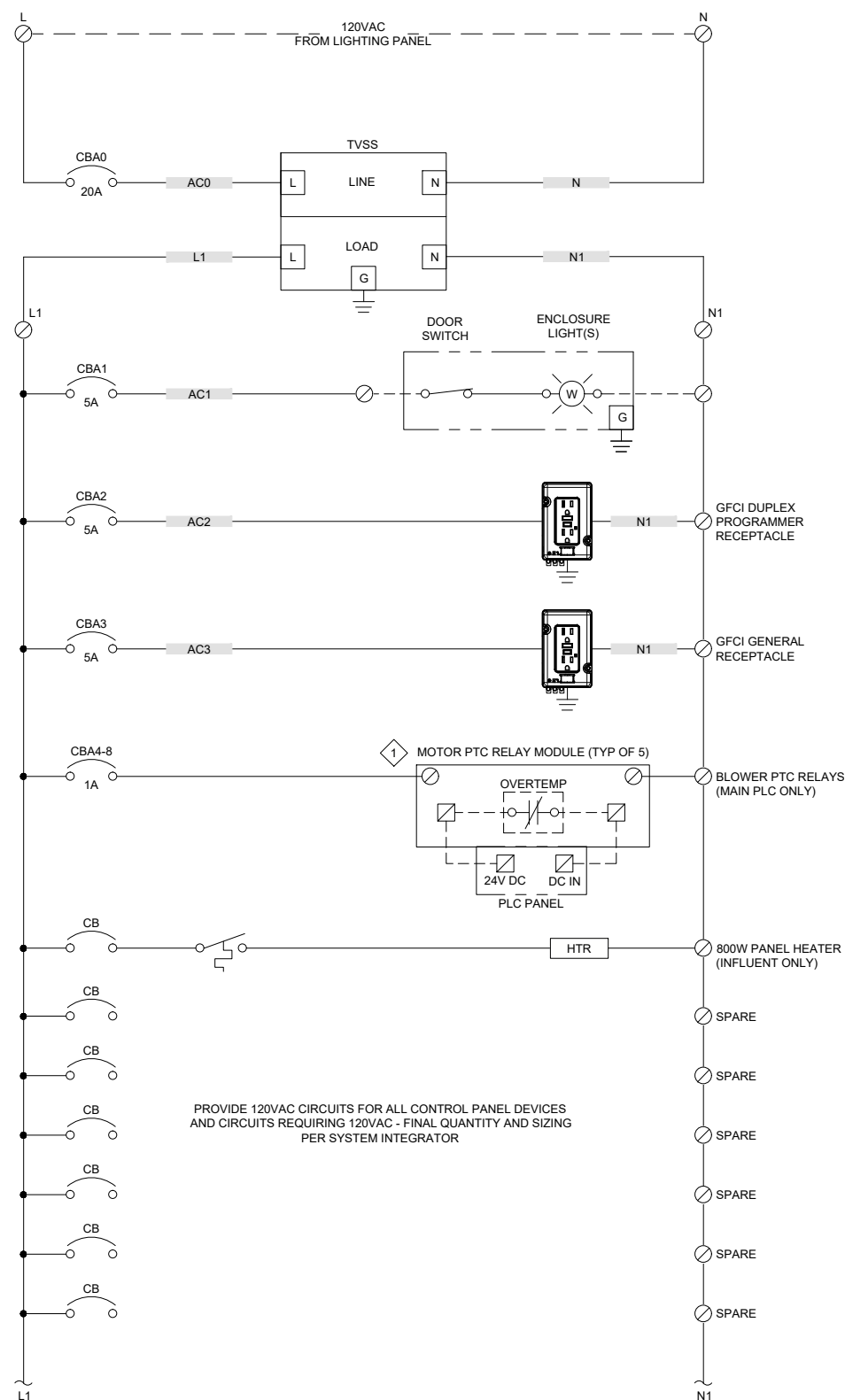
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WASHINGTON
SAN JUAN COUNTY
WASTEWATER TREATMENT PLANT UPGRADE – PHASE I

MAIN PLC CONTROL PANEL – POWER WIRING DIAGRAM

09-13-2023
DATEE7.2
SHEETSCALE
AS SHOWN

79 OF 96



GENERAL NOTES:

1. SYSTEM INTEGRATOR SHALL PROVIDE DETAILED CONTROL PANEL DESIGN AND DOCUMENTATION FOR A COMPLETE AND OPERATIONAL SYSTEM. ONLY MAJOR PANEL COMPONENTS ARE SPECIFIED.

KEY NOTES:

- 1 BLOWER SUPPLIER SHALL PROVIDE PTC RELAY MODULE FOR INSTALLATION IN MAIN PLC CONTROL PANEL. SYSTEM INTEGRATOR TO INSTALL RELAY MODULE IN CONTROL PANEL, PROVIDE 120V POWER CIRCUIT TO EACH, AND WIRE OVERTEMP SIGNAL TO PLC TO SHUT DOWN BLOWER MOTOR ON HIGH TEMPERATURE.

MAIN PLC AND INFLUENT CONTROL PANEL - POWER WIRING DIAGRAM (TYP)

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WASHINGTON
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WASTEWATER TREATMENT PLANT UPGRADE – PHASE I

MAIN PLC CONTROL PANEL – DIGITAL INPUTS

DATE: 09-12-2022
SCALE: AS SHOWN
JOB NUMBER: 2020-070

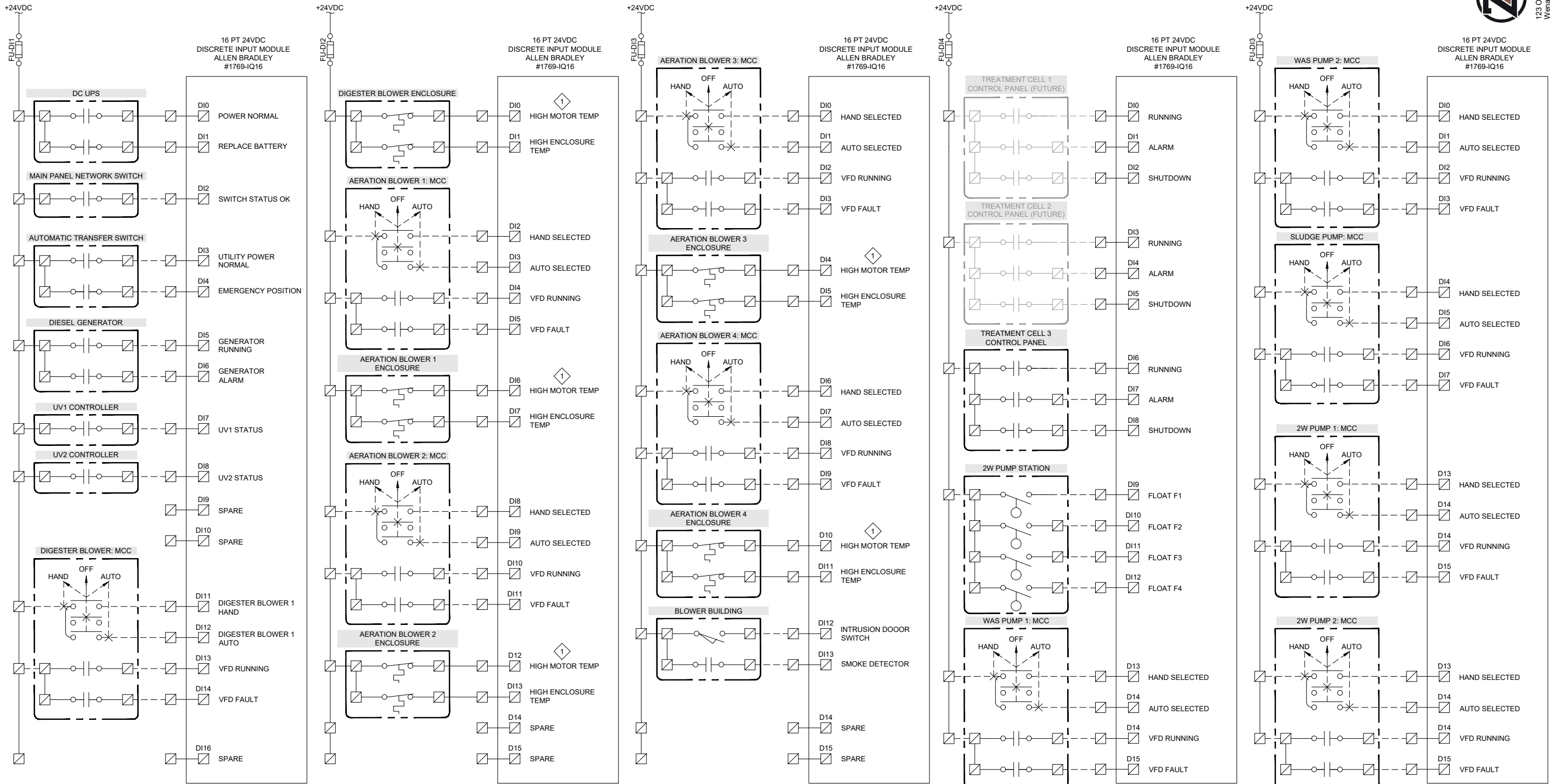
SHEET: E7.3
PAGE: 80 OF 96

KEY NOTES:

1. BLOWER SUPPLIER SHALL PROVIDE PTC RELAY MODULE FOR INSTALLATION IN MAIN PLC CONTROL PANEL. SYSTEM INTEGRATOR TO INSTALL RELAY MODULE IN CONTROL PANEL, PROVIDE 120V POWER CIRCUIT TO EACH, AND WIRE OVERTEMP SIGNAL TO PLC TO SHUT DOWN BLOWER MOTOR ON HIGH TEMPERATURE.

GENERAL NOTES:

1. SYSTEM INTEGRATOR SHALL PROVIDE DETAILED CONTROL PANEL DESIGN AND DOCUMENTATION FOR A COMPLETE AND OPERATIONAL SYSTEM. ONLY MAJOR PANEL COMPONENTS ARE SPECIFIED.



MAIN PLC CONTROL PANEL - DIGITAL INPUTS
SCALE: NONE



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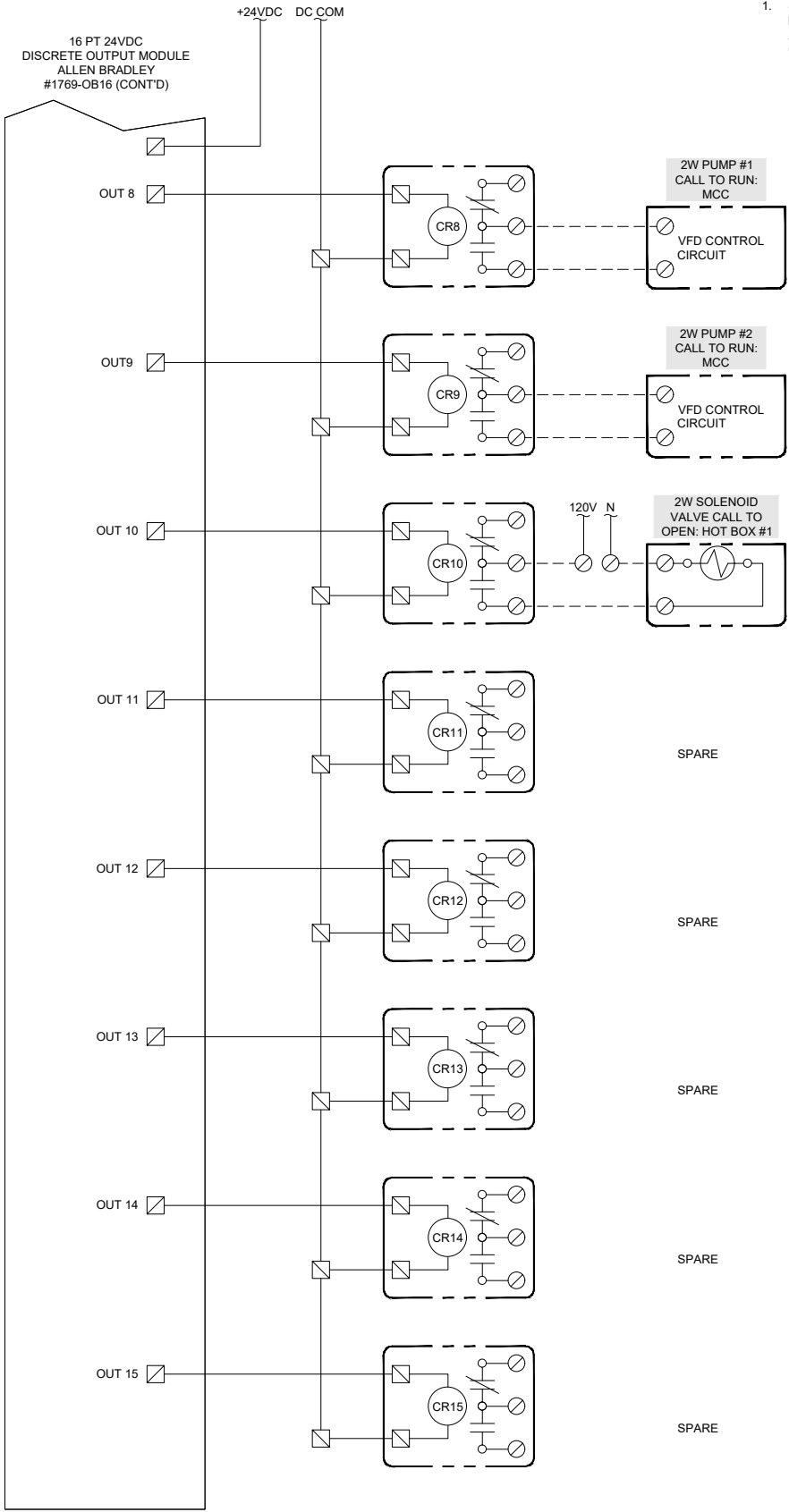
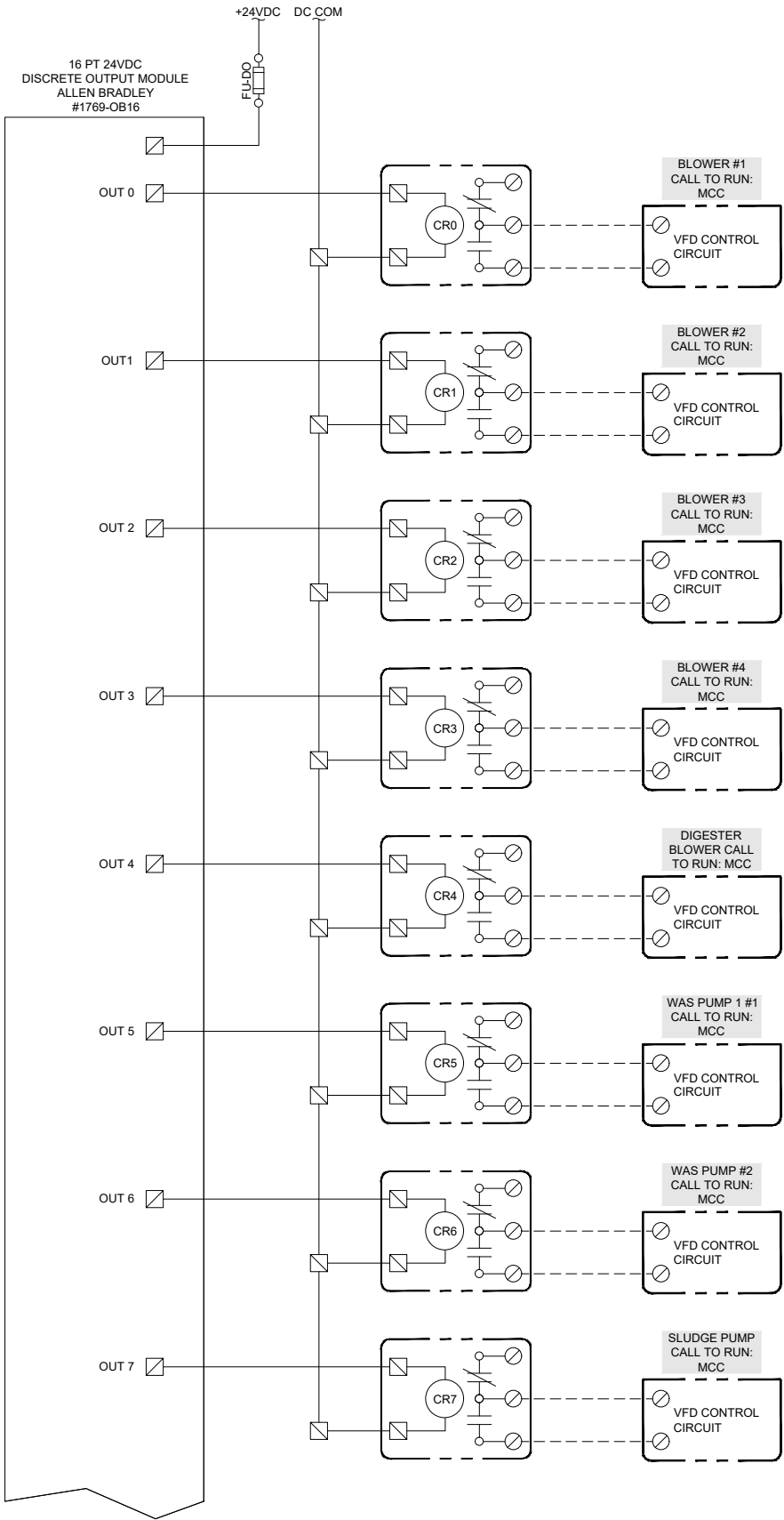
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09-12-2022

GENERAL NOTES:

1. SYSTEM INTEGRATOR SHALL PROVIDE DETAILED CONTROL PANEL DESIGN AND DOCUMENTATION FOR A COMPLETE AND OPERATIONAL SYSTEM. ONLY MAJOR PANEL COMPONENTS ARE SPECIFIED.



MAIN PLC CONTROL PANEL - DIGITAL OUTPUTS
SCALE: NONE



BID SET

EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON
SAN JUAN COUNTY
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

MAIN PLC CONTROL PANEL - DIGITAL OUTPUTS

DATE
09-12-2022

SCALE
AS SHOWN

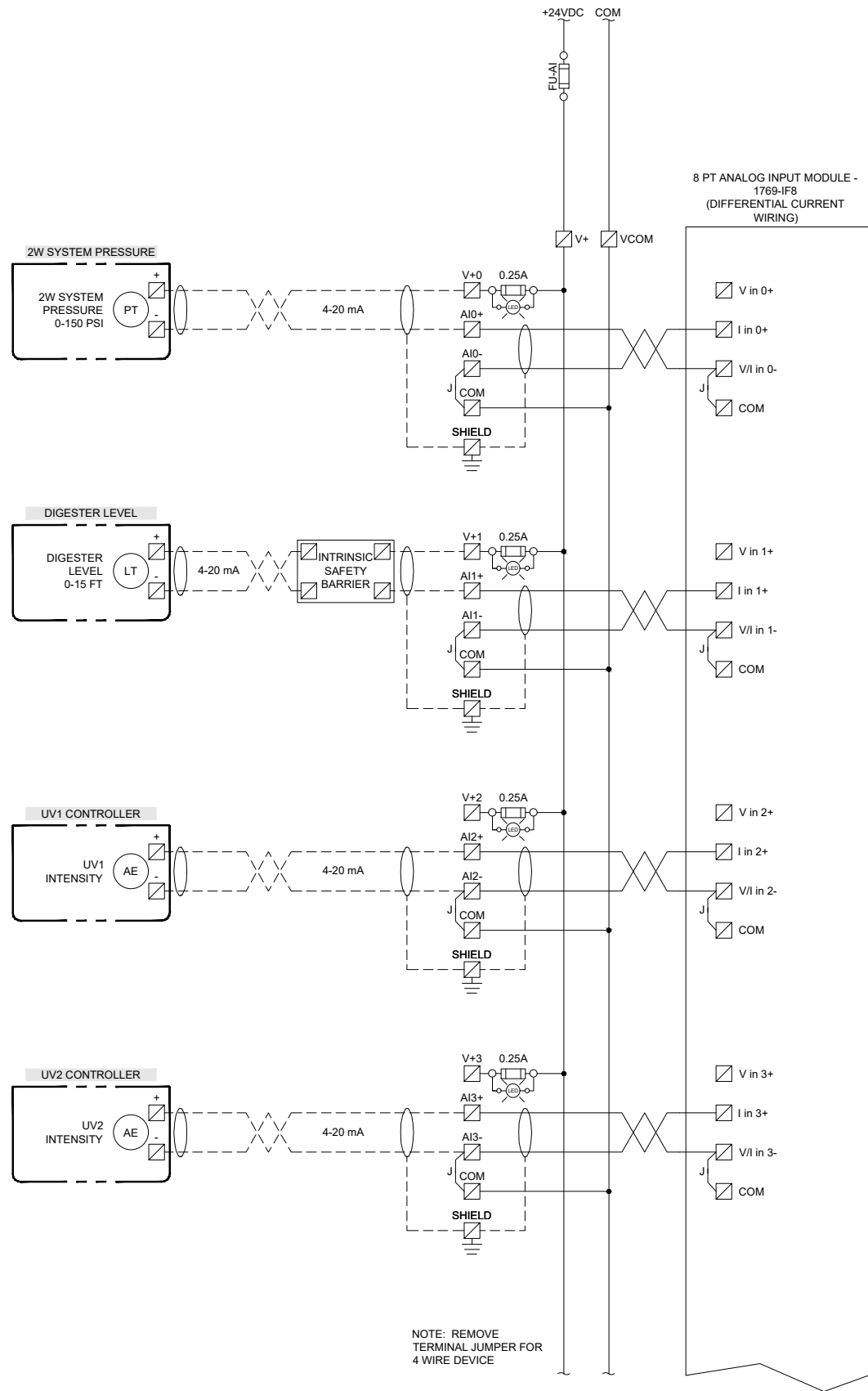
JOB NUMBER
2020-070

SHEET
E7.4

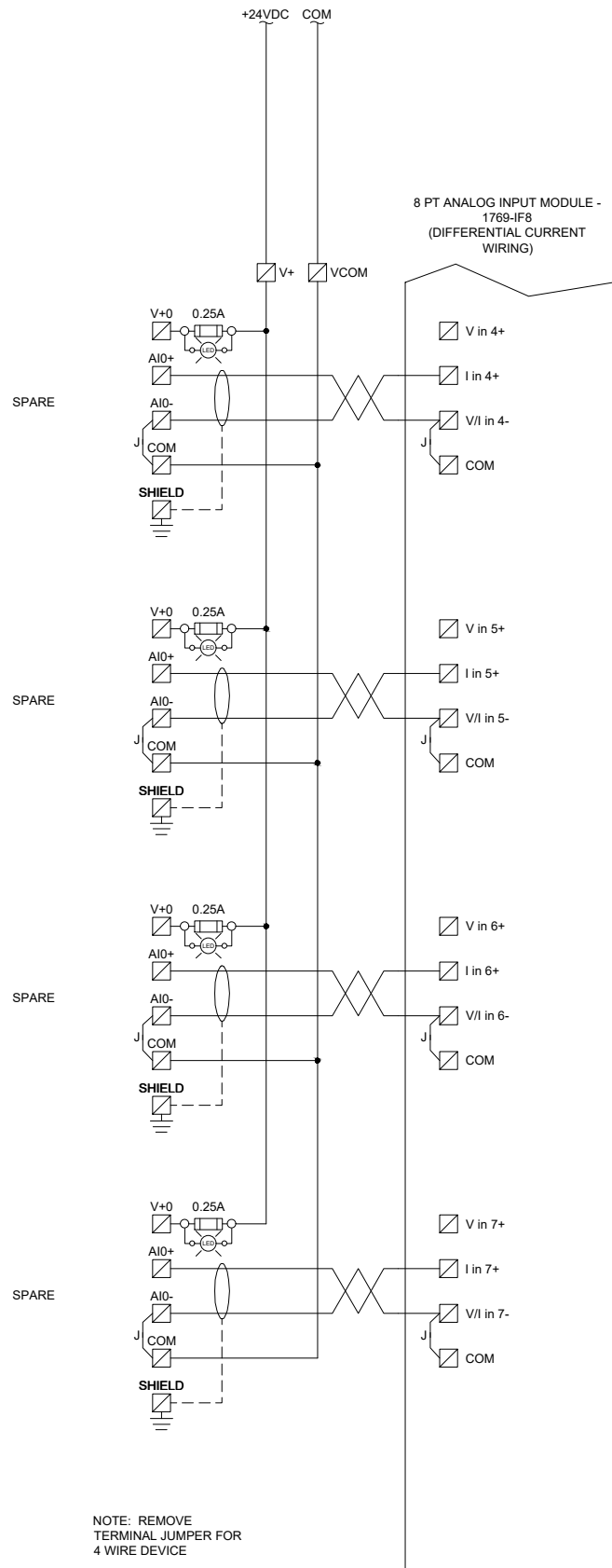
PAGE
81 OF 96

GENERAL NOTES:

1. THESE SCHEMATICS SHOW FUNCTIONAL REQUIREMENTS OF THE PLC CONTROL PANEL. SYSTEM INTEGRATOR SHALL PROVIDE DETAILED CONTROL PANEL DESIGN AND DOCUMENTATION FOR A COMPLETE AND OPERATIONAL SYSTEM.



MAIN PLC CONTROL PANEL - ANALOG INPUTS
SCALE: NONE



NO.	REVISIONS	BY	DATE



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EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON
SAN JUAN COUNTY
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

MAIN PLC CONTROL PANEL - ANALOG INPUTS - SHEET 1

DATE	09-12-2022
SCALE	AS SHOWN
JOB NUMBER	2020-070

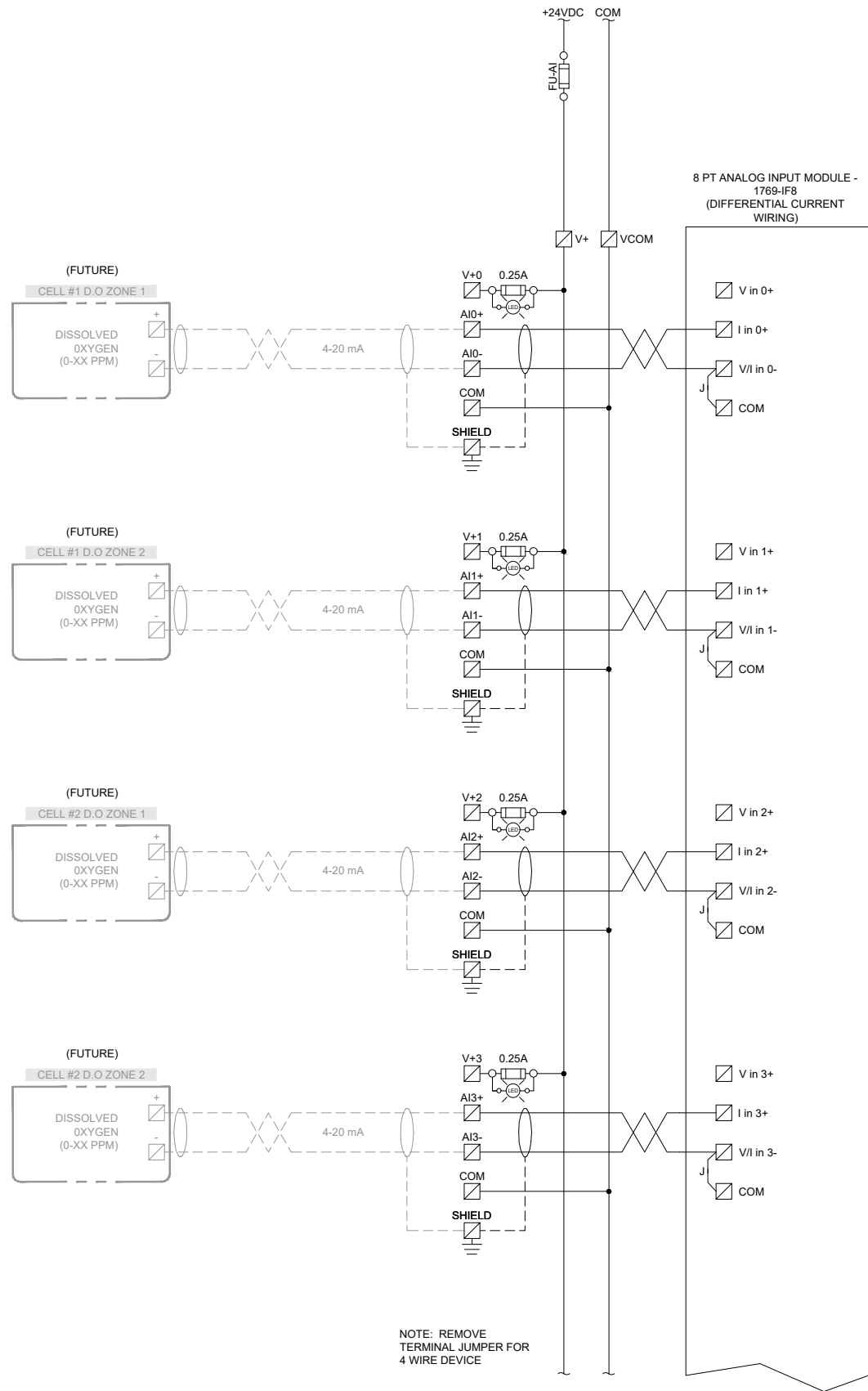
SHEET	E7.5
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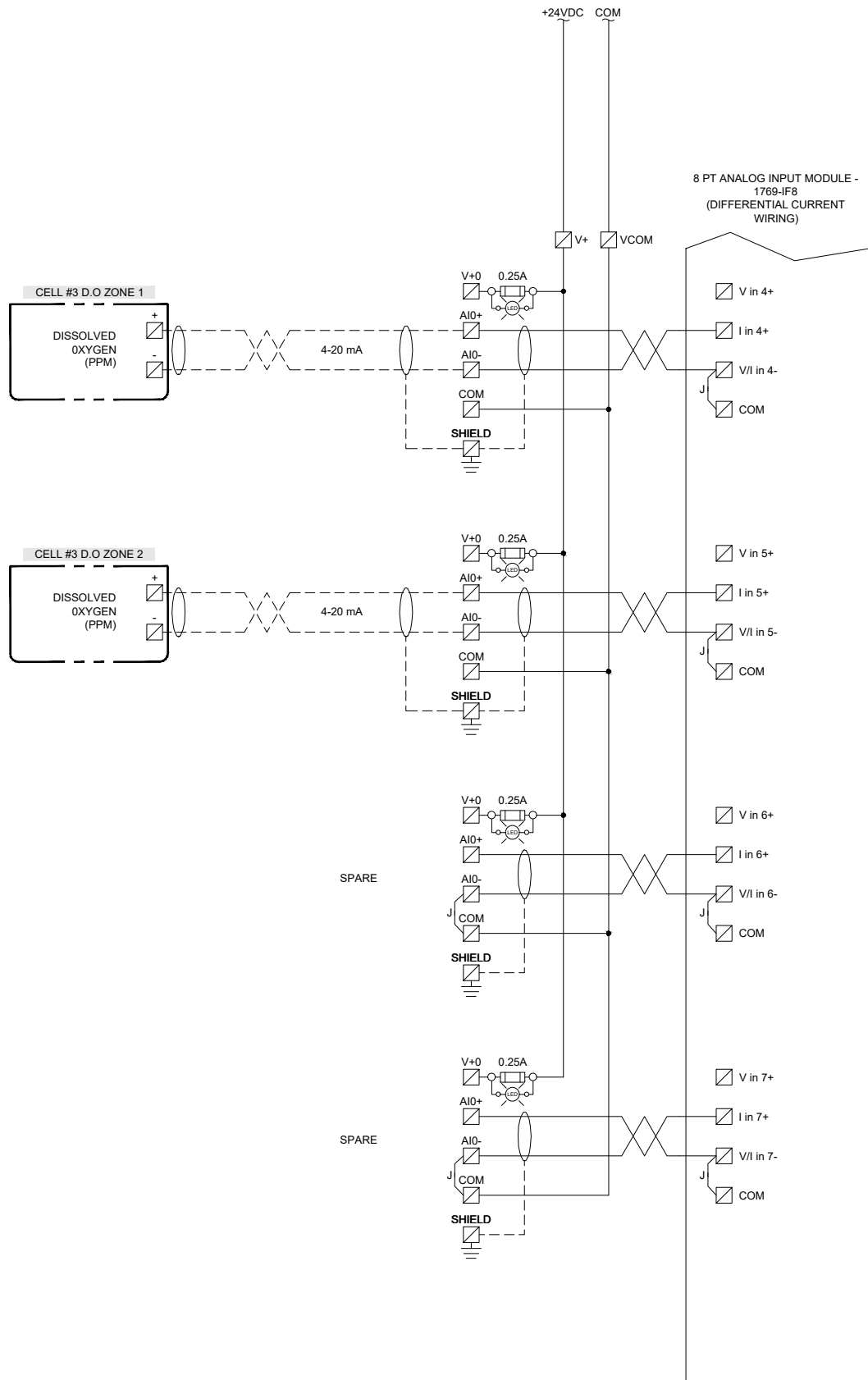
BID SET

GENERAL NOTES:

1. THESE SCHEMATICS SHOW FUNCTIONAL REQUIREMENTS OF THE PLC CONTROL PANEL. SYSTEM INTEGRATOR SHALL PROVIDE DETAILED CONTROL PANEL DESIGN AND DOCUMENTATION FOR A COMPLETE AND OPERATIONAL SYSTEM.



NOTE: REMOVE
TERMINAL JUMPER FOR
4 WIRE DEVICE



MAIN PLC CONTROL PANEL - ANALOG INPUTS
SCALE: NONE

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EASTSOUND SEWER AND WATER DISTRICT

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WASTEWATER TREATMENT PLANT UPGRADE – PHASE I

MAIN PLC CONTROL PANEL – ANALOG INPUTS – SHEET 2

DATE	09-12-2022
SCALE	AS SHOWN
JOB NUMBER	2020-070

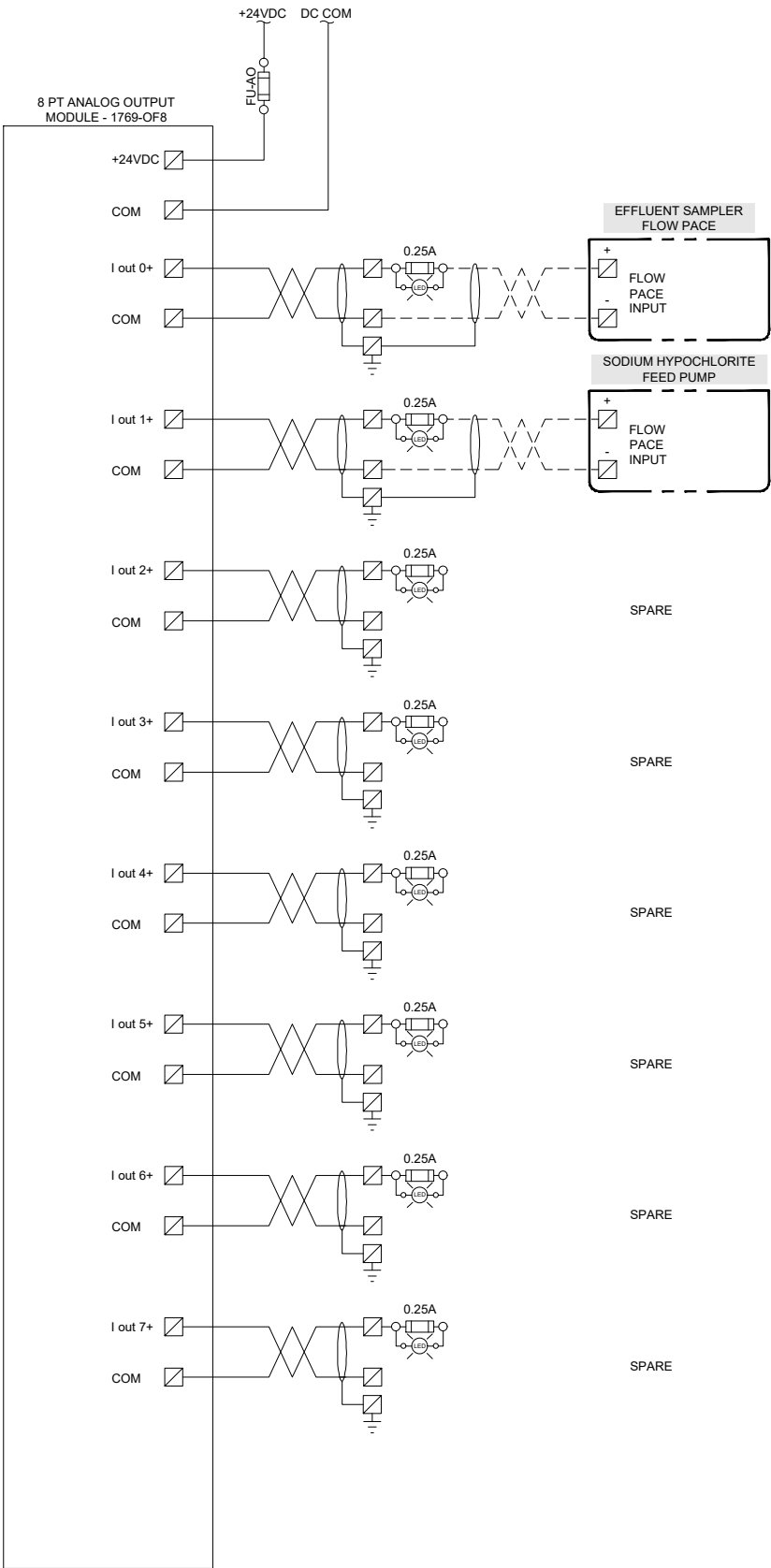
SHEET	E7.6
PAGE	83 OF 96



BID SET

GENERAL NOTES:

1. THESE SCHEMATICS SHOW FUNCTIONAL REQUIREMENTS OF THE PLC CONTROL PANEL. SYSTEM INTEGRATOR SHALL PROVIDE DETAILED CONTROL PANEL DESIGN AND DOCUMENTATION FOR A COMPLETE AND OPERATIONAL SYSTEM.



MAIN PLC CONTROL PANEL - ANALOG OUTPUTS
SCALE: NONE

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WASTEWATER TREATMENT PLANT UPGRADE – PHASE I

MAIN PLC CONTROL PANEL – ANALOG OUTPUTS

DATE	09-12-2022
SCALE	AS SHOWN
JOB NUMBER	2020-070

SHEET	E7.7
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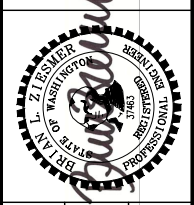
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EASTSOUND SEWER AND WATER DISTRICT

SAN JUAN COUNTY

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WASTEWATER TREATMENT PLANT UPGRADE – PHASE I

INFLUENT PUMP STATION CONTROL PANEL ELEVATIONS

SHEET	DATE	SCALE	JOB NUMBER
E8.1	09-12-2022	AS SHOWN	2020-070
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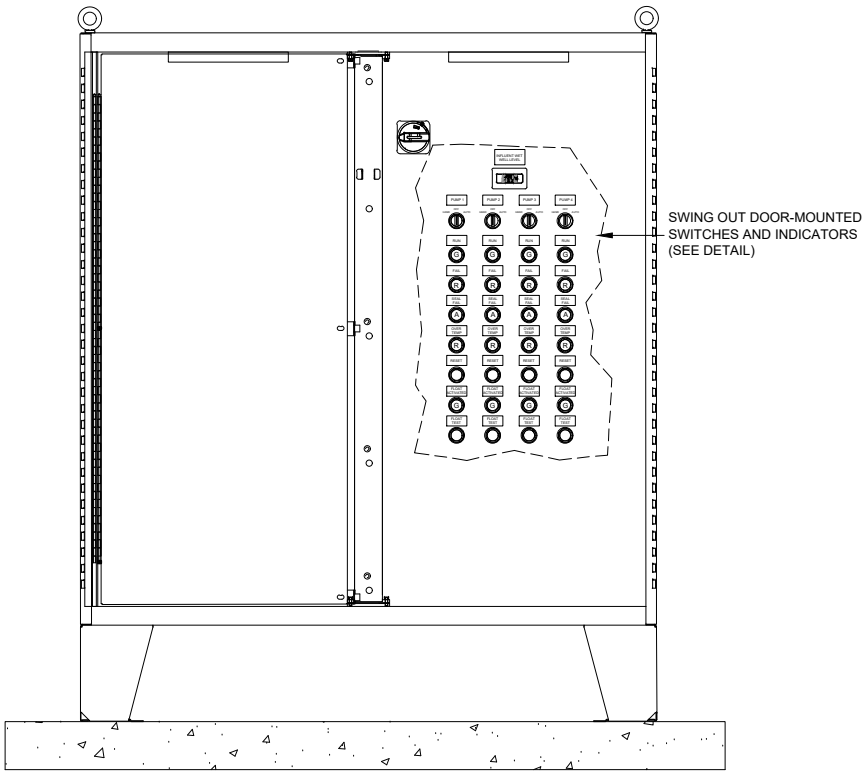


GENERAL NOTES:

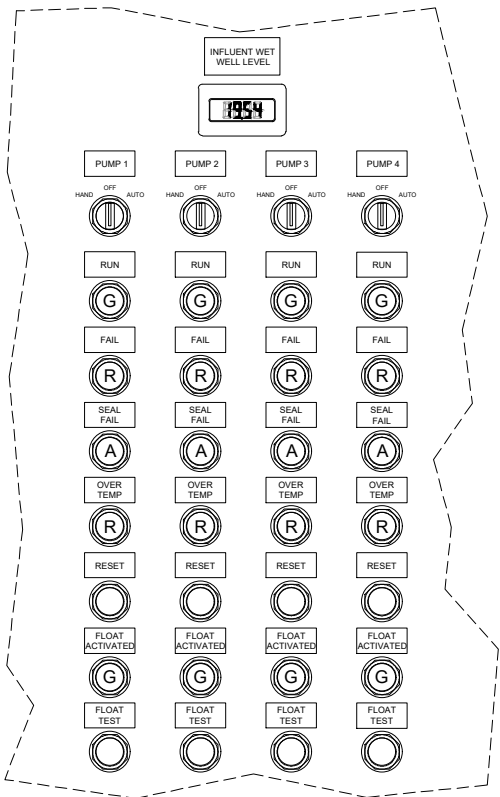
1. SYSTEM INTEGRATOR SHALL PROVIDE DETAILED CONTROL PANEL DESIGN AND DOCUMENTATION FOR A COMPLETE AND OPERATIONAL SYSTEM. ONLY MAJOR PANEL COMPONENTS ARE SPECIFIED.

KEY NOTES:

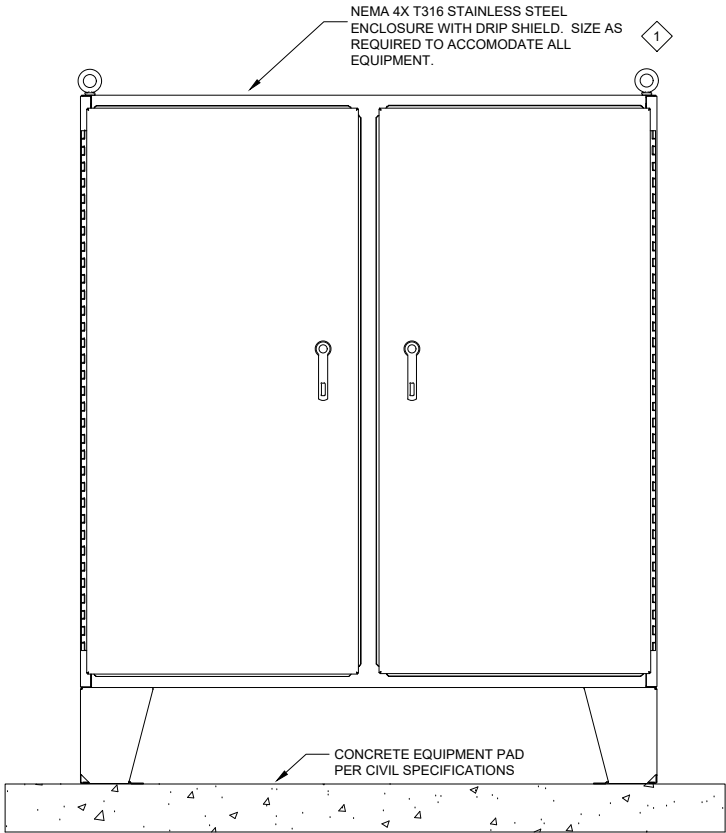
1. PROVIDE INFLUENT PUMP STATION CONTROL PANEL FOR LOCAL CONTROL OF INFLUENT PUMPS. SEAL FAIL AND OVERTEMP WIRING, AND INSTRUMENTATION. MOUNT CONTROLS AND INDICATORS ON INTERNAL SWING OUT DOOR, LOCKABLE OUTSIDE DOOR. SYSTEM INTEGRATOR SHALL SIZE CONTROL PANEL TO ACCOMMODATE ALL CONTROLS AND EQUIPMENT REQUIRED.
2. PROVIDE FLOW METER TRANSMITTERS FOR THE FOLLOWING LOCATIONS: INFLUENT, CELL 1 (FUTURE SPACE), CELL 2, CELL 3, PLANT DRAIN. SYSTEM INTEGRATOR SHALL SIZE CONTROL PANEL TO ACCOMMODATE ALL CONTROLS AND EQUIPMENT REQUIRED.
3. INSTALL SEAL FAIL AND OVERTEMP RELAYS AS PROVIDED BY PUMP MANUFACTURER; MINI-CAS OR EQUAL. SYSTEM INTEGRATOR SHALL COORDINATE WITH PUMP SUPPLIER FOR RELAYS.
4. ALL FIELD CONDUITS NOT SHOWN. CONTRACTOR SHALL ROUTE ALL RACEWAY SHOWN ON THE PLANS TO THE INFLUENT CONTROL PANEL.



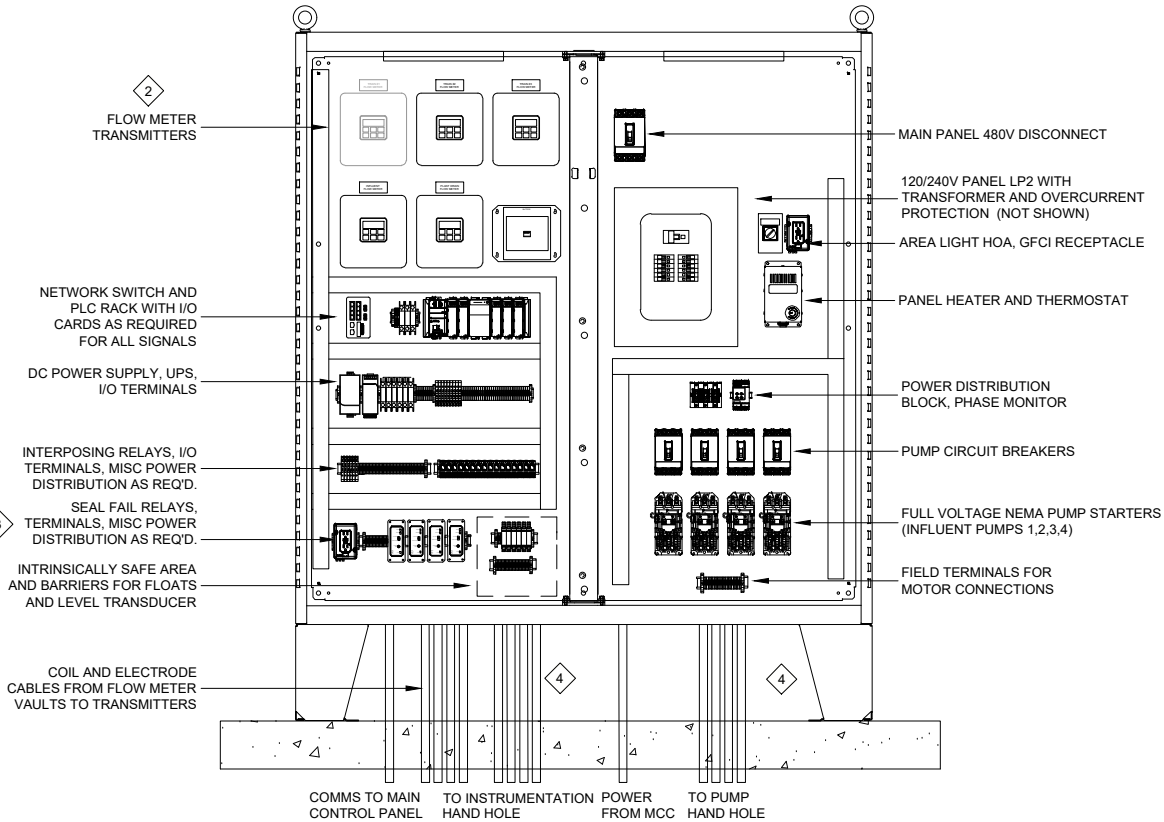
INFLUENT PUMP STATION PANEL - INTERIOR DOOR
SCALE: NONE



SWING OUT INTERIOR DOOR DETAIL
SCALE: NONE



INFLUENT PUMP STATION PANEL - EXTERIOR
SCALE: NONE



INFLUENT PUMP STATION PANEL - INTERIOR BACKPANEL
SCALE: NONE



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GENERAL NOTES:
1. SYSTEM INTEGRATOR SHALL PROVIDE DETAILED CONTROL PANEL DESIGN AND DOCUMENTATION FOR A COMPLETE AND OPERATIONAL SYSTEM. ONLY MAJOR PANEL COMPONENTS ARE SPECIFIED.



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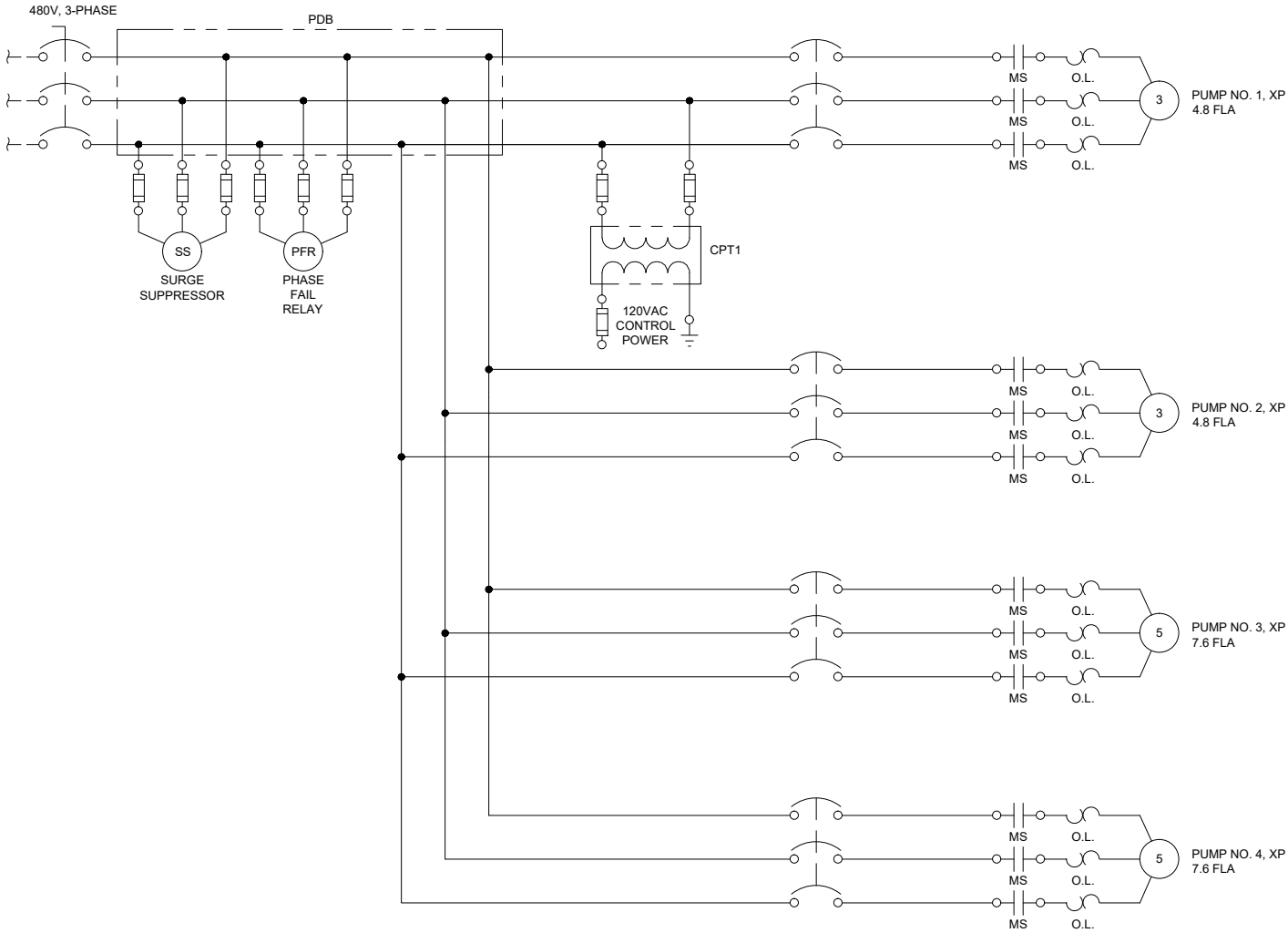
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	WASTEWATER TREATMENT PLANT UPGRADE – PHASE I
	INFLUENT PUMP STATION CONTROL PANEL–POWER WIRING DIAGRAM

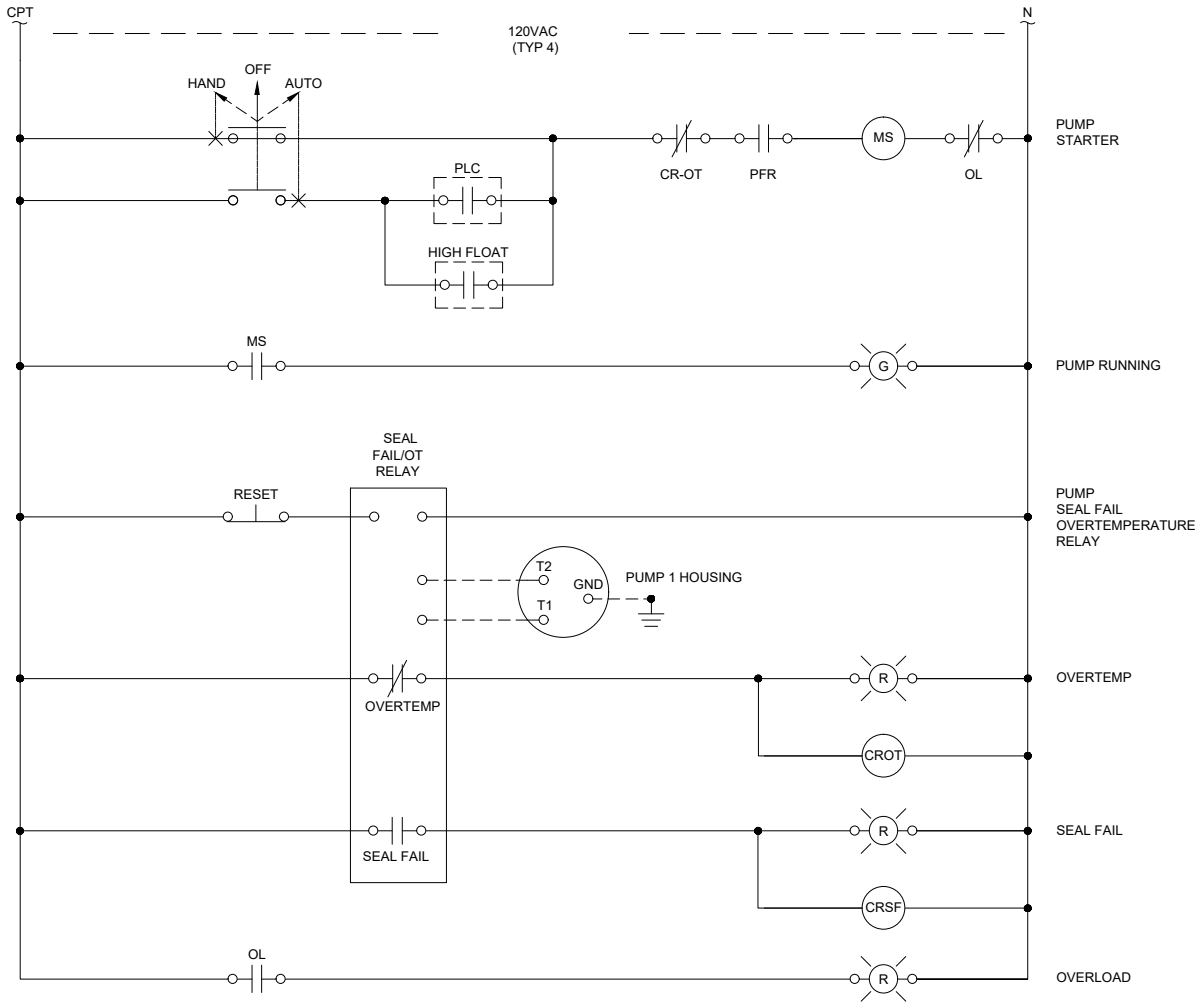
SHEET	DATE	SCALE	JOB NUMBER
E8.2	09-12-2022	AS SHOWN	2020-070
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86	OF	96	



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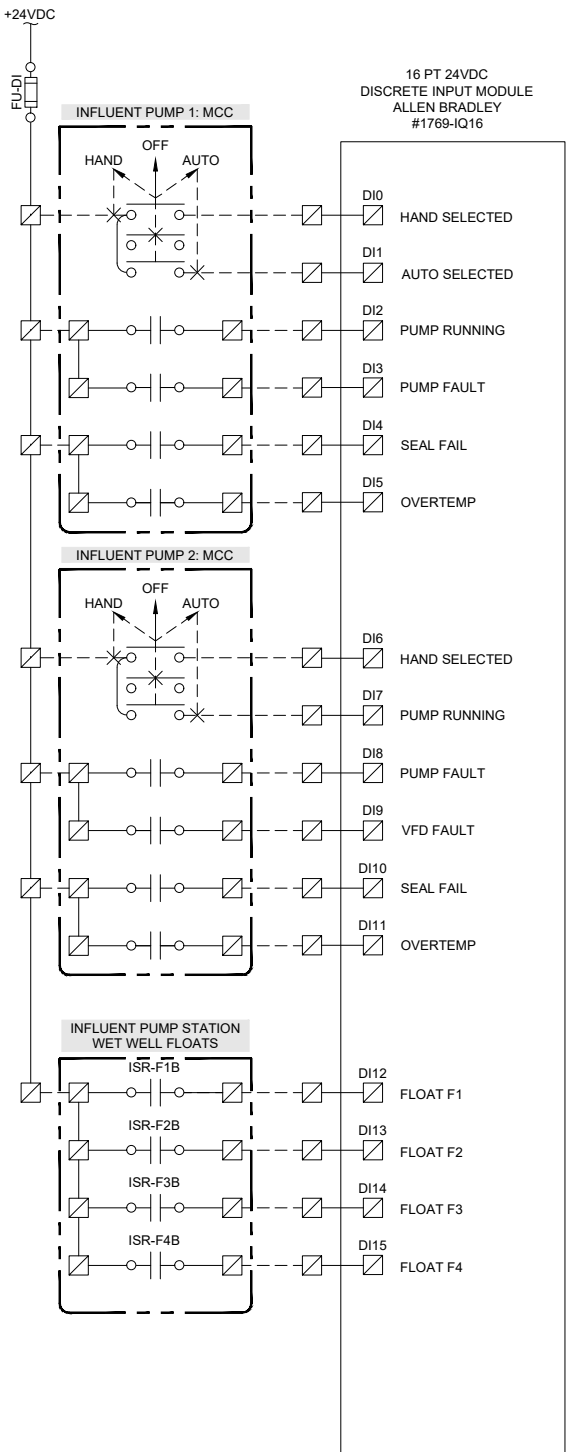
INFLUENT PUMP STATION CONTROL PANEL - SCHEMATIC POWER WIRING DIAGRAM
SCALE: NONE



PUMP CONTROL WIRING DIAGRAM (TYPICAL FOR EACH OF 4 PUMPS)
SCALE: NONE

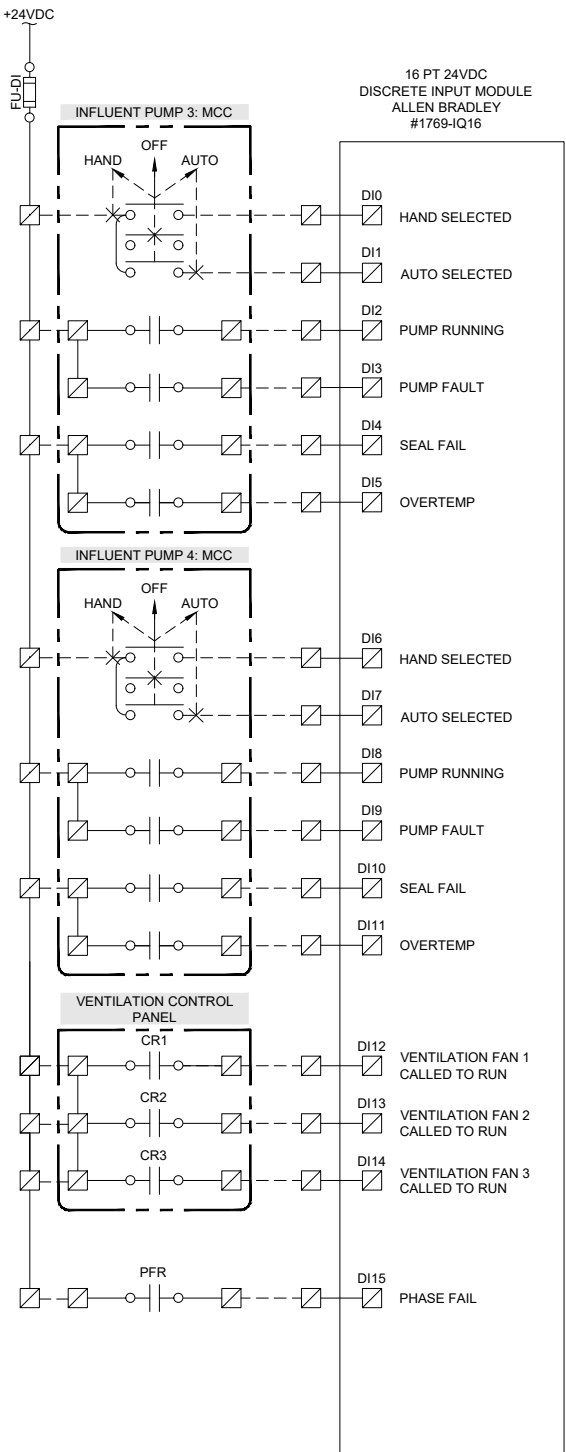
GENERAL NOTES:

- THESE SCHEMATICS SHOW FUNCTIONAL REQUIREMENTS OF THE PLC CONTROL PANEL. SYSTEM INTEGRATOR SHALL PROVIDE DETAILED CONTROL PANEL DESIGN AND DOCUMENTATION FOR A COMPLETE AND OPERATIONAL SYSTEM.
- PROVIDE ADDITIONAL I/O CARDS AS NECESSARY FOR MINIMUM 20% SPARE INPUTS, NOT SHOWN.



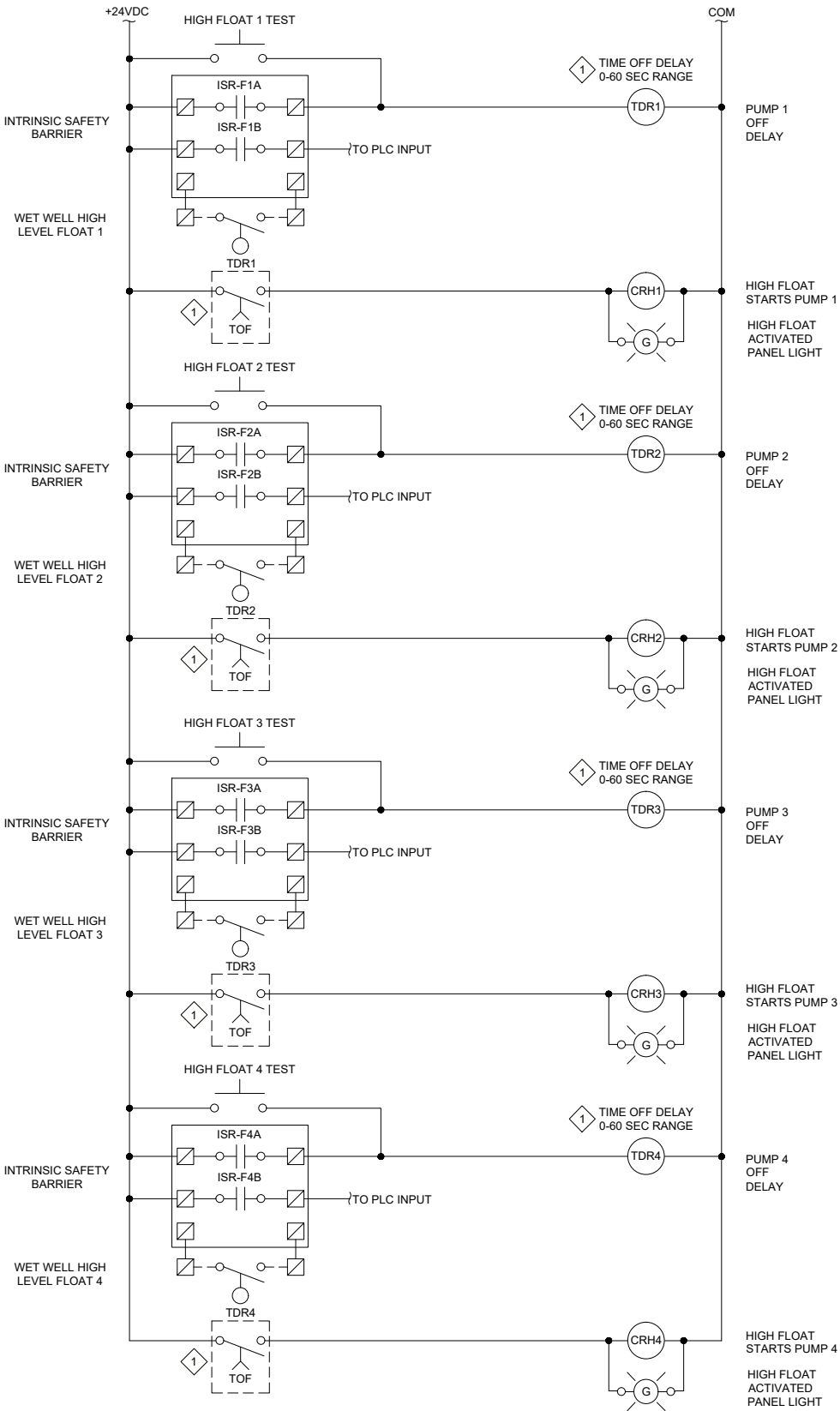
INFLUENT PUMP STATION CONTROL PANEL - DIGITAL INPUTS

SCALE: NONE



KEY NOTES:

- TIME OFF DELAY RELAYS SET FOR TRUE-OFF DELAY. WHEN INPUT POWER IS TURNED ON, THE OUTPUT CONTACT CHANGES STATE. WHEN POWER IS REMOVED, THE TIME DELAY BEGINS. THE OUTPUT CONTACT RETURNS TO NORMAL STATE AT THE END OF THE TIME DELAY. FIELD-ADJUST TIME OFF DELAY BASED ON WET WELL PUMP DOWN RATES.



INFLUENT PUMP STATION - HIGH LEVEL FLOAT CONTROL CIRCUIT

SCALE: NONE



BID SET

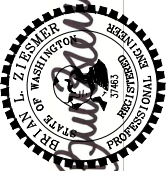
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EASTSOUND SEWER AND WATER DISTRICT

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WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

INFLUENT PUMP STATION CONTROL PANEL - DIGITAL INPUTS

DATE

09-12-2022

SHEET

E8.3

SCALE

AS SHOWN

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2020-070

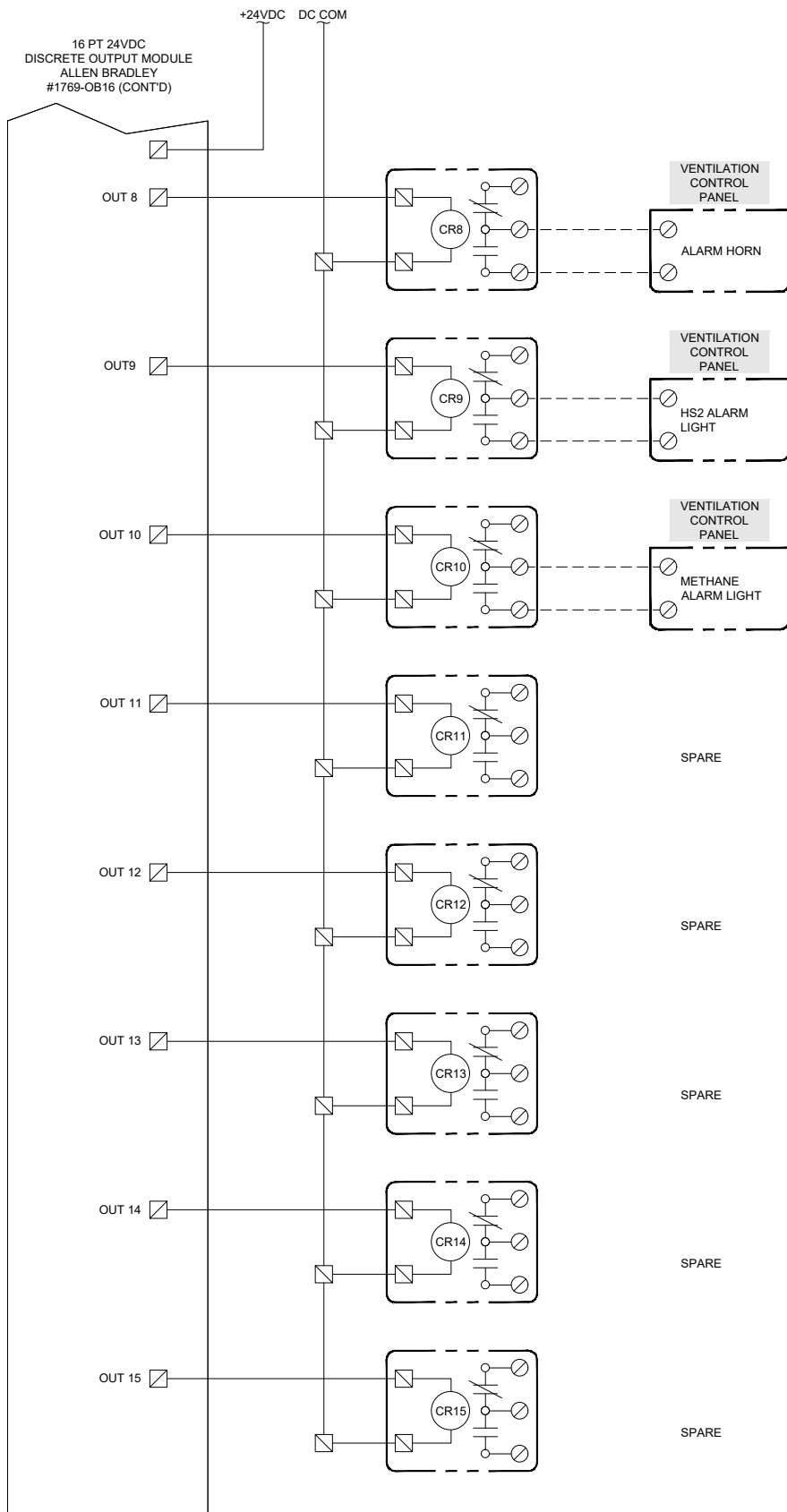
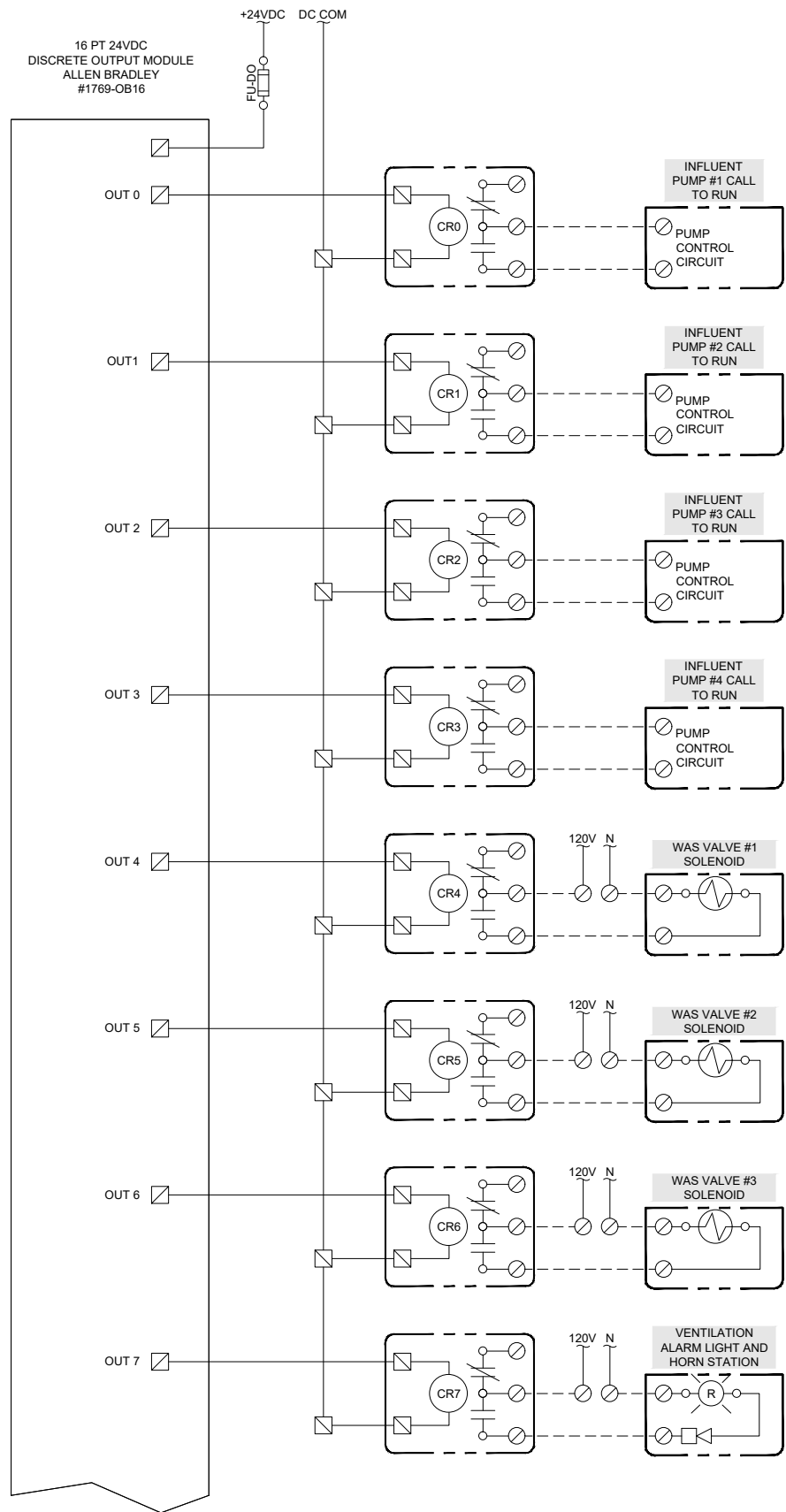
OF

96

09-12-2022

GENERAL NOTES:

1. THESE SCHEMATICS SHOW FUNCTIONAL REQUIREMENTS OF THE PLC CONTROL PANEL. SYSTEM INTEGRATOR SHALL PROVIDE DETAILED CONTROL PANEL DESIGN AND DOCUMENTATION FOR A COMPLETE AND OPERATIONAL SYSTEM.



INFLUENT PUMP STATION CONTROL PANEL - DIGITAL OUTPUTS

SCALE: NONE

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SAN JUAN COUNTY WASHINGTON

WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

INFLUENT PUMP STATION CONTROL PANEL - DIGITAL OUTPUTS

DATE	09-12-2022
SCALE	AS SHOWN
JOB NUMBER	2020-070

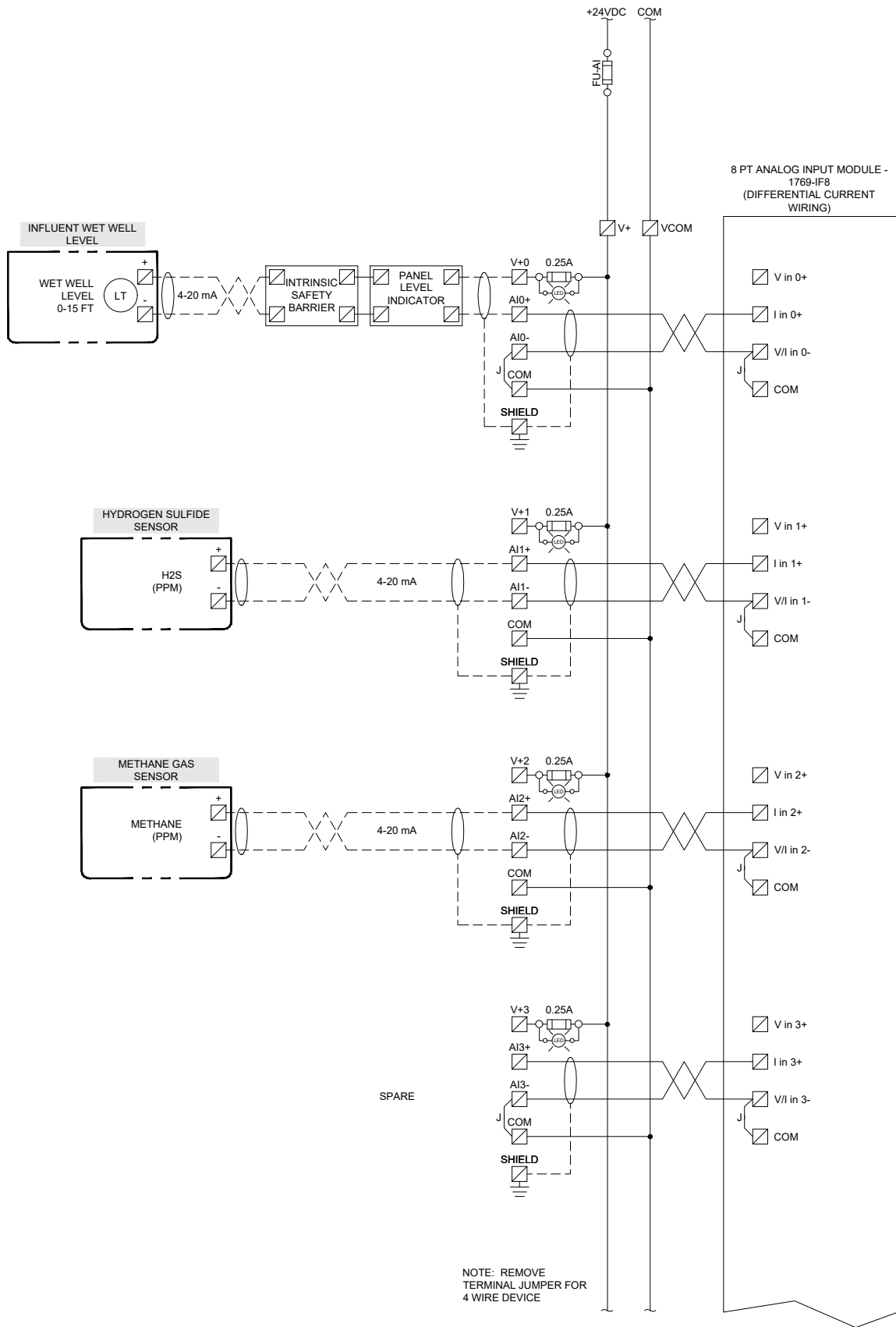
SHEET	E8.4
PAGE	88 OF 96



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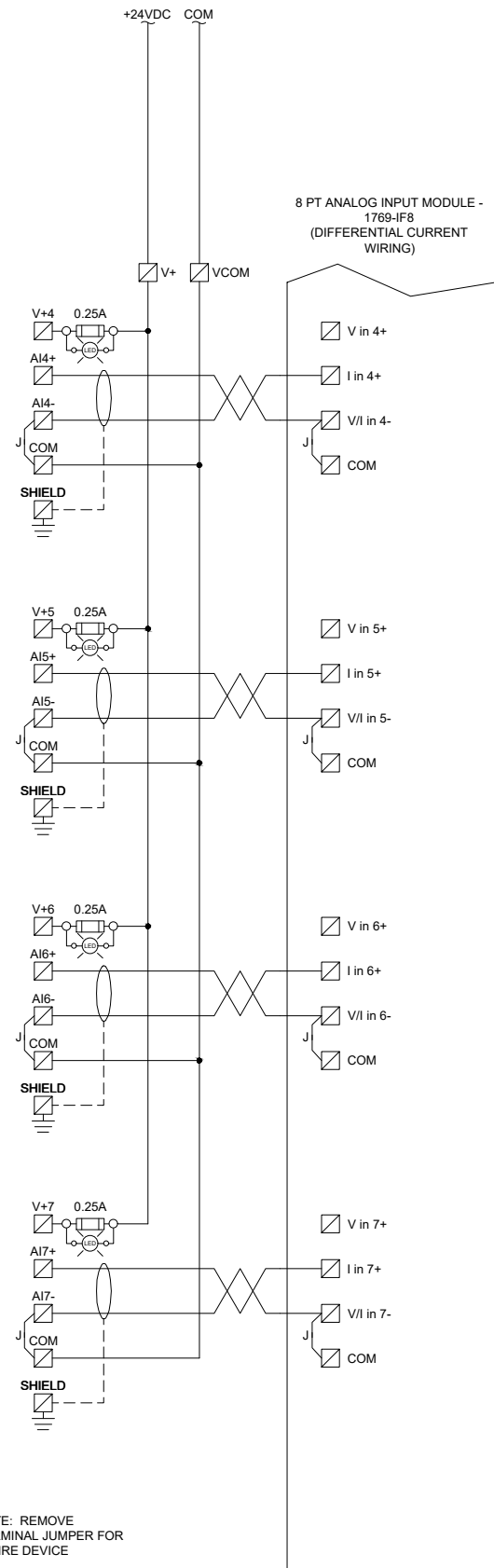
GENERAL NOTES:

1. THESE SCHEMATICS SHOW FUNCTIONAL REQUIREMENTS OF THE PLC CONTROL PANEL. SYSTEM INTEGRATOR SHALL PROVIDE DETAILED CONTROL PANEL DESIGN AND DOCUMENTATION FOR A COMPLETE AND OPERATIONAL SYSTEM.



INFLUENT PUMP STATION CONTROL PANEL - ANALOG INPUTS

SCALE: NONE



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EASTSOUND SEWER AND WATER DISTRICT

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WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

INFLUENT PUMP STATION CONTROL PANEL - ANALOG INPUTS

DATE	09-12-2022
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SHEET E8.5

SCALE	AS SHOWN
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JOB NUMBER	2020-070
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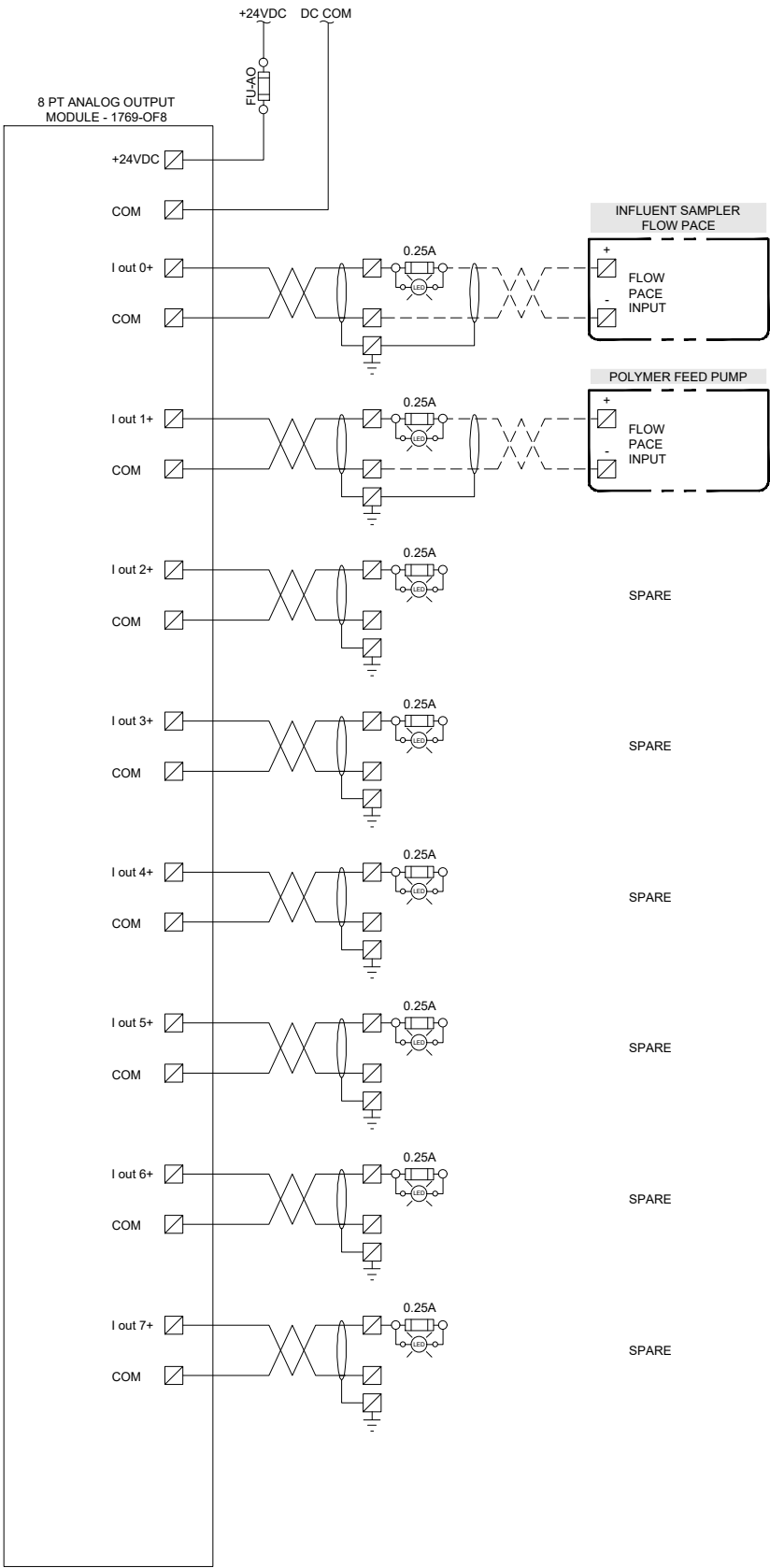
PAGE 89 OF 96



BID SET

GENERAL NOTES:

1. THESE SCHEMATICS SHOW FUNCTIONAL REQUIREMENTS OF THE PLC CONTROL PANEL. SYSTEM INTEGRATOR SHALL PROVIDE DETAILED CONTROL PANEL DESIGN AND DOCUMENTATION FOR A COMPLETE AND OPERATIONAL SYSTEM.



INFLUENT PUMP STATION CONTROL PANEL - ANALOG OUTPUTS
SCALE: NONE

NO.	REVISIONS	BY	DATE



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WASTEWATER TREATMENT PLANT UPGRADE – PHASE I
INFLUENT PUMP STATION CONTROL PANEL – ANALOG OUTPUTS

DATE	09-12-2022
SCALE	AS SHOWN
JOB NUMBER	2020-070

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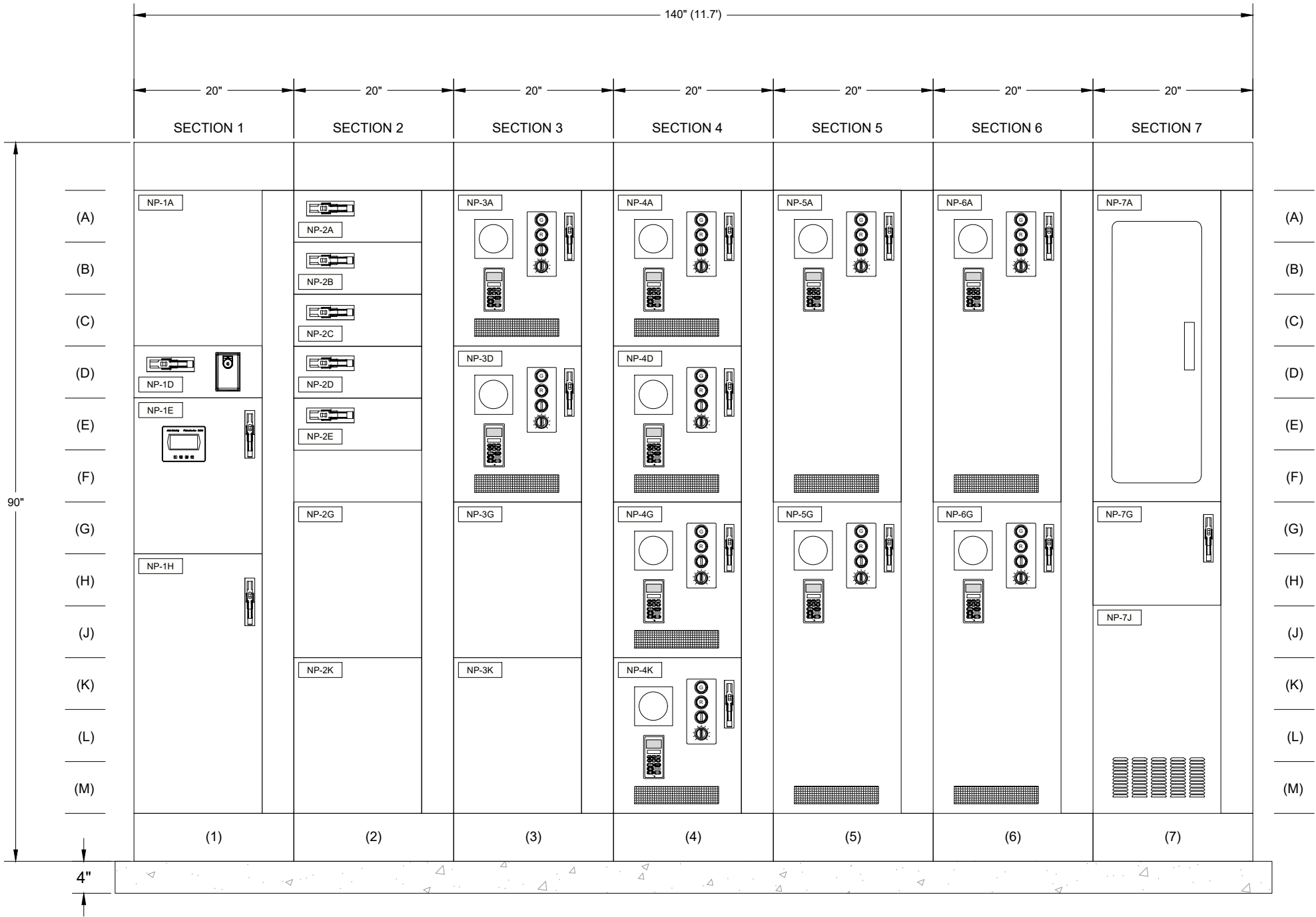


BID SET

GENERAL NOTES:

1. CONTRACTOR SHALL VERIFY AND PROVIDE FINAL MCC EQUIPMENT LAYOUT TO MEET PROJECT REQUIREMENTS. MOTOR CONTROL CENTER SHALL BE ALLEN BRADLEY CENTERLINE 2100 INTELLICENTER MODEL.

MCC NAMEPLATE SCHEDULE			
EQ ID	NAMEPLATE LABEL (LINE 1 / LINE 2 / LINE 3)	EQ ID	NAMEPLATE LABEL (LINE 1 / LINE 2 / LINE 3)
NP-1A	MCC NETWORK / ETHERNET SWITCH	NP-3G	SPACE
NP-1D	SURGE PROTECTOR	NP-3K	SPACE
NP-1E	POWER MONITOR	NP-4A	SLUDGE PUMP / 3 HP
NP-1H	MAIN BREAKER DISCONNECT / 400 AMP	NP-4D	WAS PUMP NO.1 / 3 HP
NP-2A	TRAIN NO.1 / PANEL FEEDER (FUTURE)	NP-4G	WAS PUMP NO.2 / 3 HP
NP-2B	TRAIN NO.2 / PANEL FEEDER (FUTURE)	NP-4K	DIGESTER BLOWER / 5 HP
NP-2C	TRAIN NO.3 / PANEL FEEDER	NP-5A	AERATION BLOWER NO.1 / 20 HP
NP-2D	DEWATERING SKID / FEEDER	NP-5G	AERATION BLOWER NO.2 / 20 HP
NP-2E	INFLUENT PUMP STATION/FEEDER	NP-6A	AERATION BLOWER NO.3 / 20 HP
NP-2G	SPACE	NP-6G	AERATION BLOWER NO.4 / 20 HP
NP-2K	SPACE	NP-7A	LIGHTING PANEL / 120-240V 1 PH
NP-3A	2W PUMP NO.1 / 5 HP	NP-7G	TRANSFORMER DISCONNECT
NP-3D	2W PUMP NO.2 / 5 HP	NP-7J	TRANSFORMER / 480-120/240V / 20 kVA



MOTOR CONTROL CENTER (MCC) ELEVATION
NOT TO SCALE

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WASTEWATER TREATMENT PLANT UPGRADE – PHASE I
BLOWER BUILDING MOTOR CONTROL CENTER ELEVATION

DATE: 09-12-2022
SCALE: AS SHOWN
JOB NUMBER: 2020-070

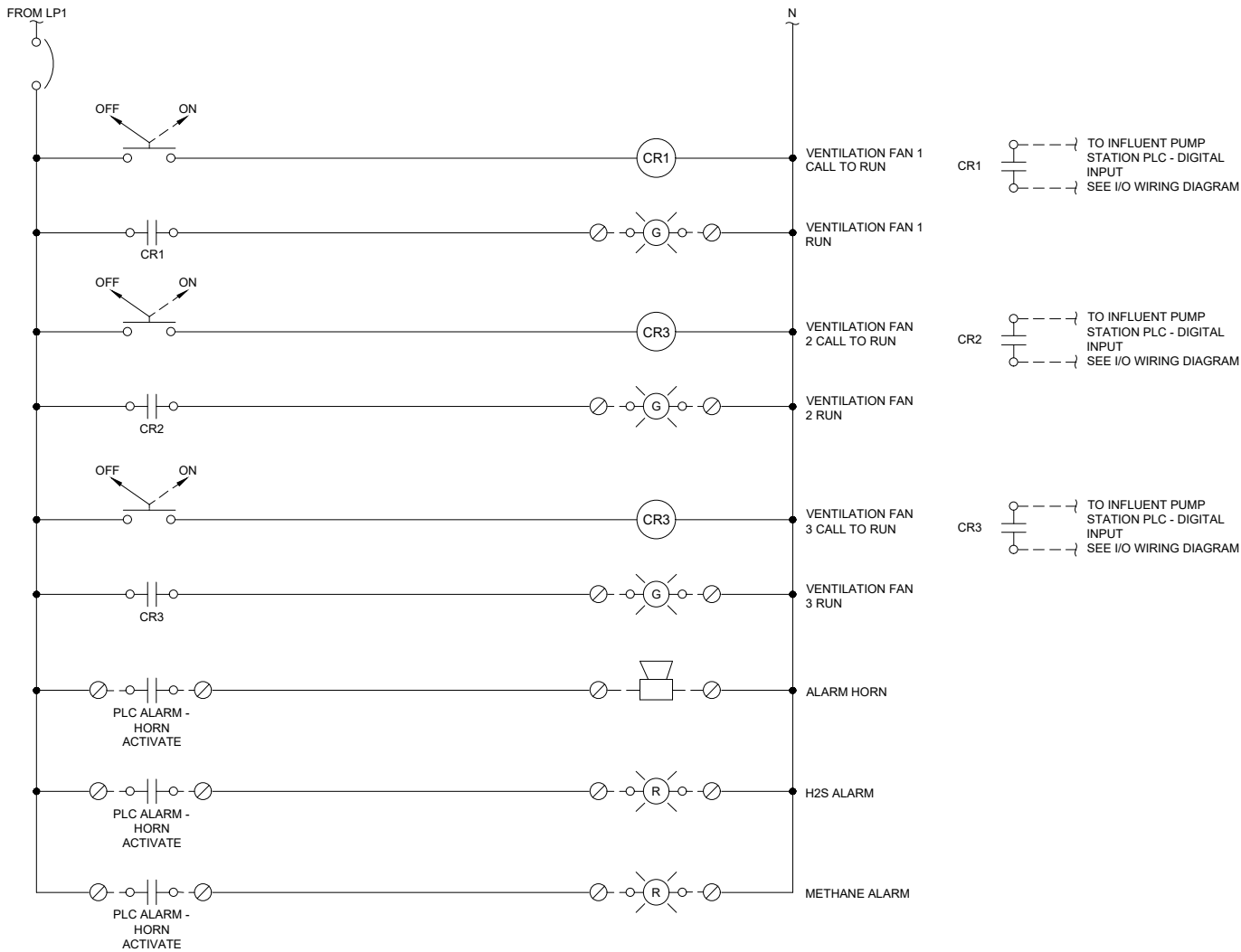
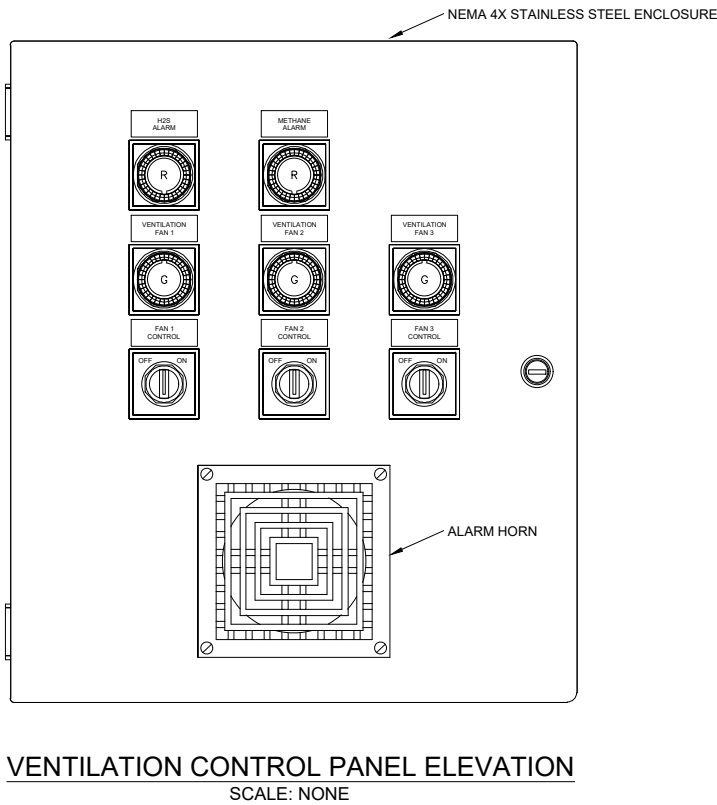
SHEET: E9.1
PAGE: 91 OF 96



BID SET

GENERAL NOTES:

1. THESE SCHEMATICS SHOW FUNCTIONAL REQUIREMENTS OF THE PLC CONTROL PANEL. SYSTEM INTEGRATOR SHALL PROVIDE DETAILED CONTROL PANEL DESIGN AND DOCUMENTATION FOR A COMPLETE AND OPERATIONAL SYSTEM.



VENTILATION CONTROL PANEL ELEVATION AND WIRING DIAGRAM
SCALE: NONE

NO.	REVISIONS	BY	DATE



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WASTEWATER TREATMENT PLANT UPGRADE – PHASE I

VENTILATION CONTROL PANEL ELEVATION AND WIRING DIAGRAM

DATE	09-12-2022
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EASTSOUND SEWER AND WATER DISTRICT

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WASTEWATER TREATMENT PLANT UPGRADE – PHASE I

ELECTRICAL DETAILS – SHEET 2

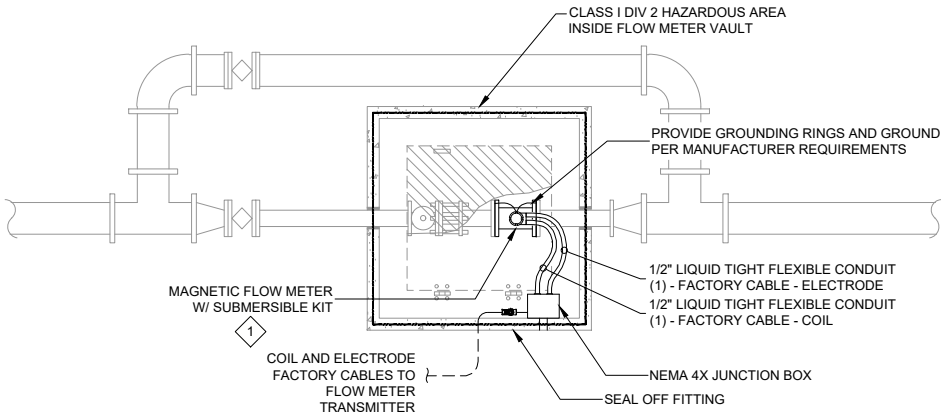
SHEET	DATE	SCALE	JOB NUMBER
E10.2	09-12-2022	AS SHOWN	2020-070
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GENERAL NOTES:

- ALL CONDUIT ROUTING IS NOT SHOWN, AND WHERE SHOWN, IS SCHEMATIC. ELECTRICAL CONTRACTOR SHALL DETERMINE THE BEST ROUTING PATH AND CIRCUIT COMBINATIONS BASED ON FIELD CONDITIONS AND ELECTRICAL CODES.
- EQUIPMENT LOCATIONS AND ARRANGEMENT ARE SCHEMATIC. CONTRACTOR SHALL COORDINATE WITH EQUIPMENT MANUFACTURER FOR DETAILED CONNECTION REQUIREMENTS AND PROVIDE MATERIALS AND INSTALLATION FOR A COMPLETE AND OPERATIONAL SYSTEM.

KEY NOTES:

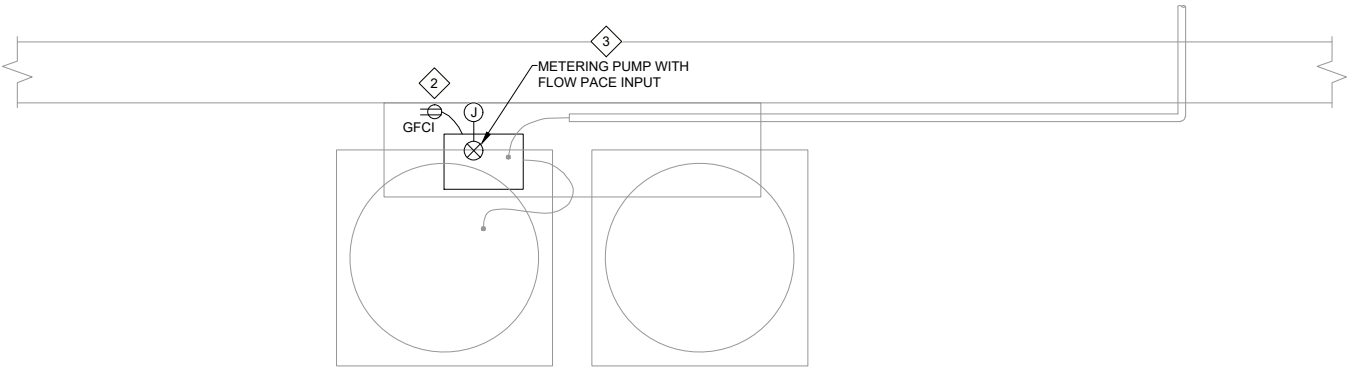
- FLOW METERS: SIZE PER CIVIL DRAWINGS. MAGNETIC FLOW METER, ENDRESS-HAUSER PROMAG W400, CLASS I DIV 2, CARBON STEEL FLANGE, 120-240VAC OR 24VAC/DC, ETHERNET/IP, POLYURETHANE LINER, 316L ELECTRODES, REMOTE HOUSING, NPT 1/2, SENSOR IP68 TYPE 6, POTTED CORROSION PROTECTION AND SUBMERSIBLE. PROVIDE (2) 316L GROUNDING RINGS FOR EACH FLOW TUBE.
- PROVIDE DEDICATED GFCI RECEPTACLE FOR METERING PUMP POWER CONNECTION. WIRE TO LIGHTING PANEL, SEE SCHEDULES.
- PROVIDE 4-20 mA OUTPUT FROM PLC TO METERING PUMP. SCALE OUTPUT PROPORTIONAL TO FLOW METER SIGNAL.



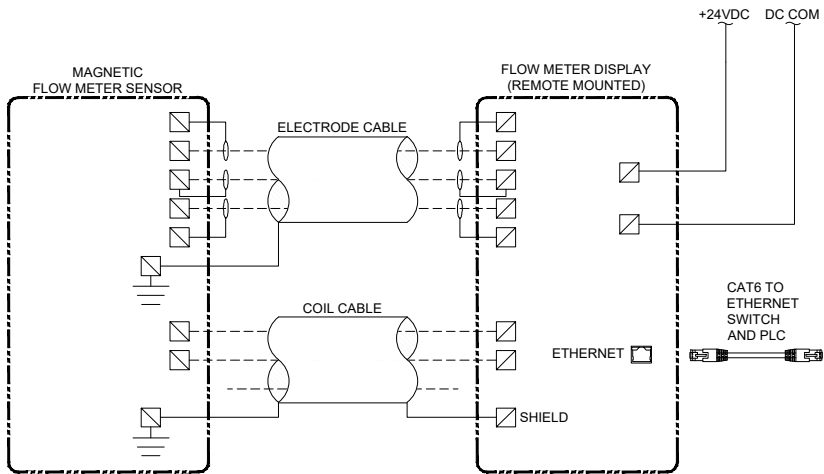
VAULT FLOW METER STANDARD DETAIL
SCALE: NONE

1 FLOW METER SUMMARY:

- INFLUENT FLOW METER. SIZE: 6". INSTALL IN VAULT WITH SUBMERSIBLE KIT. TRANSMITTER LOCATED IN INFLUENT CONTROL PANEL.
- TRAIN #1 FLOW METER. SIZE: 8". (FUTURE PROJECT - NOT IN THIS PROJECT) PROVIDE SPACE FOR FUTURE TRANSMITTER IN INFLUENT CONTROL PANEL.
- TRAIN #2 FLOW METER. SIZE: 8". INSTALL IN VAULT WITH SUBMERSIBLE KIT. TRANSMITTER LOCATED IN INFLUENT CONTROL PANEL.
- TRAIN #3 FLOW METER. SIZE: 8". INSTALL IN VAULT WITH SUBMERSIBLE KIT. TRANSMITTER LOCATED IN INFLUENT CONTROL PANEL.
- PLANT DRAIN FLOW METER. SIZE: 4". INSTALL IN VAULT WITH SUBMERSIBLE KIT. TRANSMITTER LOCATED IN INFLUENT CONTROL PANEL.
- EFFLUENT FLOW METER. SIZE: 8". INSTALL IN VAULT WITH SUBMERSIBLE KIT. TRANSMITTER LOCATED IN PANEL NEAR UV DISINFECTION CHAMBER.
- WAS FLOW METER. SIZE: 2". INSTALL ON PIPING NEAR WAS PUMPS, MOUNT TRANSMITTER TO FLOW TUBE.
- 2W FLOW METER. SIZE: 1". INSTALL ON PIPING IN HOT BOX #1, MOUNT TRANSMITTER TO FLOW TUBE.



CHEMICAL FEED EQUIPMENT ELECTRICAL DETAIL
NOT TO SCALE



GENERAL NOTES:

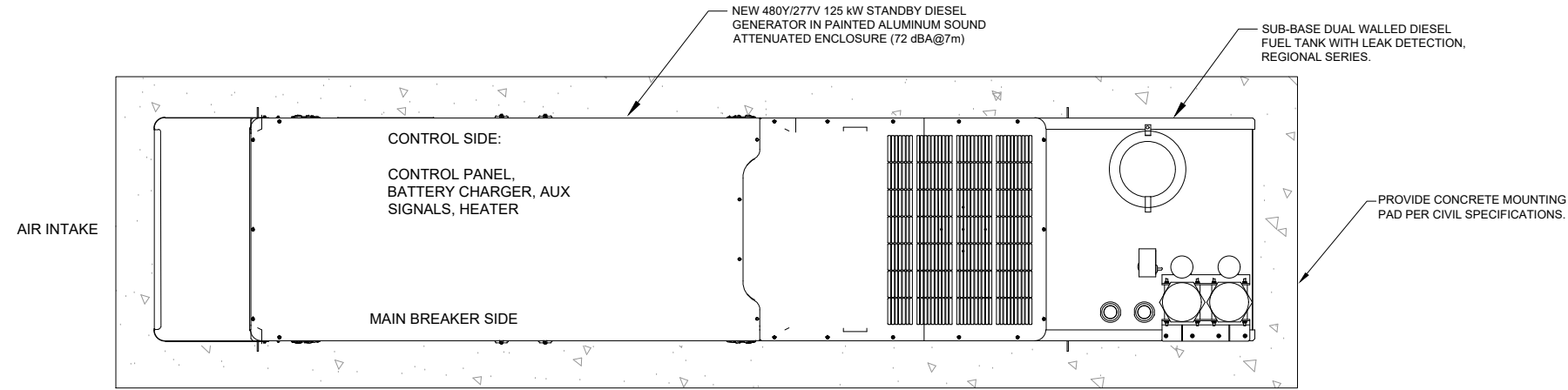
- KEEP UNSCREENED CABLE ENDS AS SHORT AS POSSIBLE.
- SEPARATE ELECTRODE AND COIL CABLES TO PREVENT INTERFERENCE.

REMOTE DISPLAY FLOW METER WIRING DETAIL
SCALE: NONE

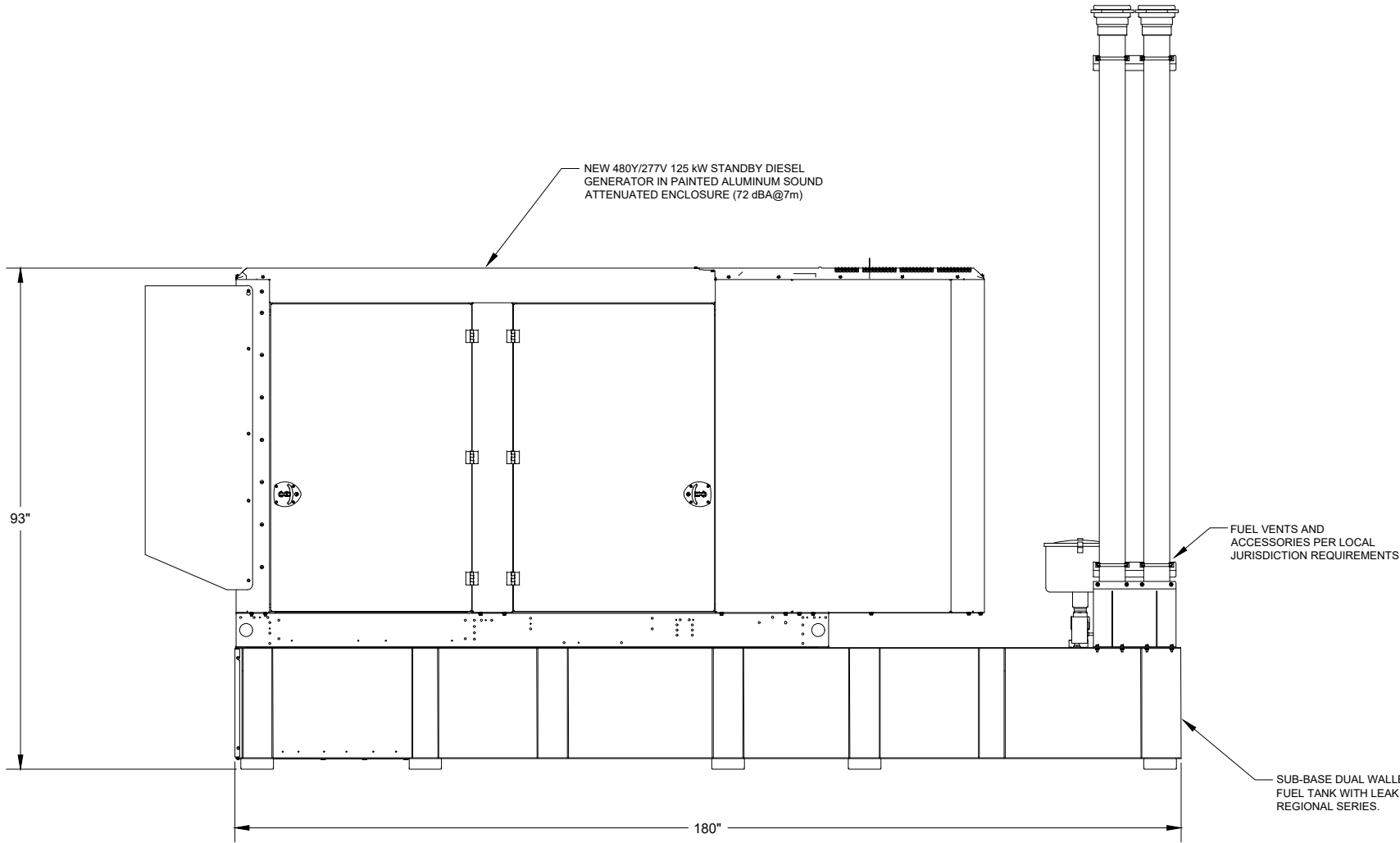


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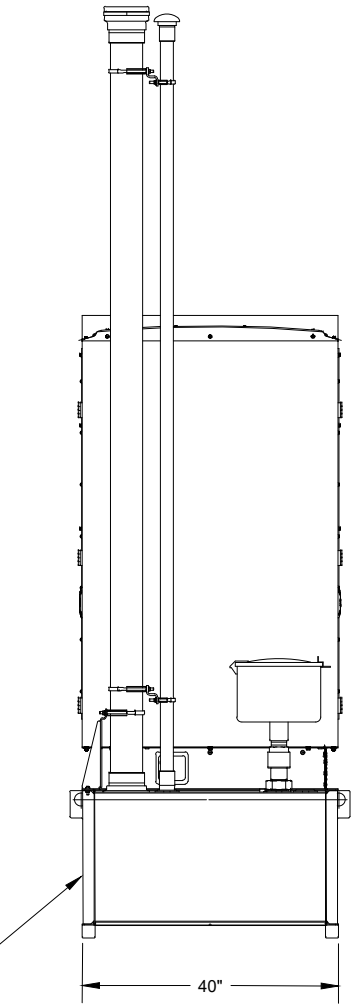
NO.	REVISIONS	BY	DATE



GENERATOR PLAN VIEW
NOT TO SCALE



GENERATOR ELEVATION - SIDE
NOT TO SCALE



GENERATOR ELEVATION - END
NOT TO SCALE

GENERAL NOTES:

1. BASIS OF DESIGN: CUMMINS MODEL C125D6D 60 Hz DIESEL, STANDBY DUTY, EPA TIER 3 EMISSIONS. 125 kW / 156 kVA STANDBY RATING, 480Y/277V THREE PHASE.
2. FUEL TANK: REGIONAL SERIES WITH PRIMARY AND SECONDARY EMERGENCY VENTS, NORMAL VENT, SPILL BOX WITH OVERFILL PREVENTION VALVE, HIGH FUEL ALARM PANEL, LOW FUEL SWITCH, LEAK DETECTION, FUEL GAUGE. 351 GALLON USABLE TANK VOLUME WITH 10.1 GAL/HR PROVIDES 35 HOURS RUNTIME AT FULL LOAD, UP TO 48 HOURS UNDER NORMAL OPERATIONAL LOADS.
3. SEE SPECIFICATIONS FOR ADDITIONAL REQUIREMENTS.



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DRAWN BY	BZ
CHECKED BY	BZ

EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON
SAN JUAN COUNTY
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

STANDBY DIESEL GENERATOR ELEVATIONS

DATE	09-12-2022
SCALE	AS SHOWN
JOB NUMBER	2020-070

SHEET
E10.3

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BID SET